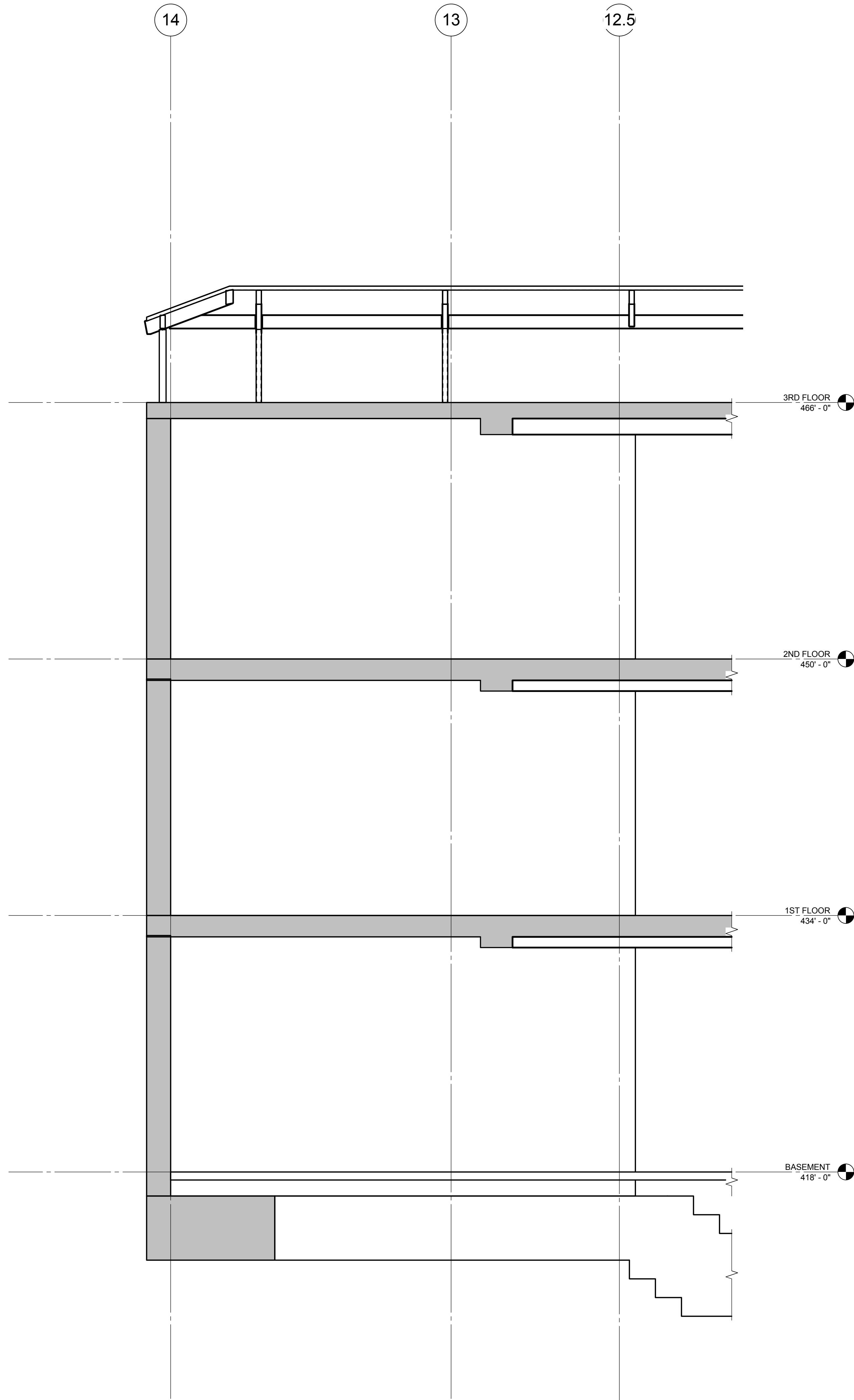
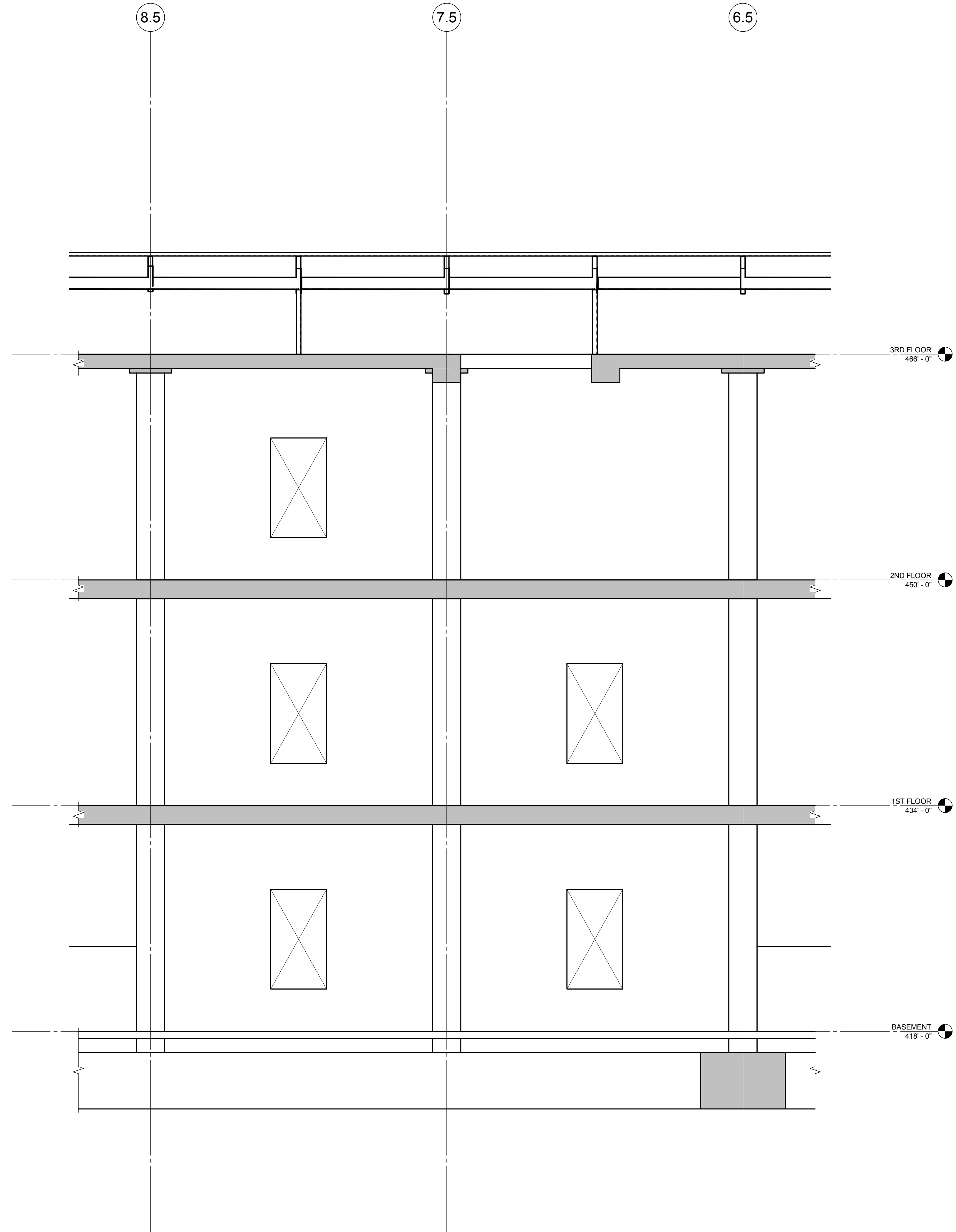




2 WALL ELEVATION NEAR GRIDLINE A
SCALE : 1/4" = 1'-0"



1 WALL ELEVATION AT GRIDLINE A
SCALE : 1/4" = 1'-0"



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CONCRETE SHEAR WALL ELEVATIONS

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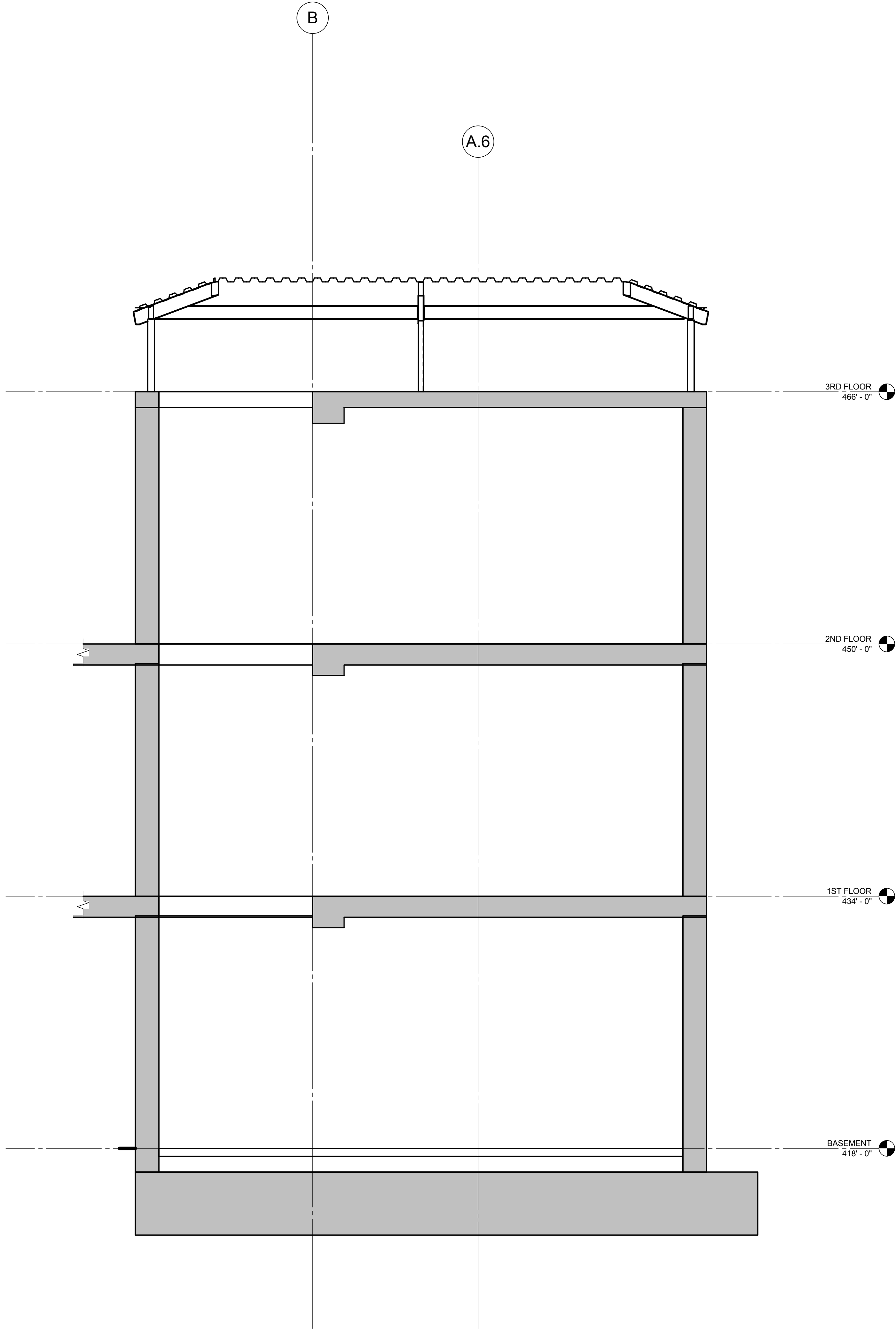
project no. 2014307.00



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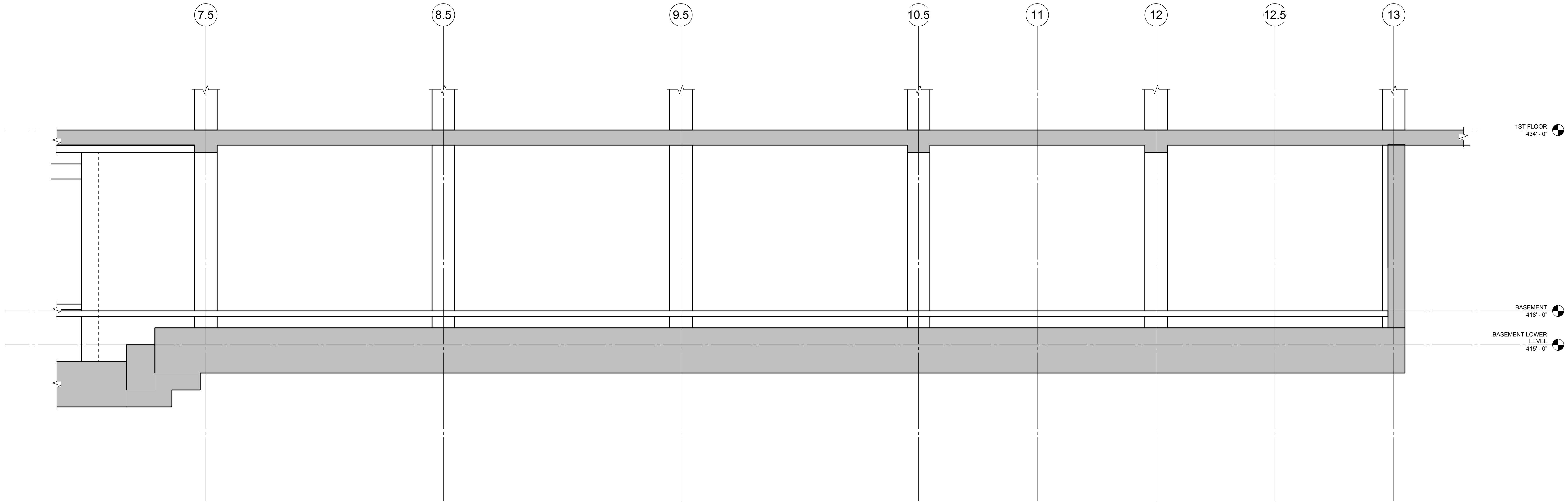
S-400

plot date: 5/7/2015 4:25:43 PM

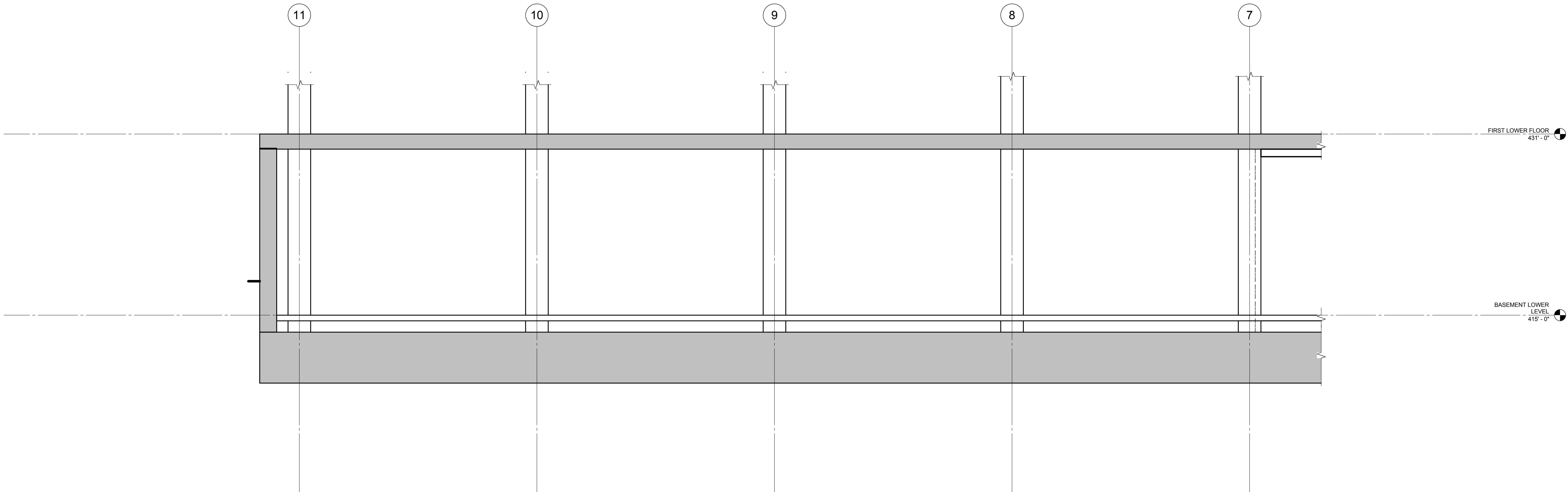


2 WALL ELEVATION AT GRIDLINE 14

SCALE : 1/4" = 1'-0"



1 WALL ELEVATION AT GRIDLINE C
SCALE : 1/4" = 1'-0"



2 WALL ELEVATION AT GRIDLINE F
SCALE : 1/4" = 1'-0"



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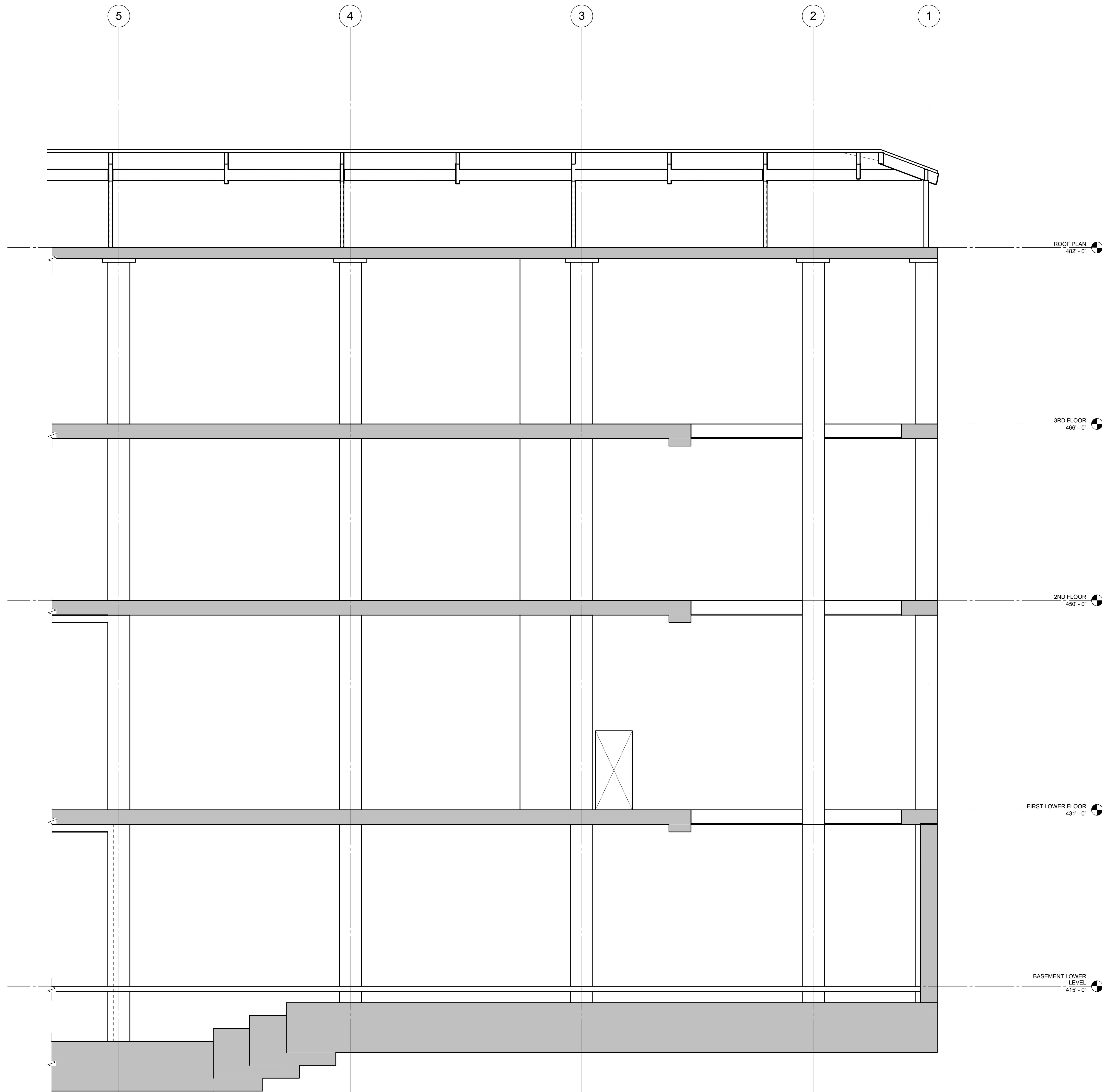
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1 WALL ELEVATION AT GRIDLINE F
SCALE: 1/4" = 1'-0"



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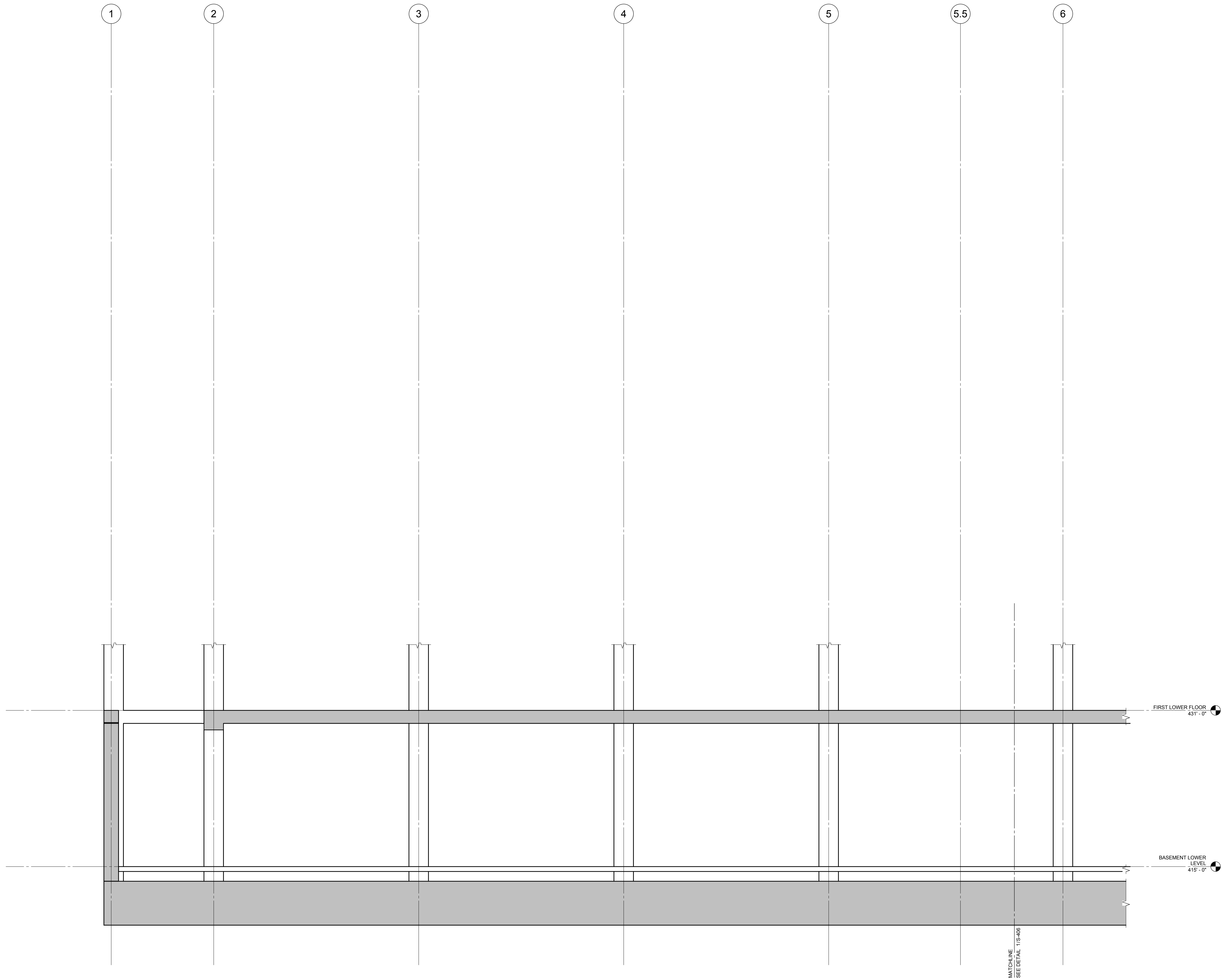
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1 WALL ELEVATION AT GRIDLINE H
SCALE : 1/4" = 1'-0"



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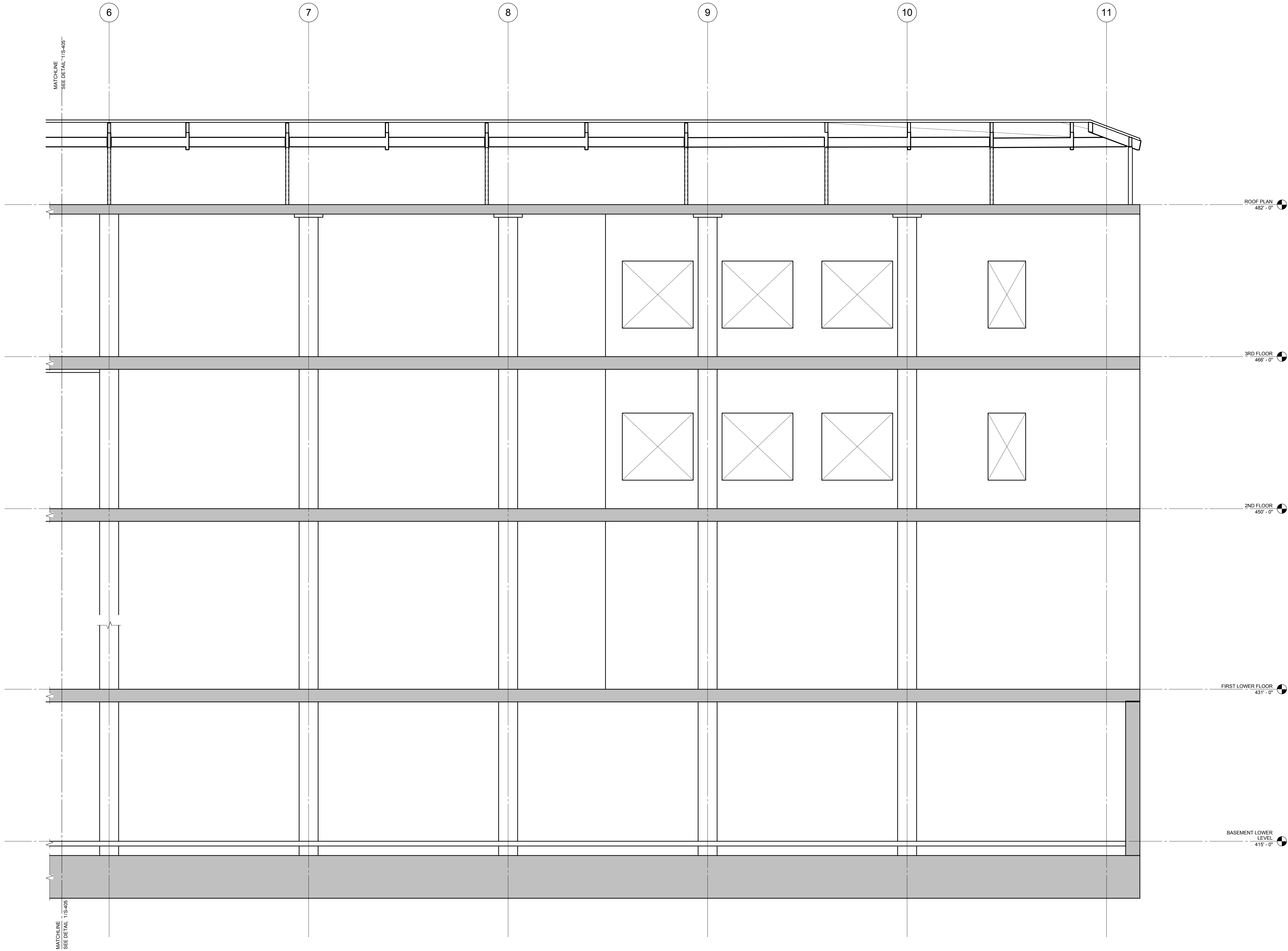
project no. 2014307.00



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1 WALL ELEVATION AT GRIDLINE H

SCALE : 1/4" = 1'-0"

**CONCRETE SHEAR WALL
ELEVATIONS**

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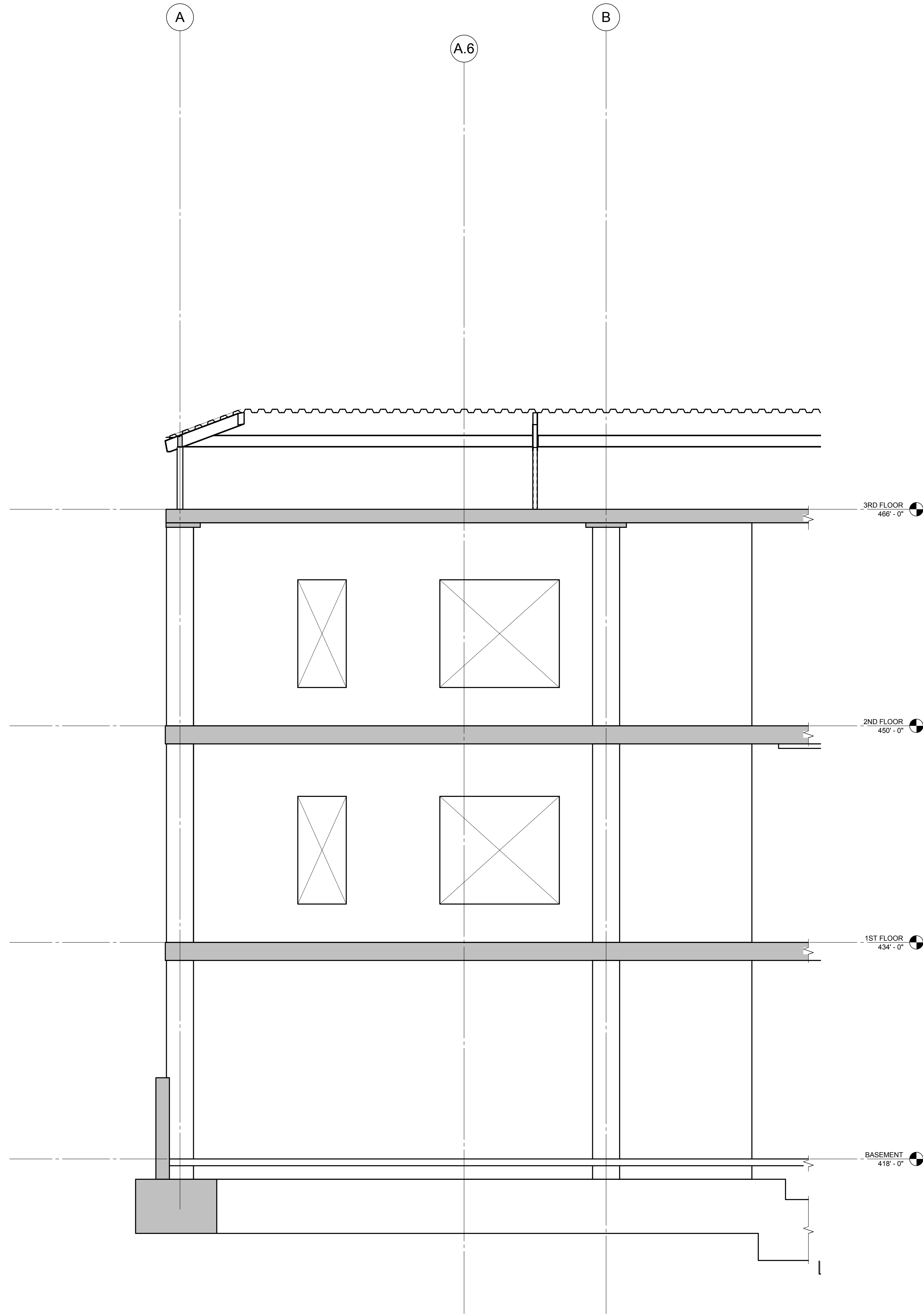
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project no. 2014307.00

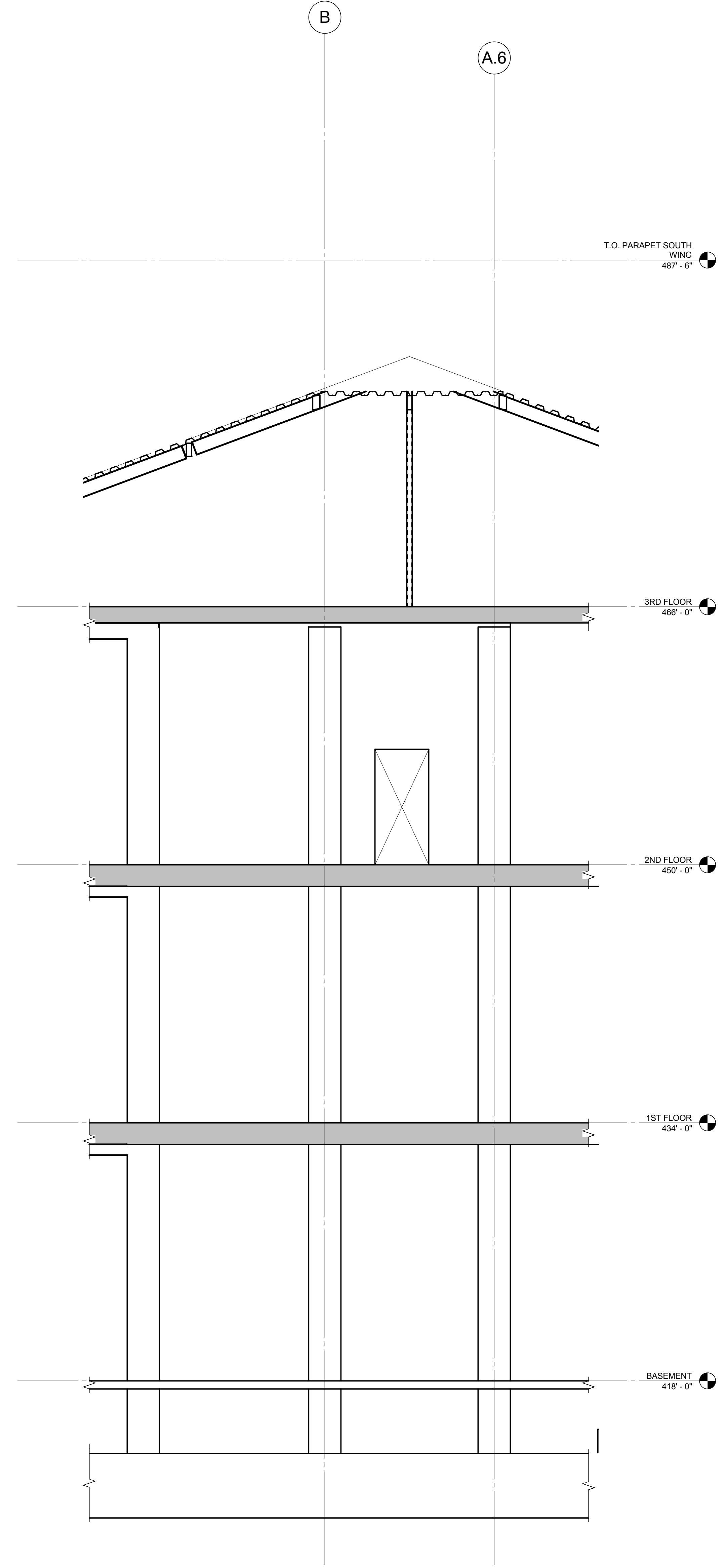


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2 WALL ELEVATION AT GRIDLINE 5.4
SCALE: 1/4" = 1'-0"



1 WALL ELEVATION AT GRIDLINE 6.5
SCALE: 1/4" = 1'-0"



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CONCRETE SHEAR WALL ELEVATIONS

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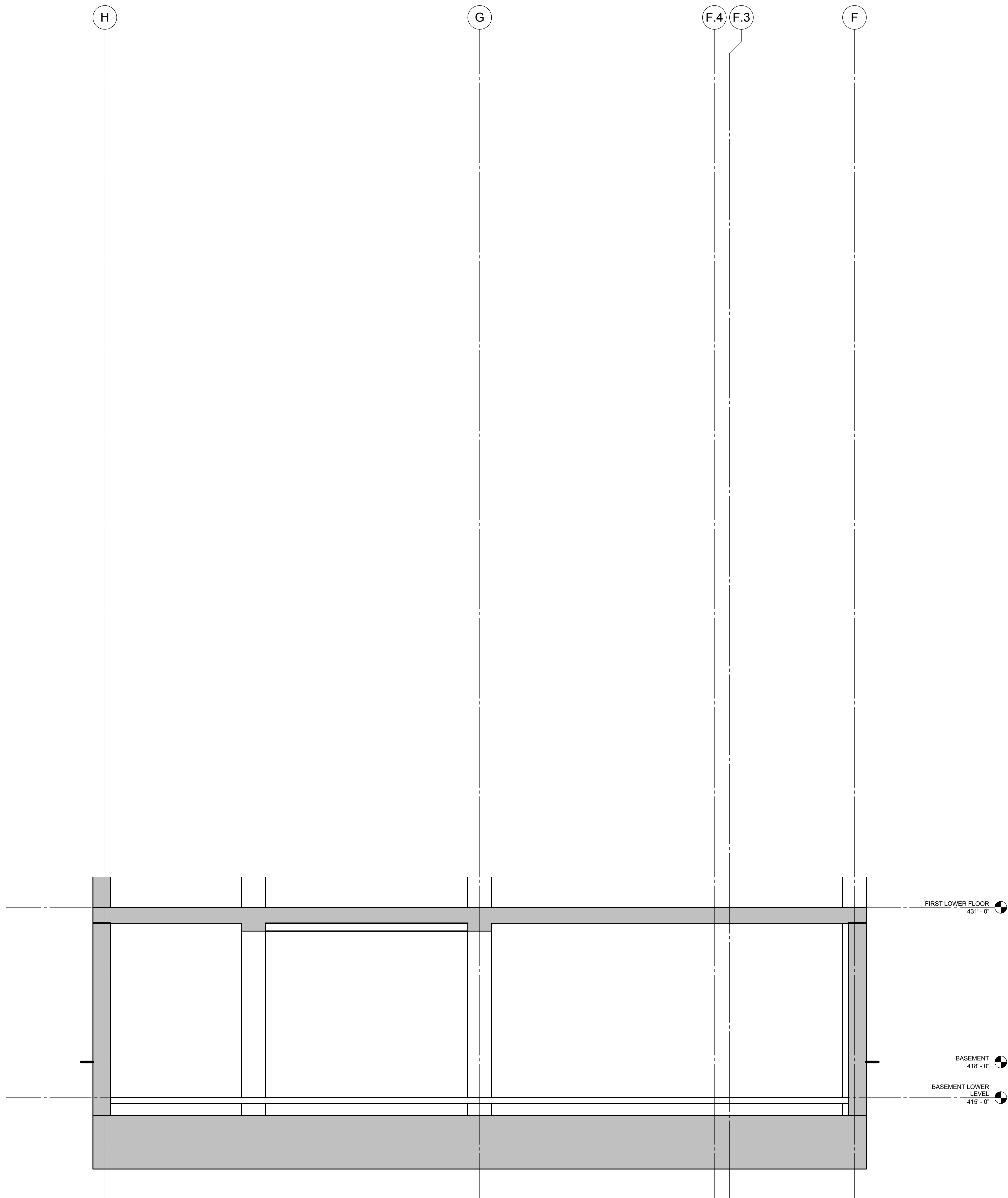
CLARK
CONSTRUCTION

ac martin
PLANNING
ARCHITECTURE
INTERIOR ARCHITECTURE
RESEARCH

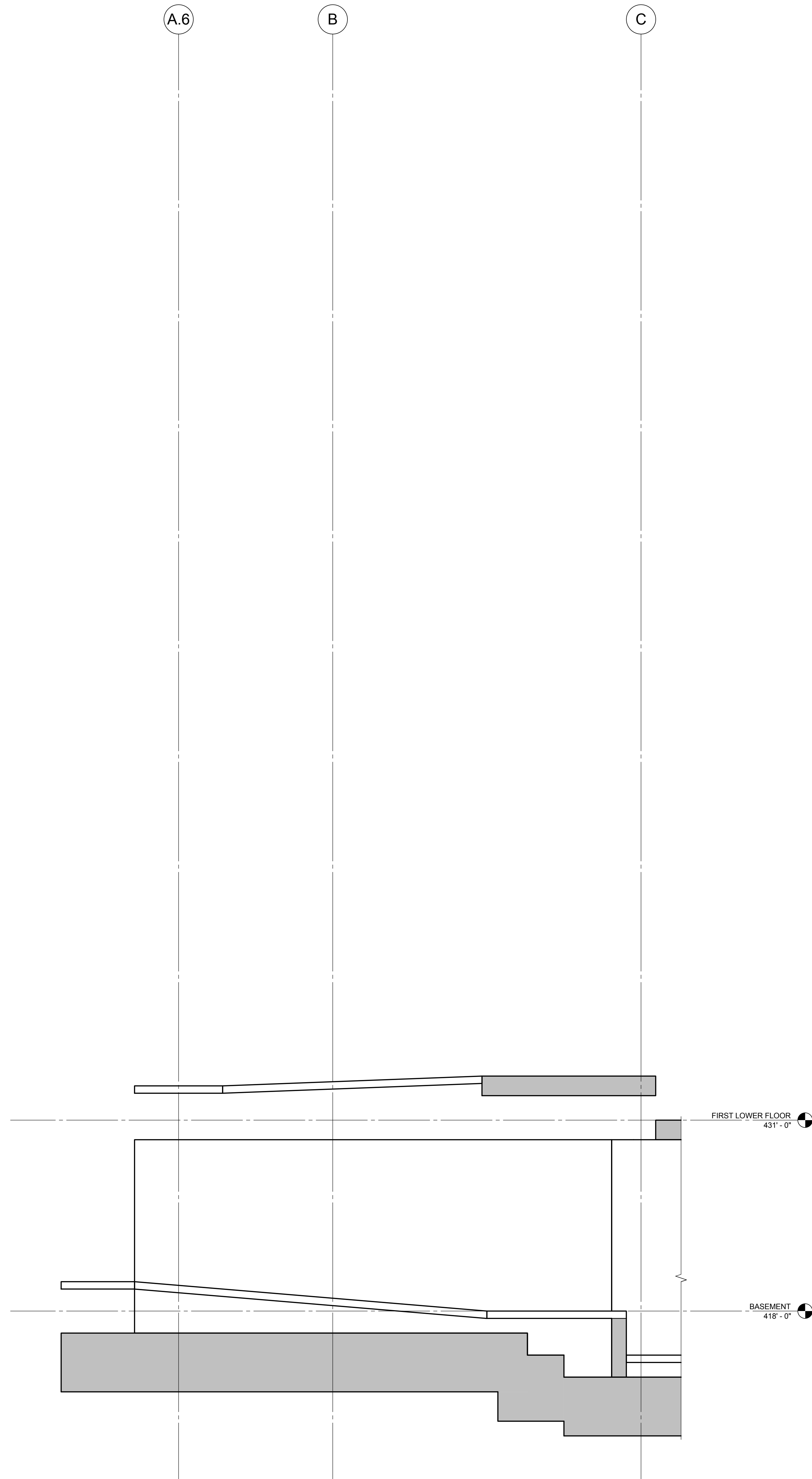
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2 WALL ELEVATION AT GRIDLINE 11
SCALE : 1/4" = 1'-0"



1 WALL ELEVATION NEAR GRIDLINE 5
SCALE : 1/4" = 1'-0"



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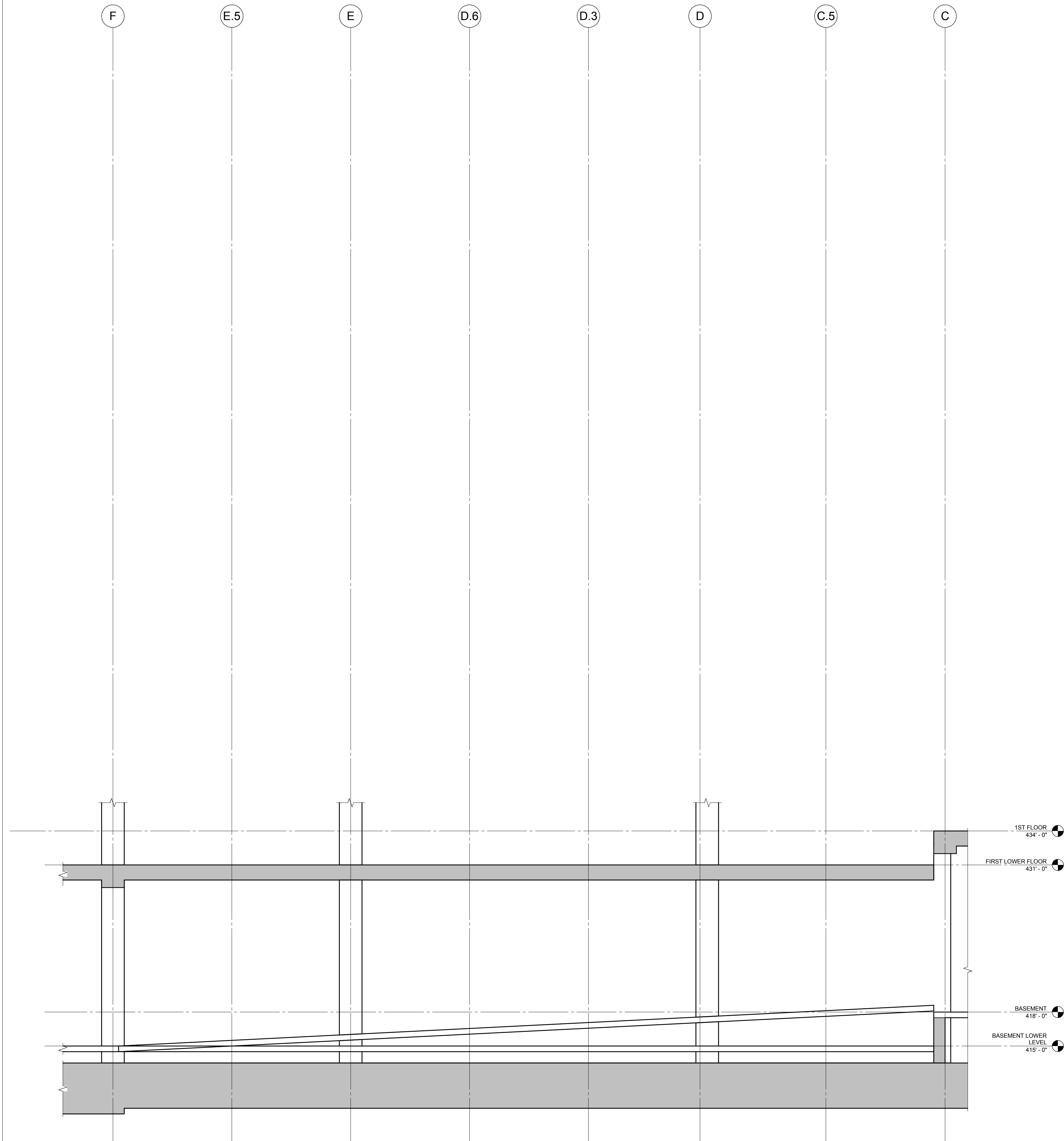
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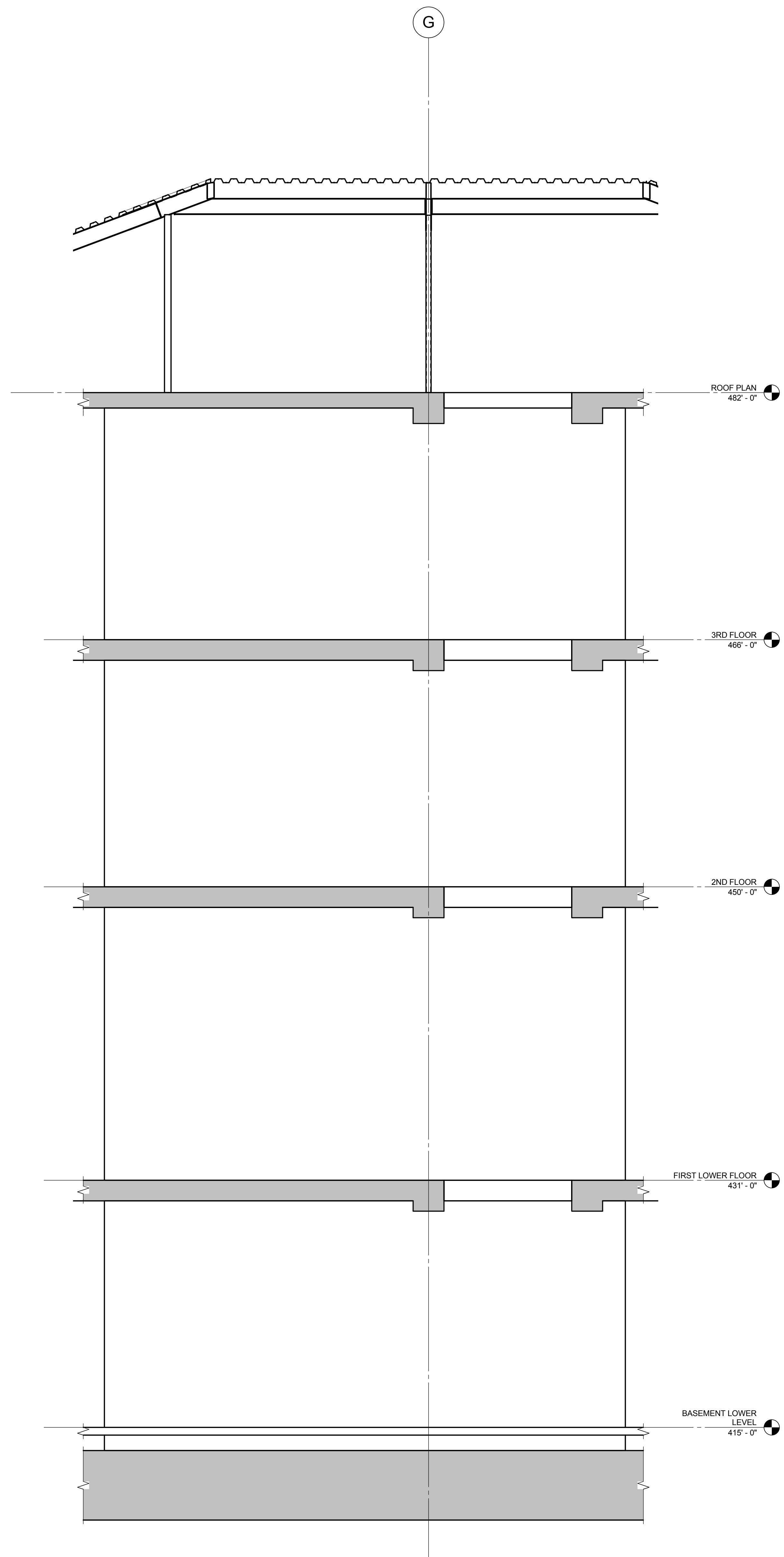
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2 WALL ELEVATION AT GRIDLINE 7
SCALE: 1/4" = 1'-0"



1 WALL ELEVATION NEAR GRIDLINE 10
SCALE: 1/4" = 1'-0"



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CONCRETE SHEAR WALL ELEVATIONS

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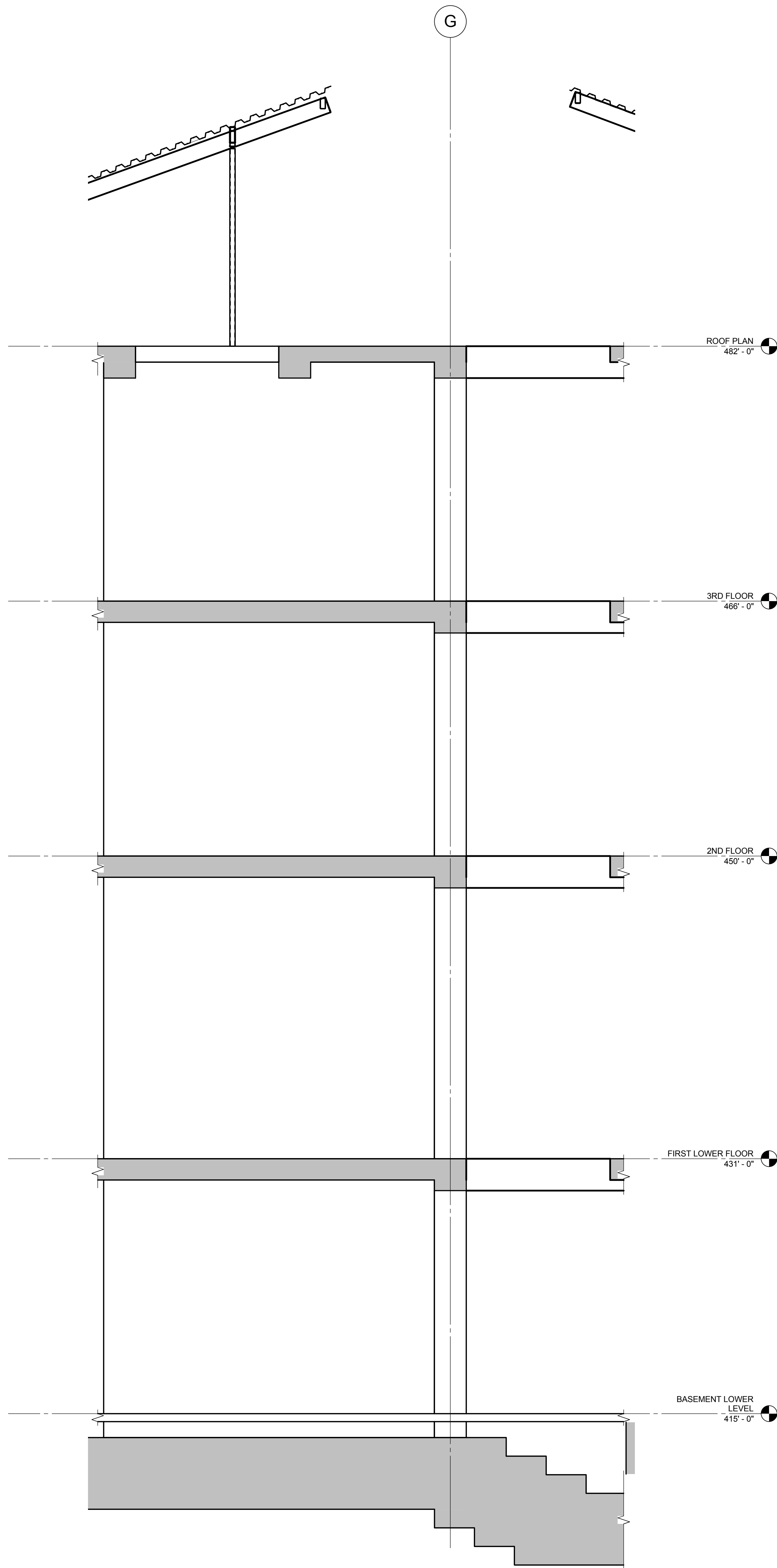
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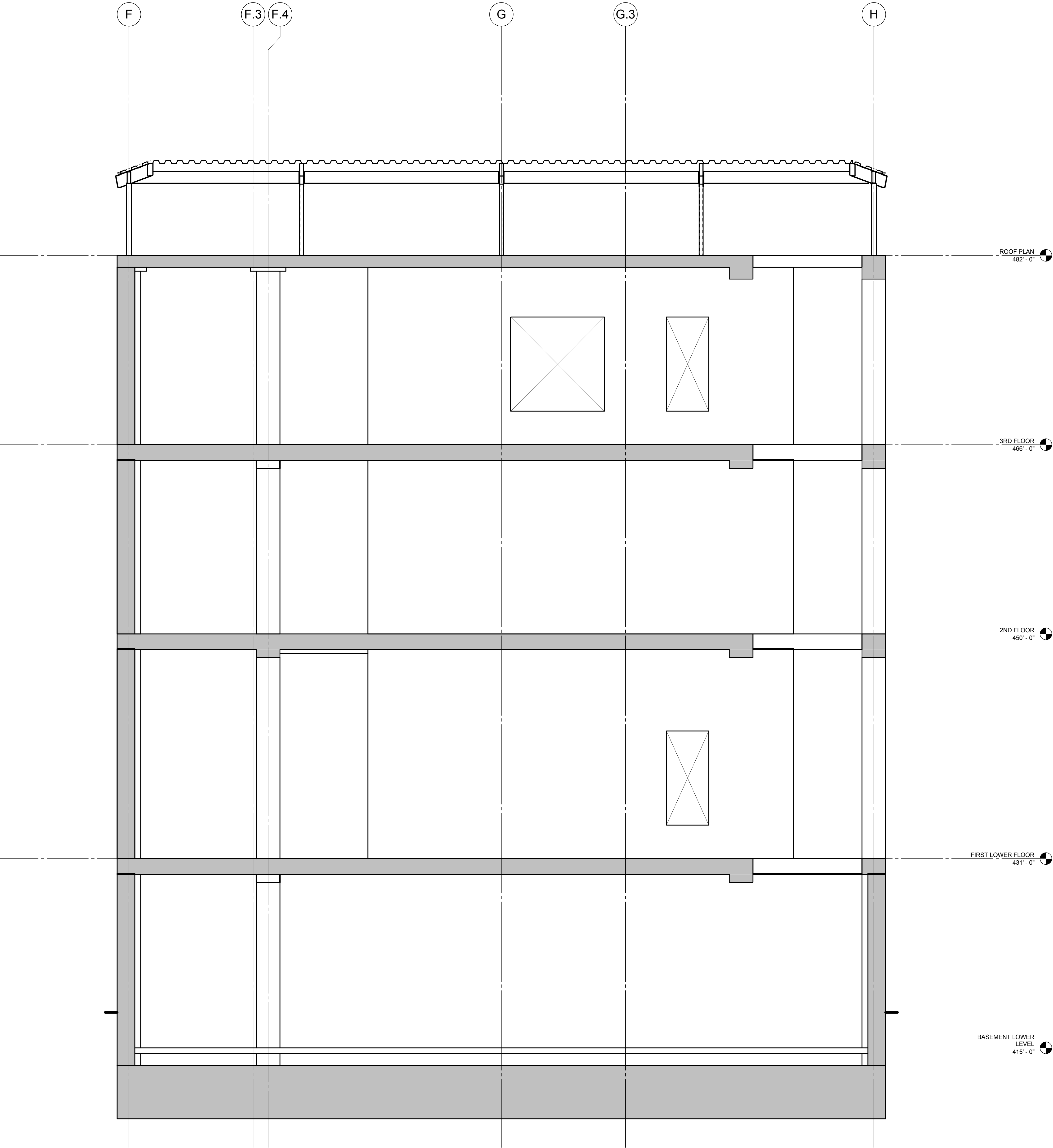
S-409

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1 WALL ELEVATION AT GRIDLINE 6
SCALE : 1/4" = 1'-0"

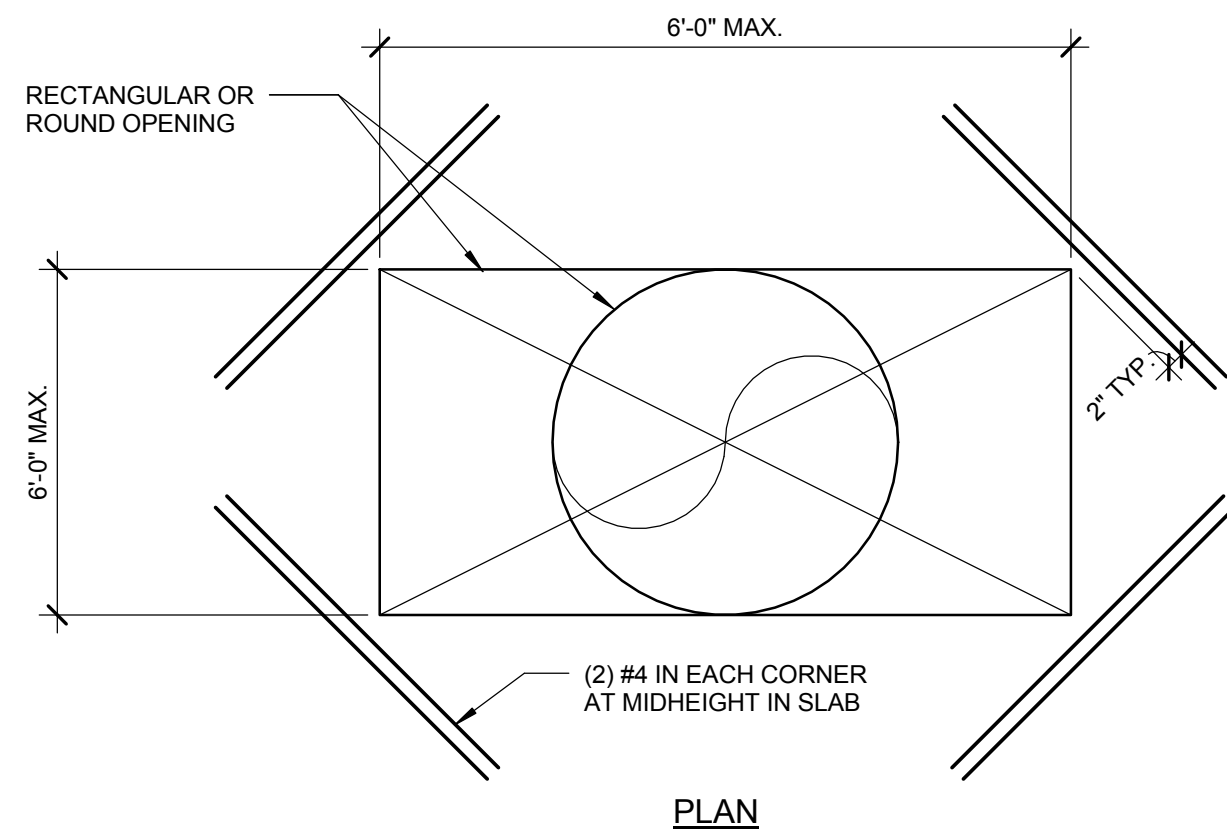




1 WALL ELEVATION AT GRIDLINE 1
SCALE: 1/4" = 1'-0"

7 OPENING IN SLAB ON GRADE

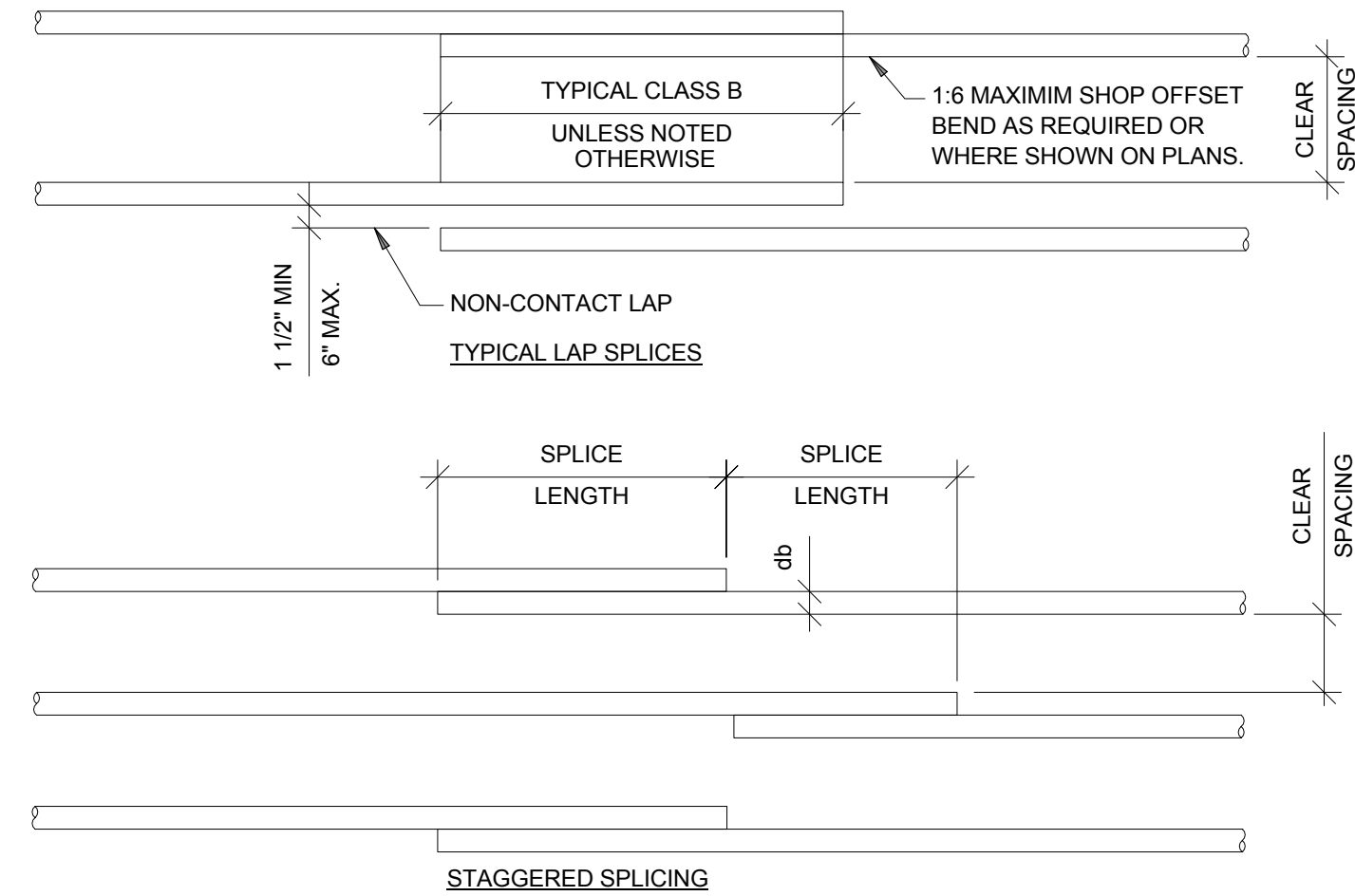
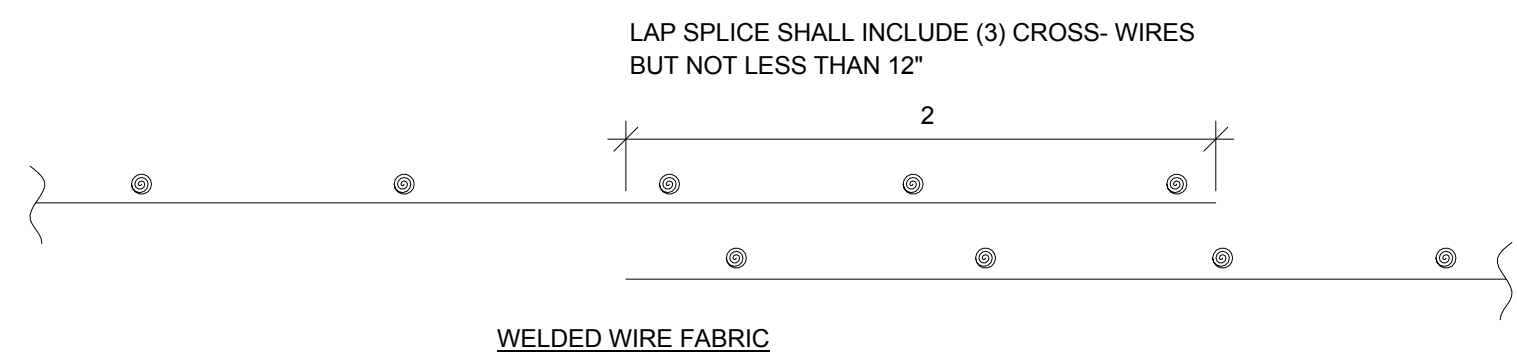
SCALE : 1" = 1'-0"



BAR DESCRIPTION AND LOCATION IN STRUCTURE	CONCRETE STRENGTH (PSI)	BAR SIZE	LAP CLASS	NON-SHEAR WALL REINFORCING BAR TENSION SPLICE LENGTH (GRADE 60)													
				#11		#10		#9		#8		#7		#6		#5	
				A	B	A	B	A	B	A	B	A	B	A	B	A	B
CLEAR SPACING OF BARS NOT LESS THAN 4db, CLEAR COVER NOT LESS THAN 4db, AND STIRRUPS AND TIES THROUGHOUT IS NOT LESS THAN THE CODE MINIMUM OR CLEAR SPACING OF BARS NOT LESS THAN 2db AND CLEAR COVER NOT LESS THAN 4db	3000 WVC	4000 & HIGHER WVC	TOP	8'-2"	10'-7"	7'-5"	9'-8"	6'-8"	8'-8"	5'-11"	7'-9"	5'-2"	6'-9"	3'-7"	4'-8"	3'-0"	3'-10"
			OTHER	6'-3"	8'-2"	5'-8"	7'-5"	5'-2"	6'-8"	4'-7"	5'-11"	4'-0"	5'-2"	2'-9"	3'-7"	2'-3"	3'-0"
	3000 LVC	4000 & HIGHER WVC	TOP	7'-1"	9'-2"	6'-5"	8'-4"	5'-9"	7'-6"	5'-2"	6'-8"	4'-6"	5'-10"	3'-1"	4'-0"	2'-7"	3'-4"
			OTHER	5'-5"	7'-1"	4'-11"	6'-5"	4'-5"	5'-9"	3'-11"	5'-2"	3'-6"	4'-6"	2'-4"	3'-1"	2'-0"	2'-7"
	3000 WVC	4000 & HIGHER WVC	TOP	11'-0"	14'-3"	9'-11"	12'-10"	8'-10"	11'-5"	7'-10"	10'-1"	6'-10"	8'-10"	4'-8"	6'-1"	3'-11"	5'-1"
			OTHER	8'-6"	11'-0"	7'-7"	9'-11"	6'-10"	8'-10"	6'-0"	7'-10"	5'-3"	6'-10"	3'-7"	4'-8"	3'-1"	3'-11"
	3000 WVC	4000 & HIGHER WVC	TOP	12'-3"	15'-11"	11'-2"	14'-6"	10'-0"	13'-0"	8'-11"	11'-7"	7'-9"	10'-1"	5'-4"	6'-11"	4'-5"	3'-11"
			OTHER	9'-5"	12'-3"	8'-7"	11'-2"	7'-8"	10'-0"	6'-10"	8'-11"	6'-0"	7'-9"	4'-1"	5'-4"	3'-5"	4'-5"
	3000 LVC	4000 & HIGHER WVC	TOP	10'-7"	13'-9"	9'-8"	12'-6"	8'-8"	11'-3"	7'-8"	10'-0"	6'-9"	8'-9"	4'-7"	6'-0"	3'-10"	5'-0"
			OTHER	8'-2"	10'-7"	7'-5"	9'-8"	6'-8"	8'-8"	5'-11"	7'-8"	5'-2"	6'-9"	3'-7"	4'-7"	3'-0"	3'-10"
	3000 WVC	4000 & HIGHER WVC	TOP	16'-5"	21'-3"	14'-9"	19'-3"	13'-2"	17'-1"	11'-8"	15'-1"	10'-3"	13'-3"	7'-0"	9'-2"	5'-11"	7'-7"
			OTHER	12'-7"	16'-5"	11'-5"	14'-9"	10'-1"	13'-2"	8'-11"	11'-8"	7'-10"	10'-3"	5'-5"	7'-0"	4'-6"	5'-11"

NOTES:

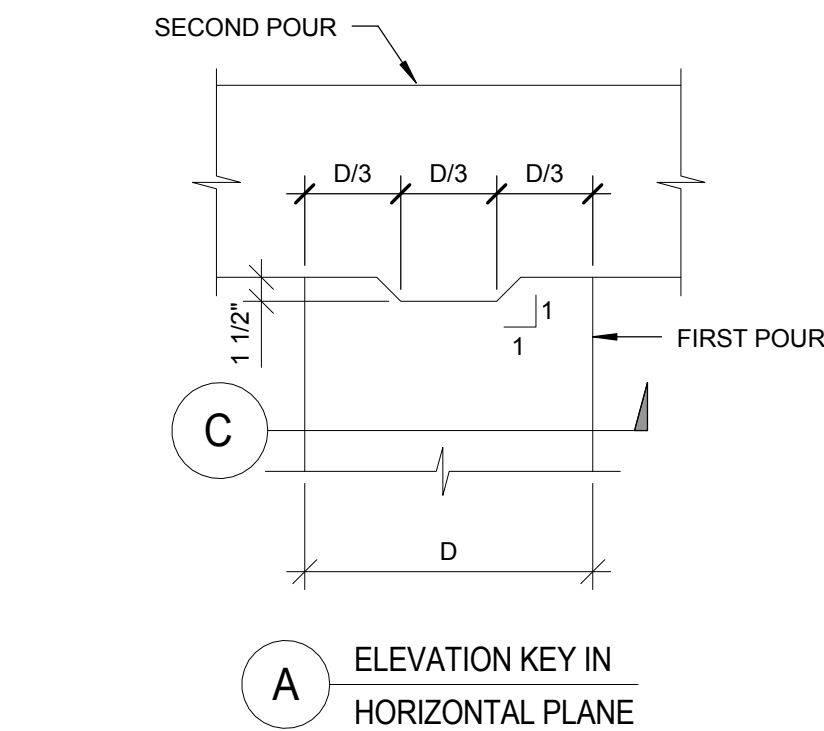
1. BAR SPLICES NOT COVERED BY THIS SCHEDULE ARE SPECIFICALLY DETAILED AND DIMENSIONED ON PLANS.
2. FOR TENSION DEVELOPMENT LENGTHS l_{dh} , USE CLASS "A" SPLICE LENGTHS.
3. ALL SPLICES SHALL BE CLASS "B" UNLESS NOTED OTHERWISE ON PLANS.
4. TOP BARS ARE HORIZONTAL REINFORCEMENT WITH MORE THAN 12" OF CONCRETE CAST BELOW BAR.
5. OTHER BARS ARE VERTICAL REINFORCEMENT, AND HORIZONTAL REINFORCEMENT WITH LESS THAN 12" OF CONCRETE CAST BELOW BAR.
6. COVER DESIGNATES CLEAR CONCRETE COVER FROM SPLICED BAR TO FACE OF MEMBER. SPACING DESIGNATES CENTER - TO - CENTER SPACING OF SPLICED BARS.
7. FOR SHEAR WALLS, DEVELOPMENT AND SPLICE LENGTHS SHALL BE 1.25 TIMES THE SCHEDULED LENGTHS.
8. FOR CMU DEVELOPMENT AND SPLICES SEE 1/GS6.92.



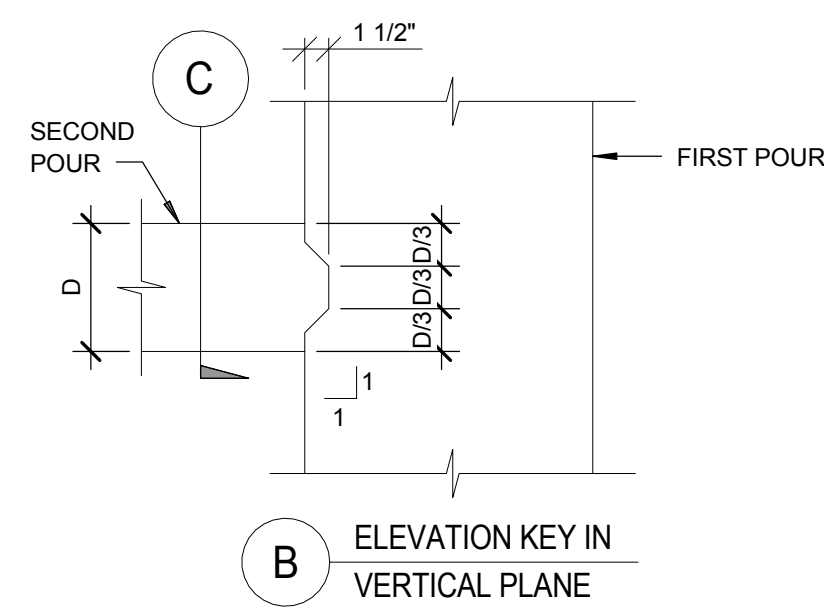
STAGGERED SPLICING

1 DEVELOPMENT AND SPLICES OF CONCRETE REINFORCING BARS (NON SHEAR WALL)

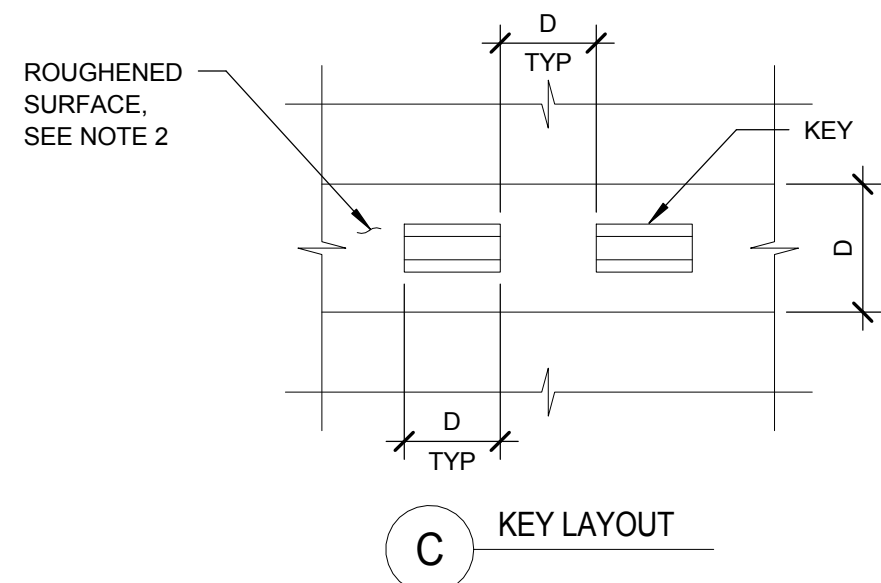
SCALE : 1" = 1'-0"



A ELEVATION KEY IN HORIZONTAL PLANE



B ELEVATION KEY IN VERTICAL PLANE



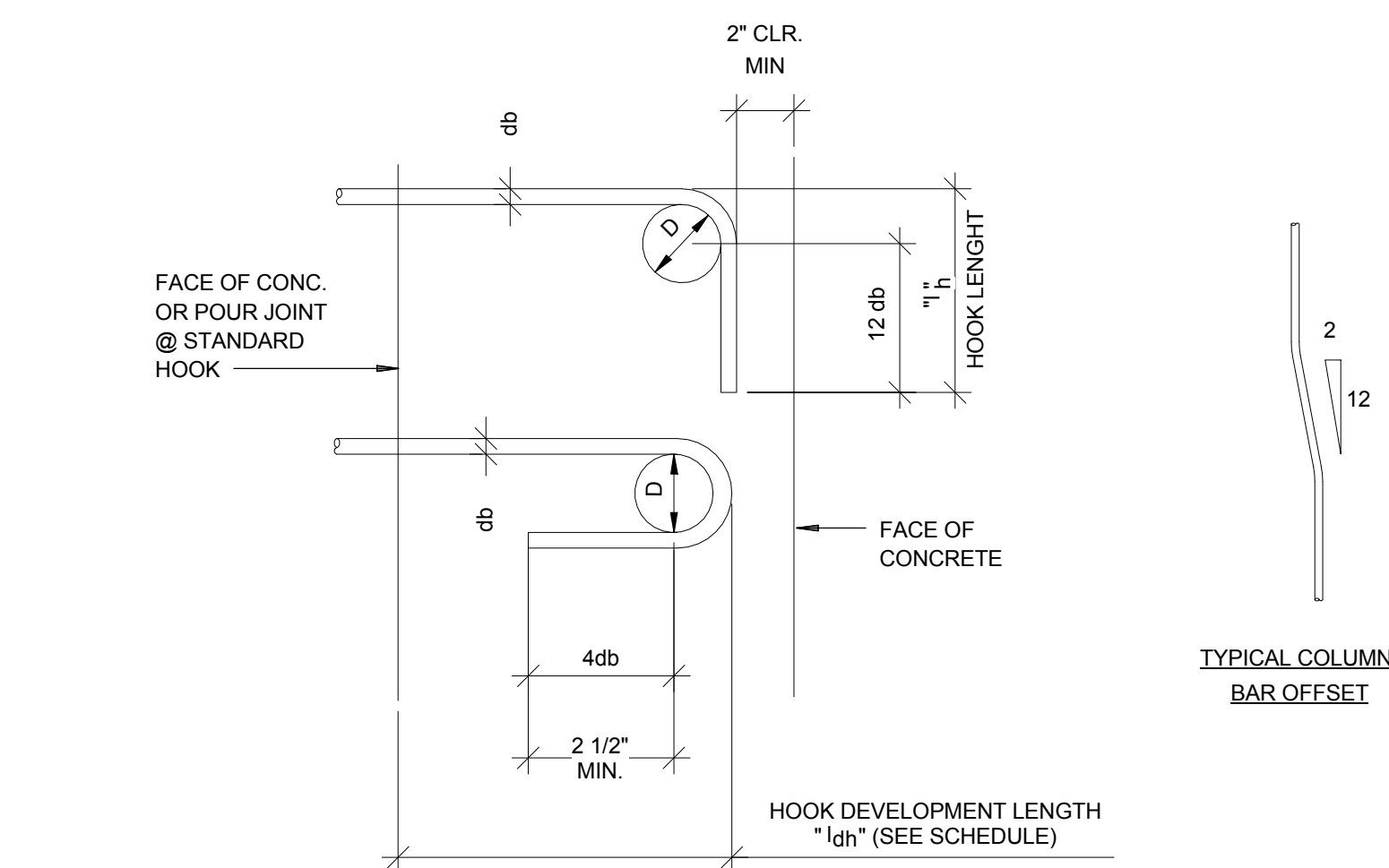
C KEY LAYOUT

NOTES:

1. CONCRETE REINFORCEMENT NOT SHOWN FOR CLARITY.
2. SURFACE OF CONCRETE CONSTRUCTION JOINTS SHALL BE CLEANED, ROUGHENED TO 1/4" AMPLITUDE AND LAITANCE REMOVED.
3. PROVIDE CONTINUOUS KEY INDICATED ON DRAWINGS.

4 STANDARD HOOK DEVELOPMENT LENGTH BENDING DETAIL

SCALE : 1" = 1'-0"



TYPICAL COLUMN BAR OFFSET

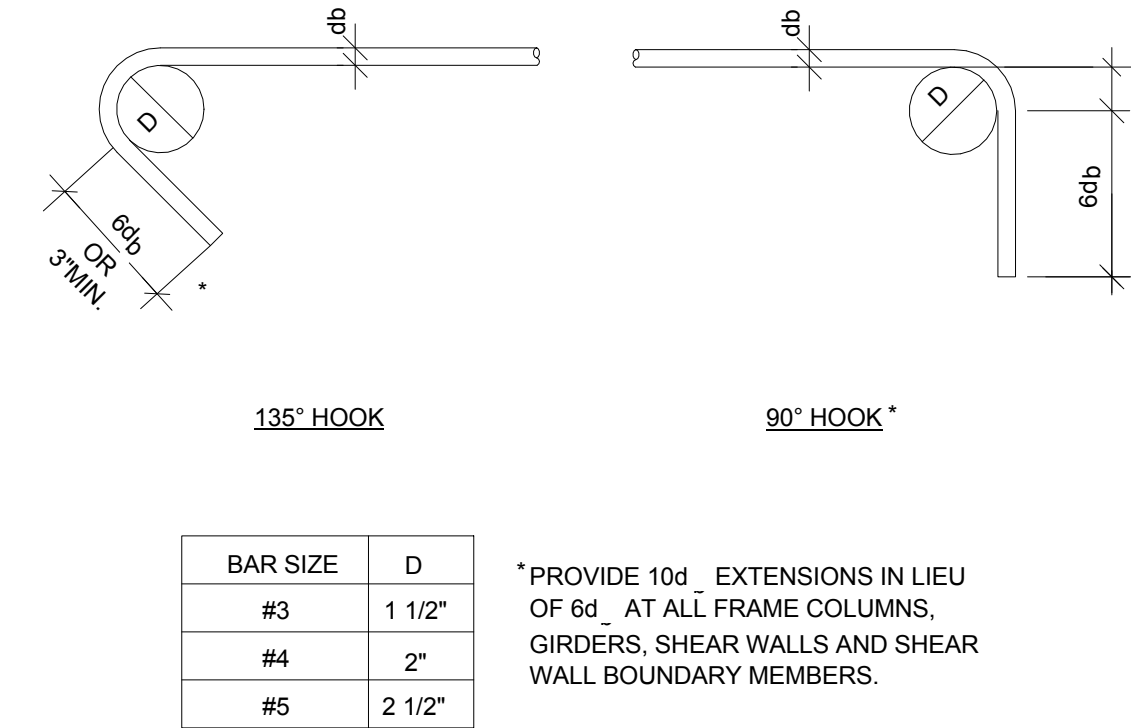
NOTES:

1. ALL HOOKED BARS SHALL EXTEND AS FAR AS POSSIBLE WITH A MINIMUM 2" END COVER AND WITH EMBEDMENT NOT LESS THAN SHOWN ON THE SCHEDULE. UNO ON PLANS.
2. MINIMUM SIDE COVER = 2 1/2"
3. D = FINISHED INSIDE BEND DIAMETER
4. db = NOMINAL BAR DIAMETER
5. SEE 4/- FOR BALANCE OF INFO.

STANDARD HOOK DEVELOPMENT LENGTH * l _{dh} *							
BAR SIZE	D	l _h	NORMAL WEIGHT			LIGHT WEIGHT	
			3000	4000	5000	3000	4000
#3	2 1/4"	5"	0'-9"	0'-8"	0'-7"	0'-11"	0'-10"
#4	3"	8"	0'-11"	0'-10"	0'-9"	1'-3"	1'-1"
#5	3 3/4"	10"	1'-2"	1'-0"	0'-11"	1'-6"	1'-4"
#6	4 1/2"	12"	1'-5"	1'-3"	1'-1"	1'-10"	1'-7"
#7	5 1/4"	1'-2"	1'-8"	1'-5"	1'-3"	2'-1"	1'-10"
#8	6"	1'-4"	1'-10"	1'-7"	1'-5"	2'-5"	2'-1"
#9	9 1/2"	1'-7 1/2"	2'-1"	1'-10"	1'-8"	2'-9"	2'-4"
#10	10 3/4"	1'-10"	2'-4"	2'-1"	1'-10"	3'-1"	2'-8"
#11	12"	2'-0 1/2"	2'-7"	2'-3"	2'-0"	3'-5"	2'-11"

2 STANDARD HOOK LENGTH SCHEDULE

SCALE : 1" = 1'-0"



BAR SIZE	D
#3	1 1/2"
#4	2"
#5	2 1/2"

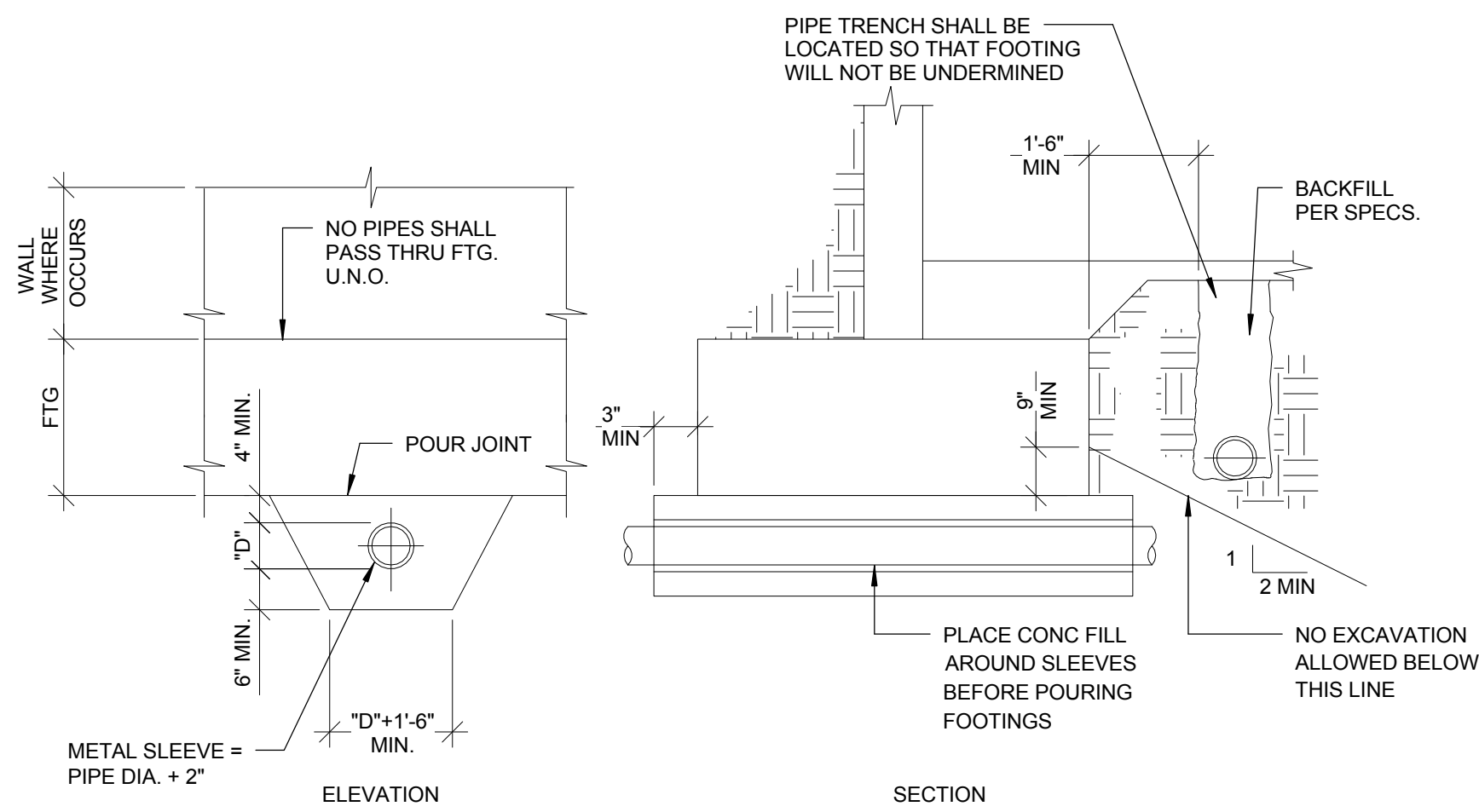
* PROVIDE 10d EXTENSIONS IN LIEU OF 6d AT ALL FRAME COLUMNS, GIRDERS, SHEAR WALLS AND SHEAR WALL BOUNDARY MEMBERS.

6 STAGGERED SHEAR KEY DETAIL

SCALE : 1" = 1'-0"

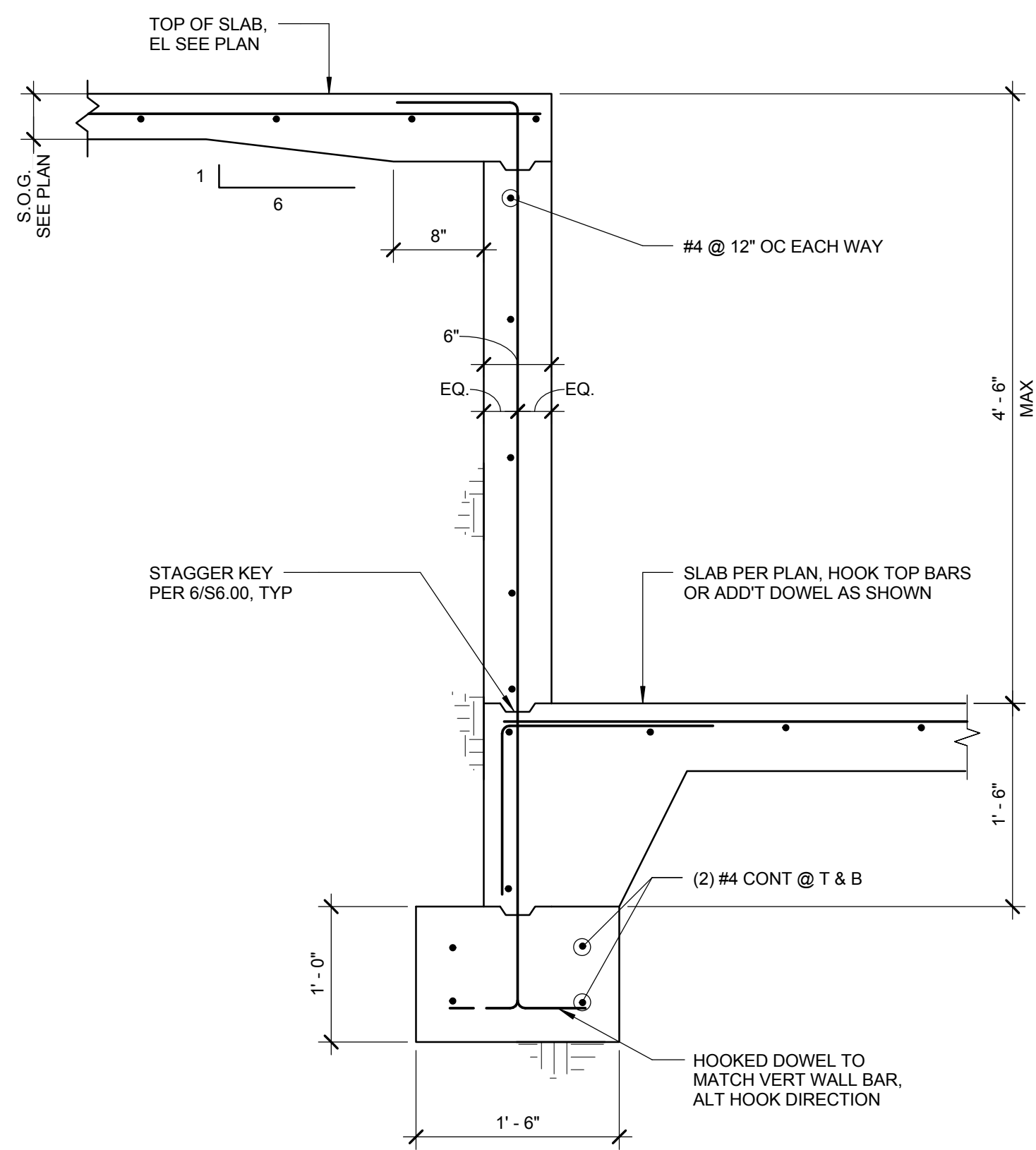
5 PIPE TRENCH/FOOTING DETAIL

SCALE : 1" = 1'-0"

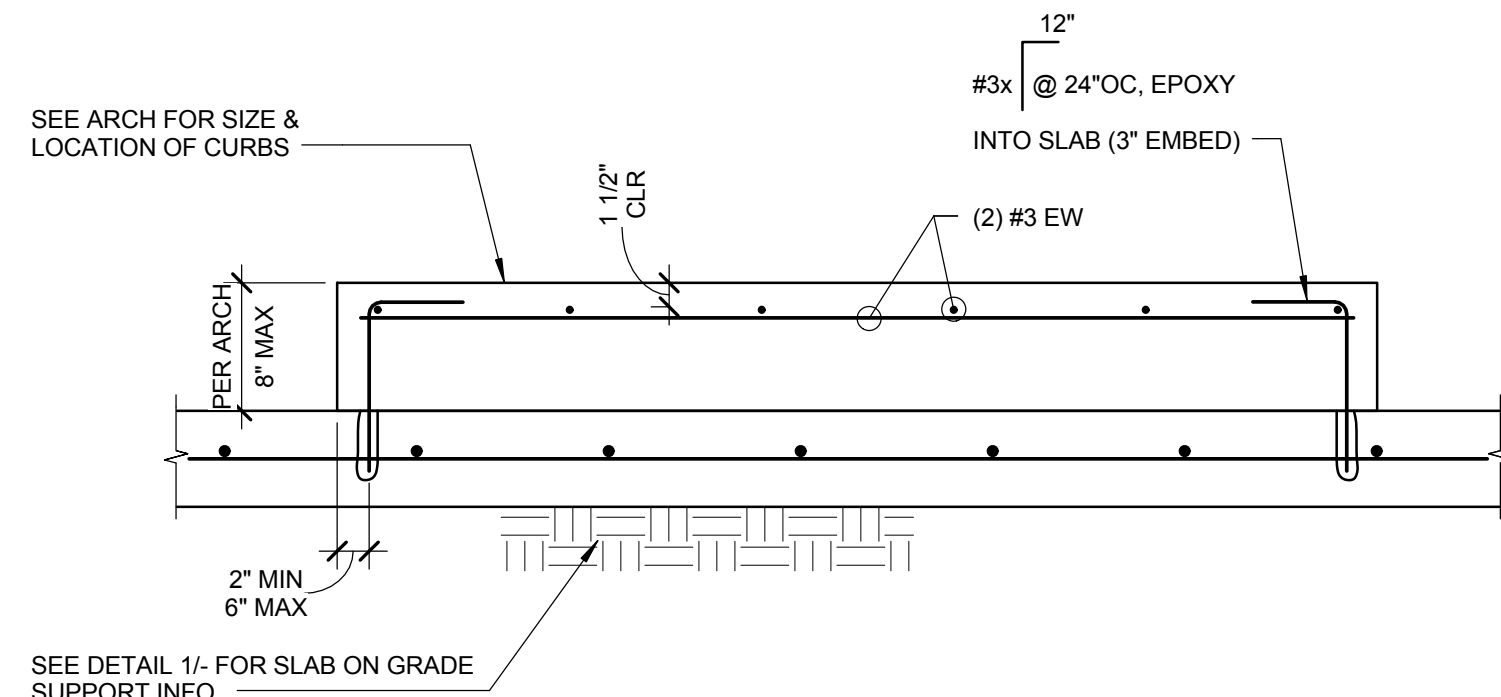


3 TIES & STIRRUPS DETAILS

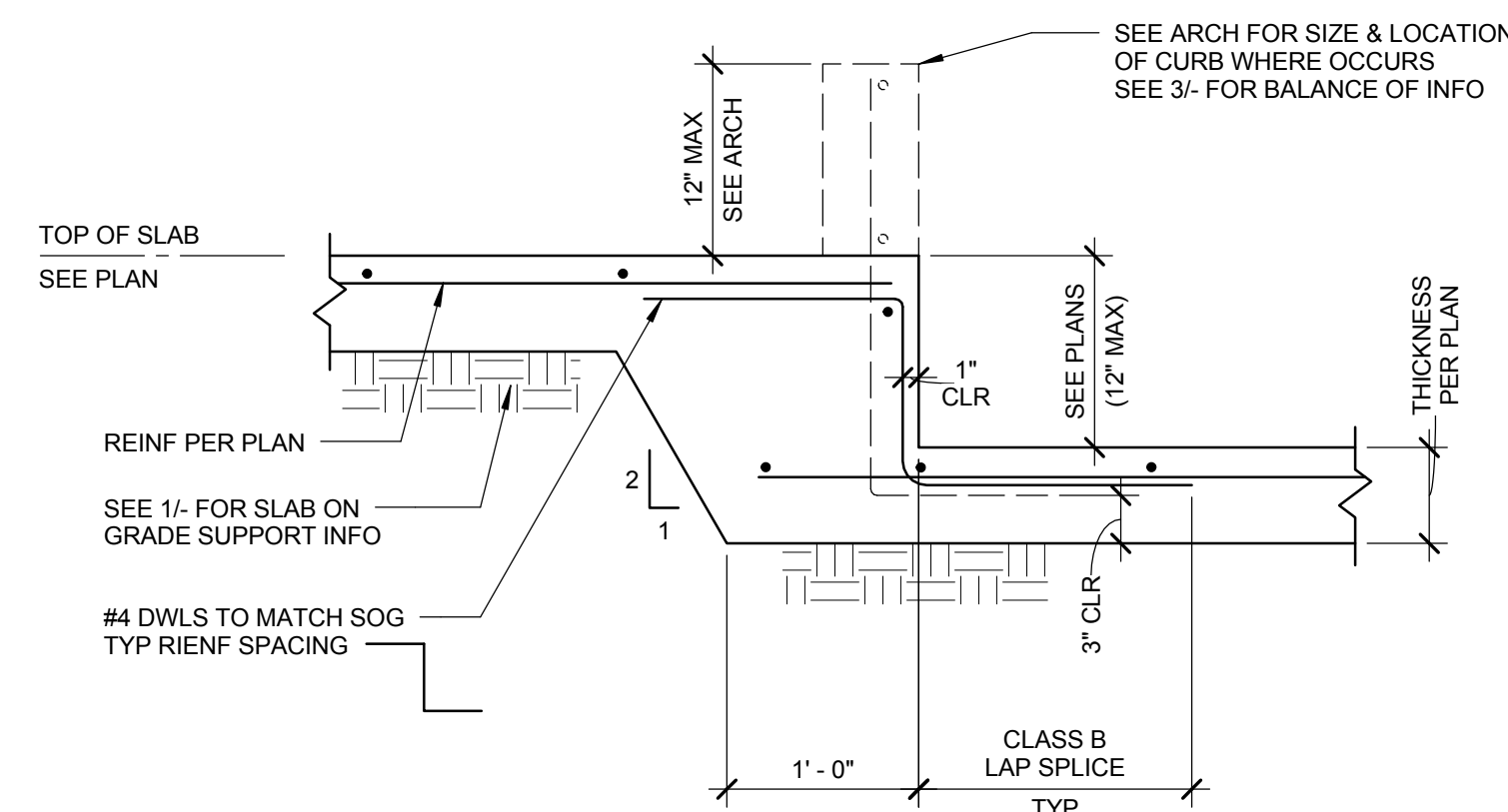
SCALE : 1" = 1'-0"



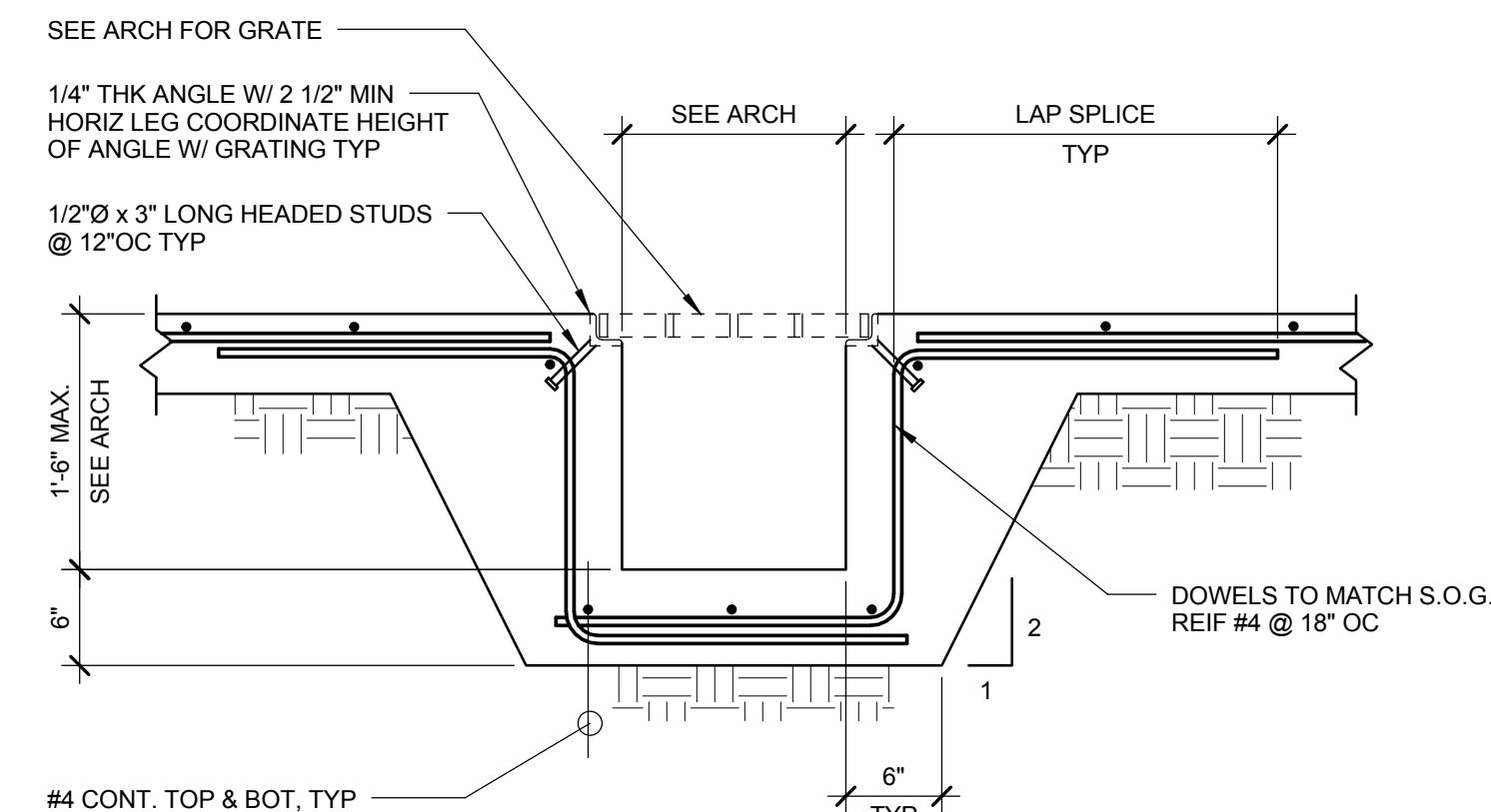
12 SECTION AT FOUNDATION WALL
SCALE : 1" = 1'-0"



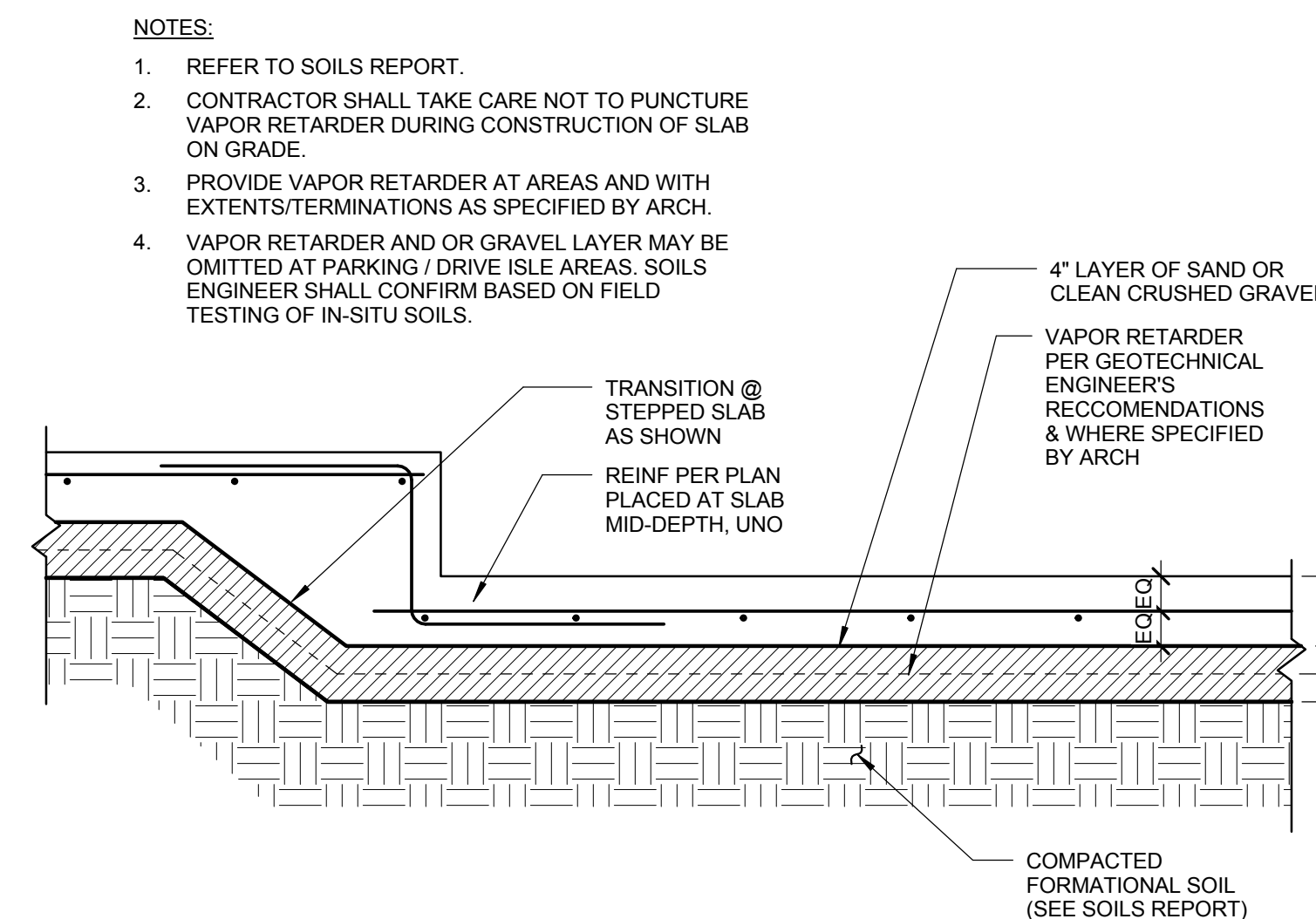
9 CONCRETE CURB AND RAISED ISLAND AT SLAB ON GRADE
SCALE : 1" = 1'-0"



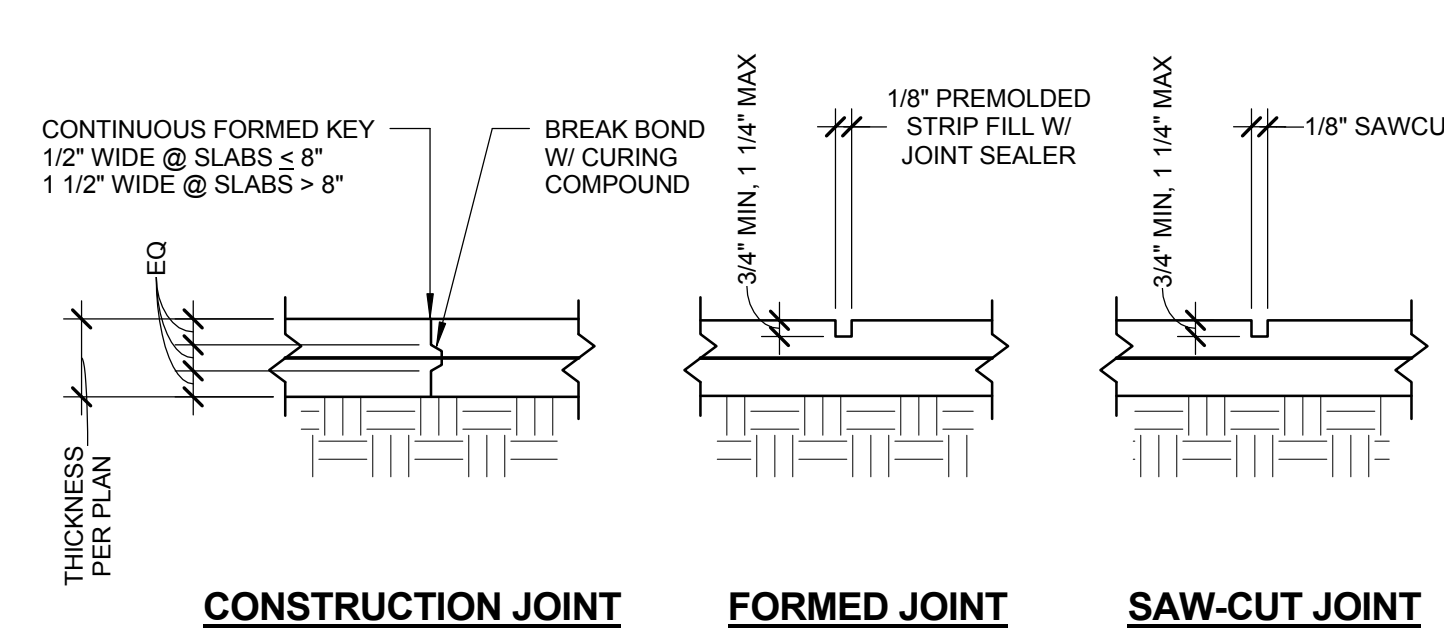
5 STEP IN SLAB ON GRADE
SCALE : 1" = 1'-0"



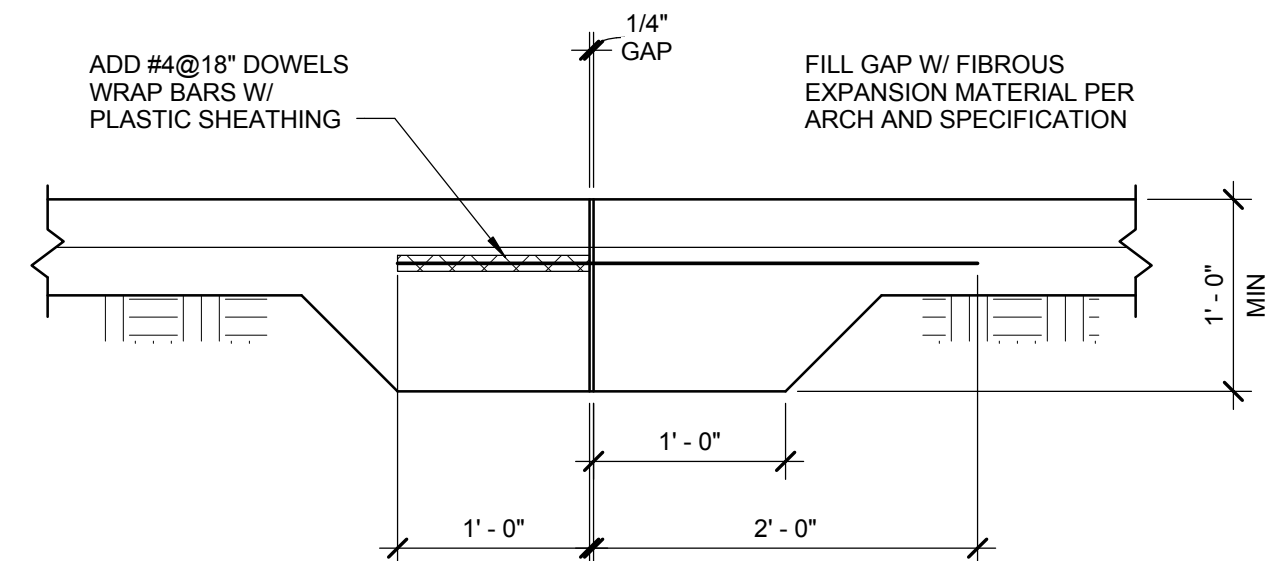
6 TRENCH DRAIN DETAIL
SCALE : 1" = 1'-0"



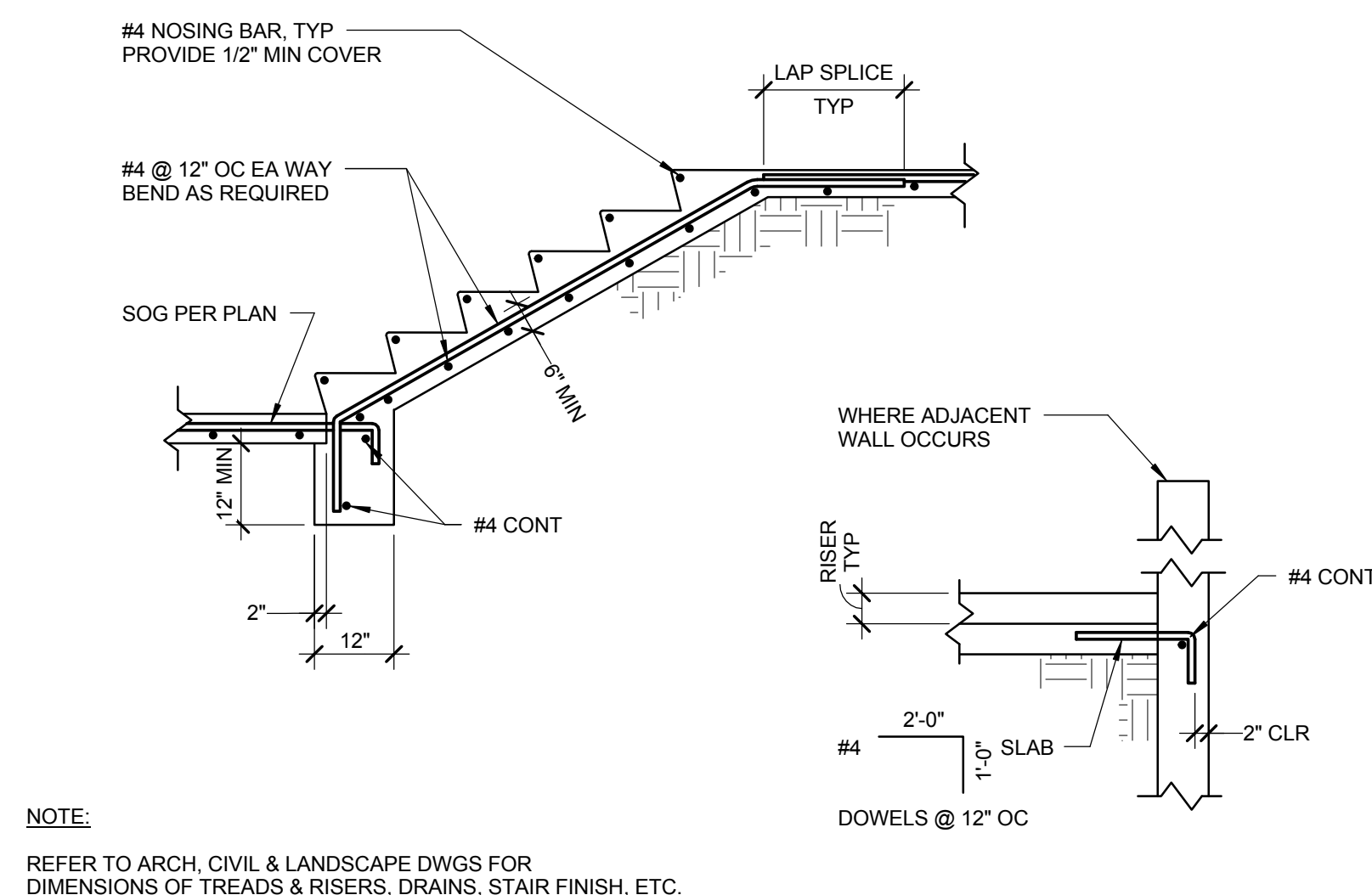
1 TYPICAL SLAB ON GRADE SUPPORT AND VAPOR BARRIER DETAIL
SCALE : 1" = 1'-0"



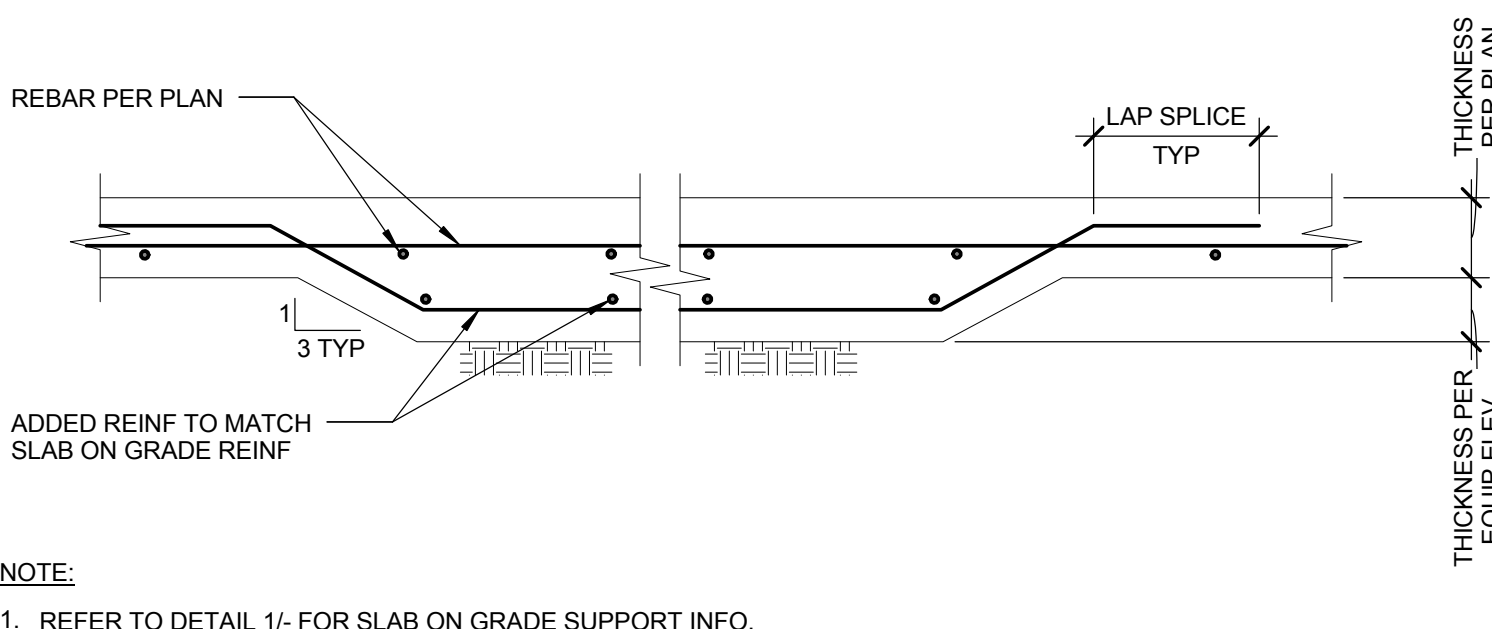
2 SLAB ON GRADE JOINTS
SCALE : 1" = 1'-0"



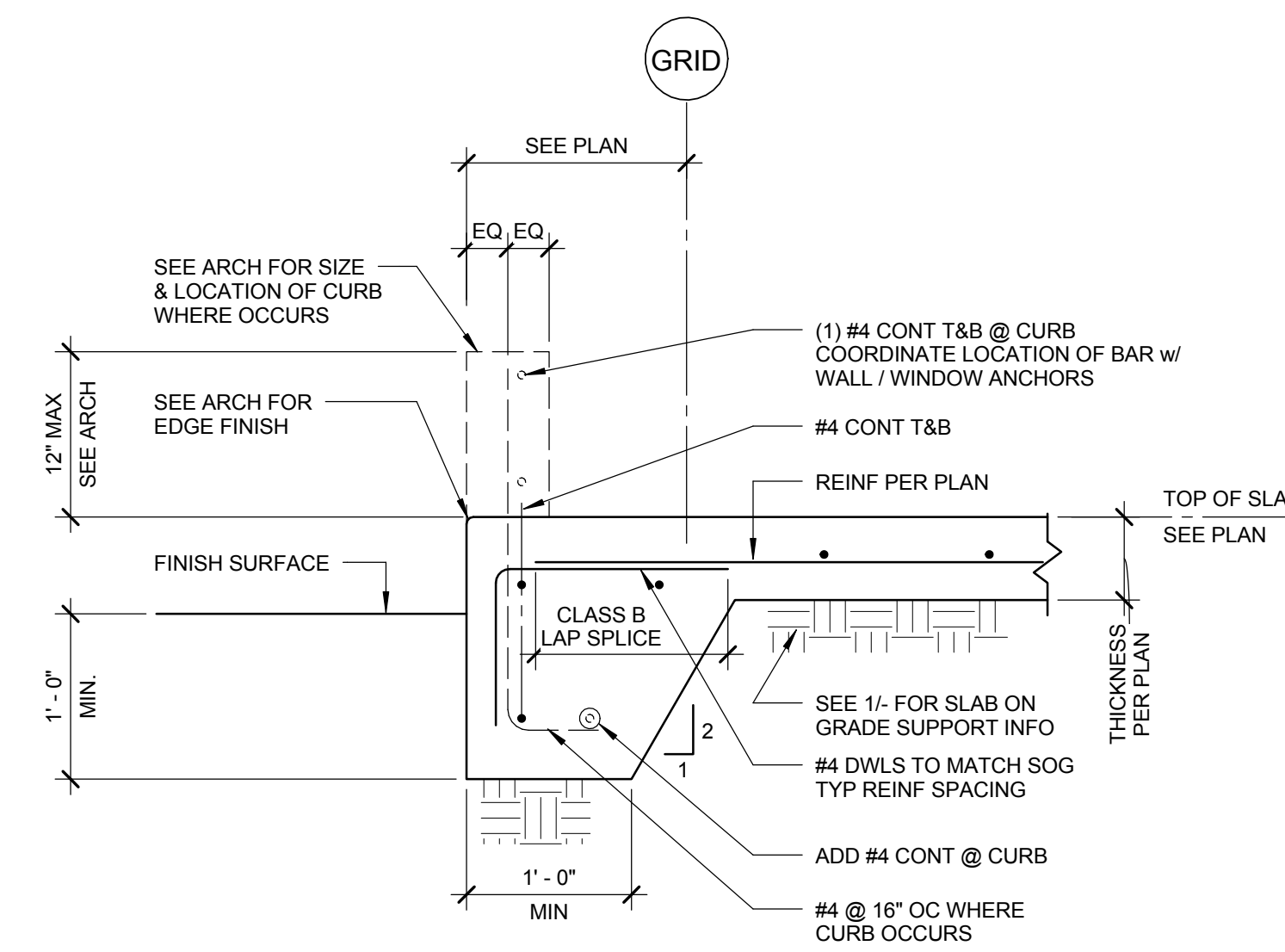
13 EXPANSION JOINT
SCALE : 1" = 1'-0"



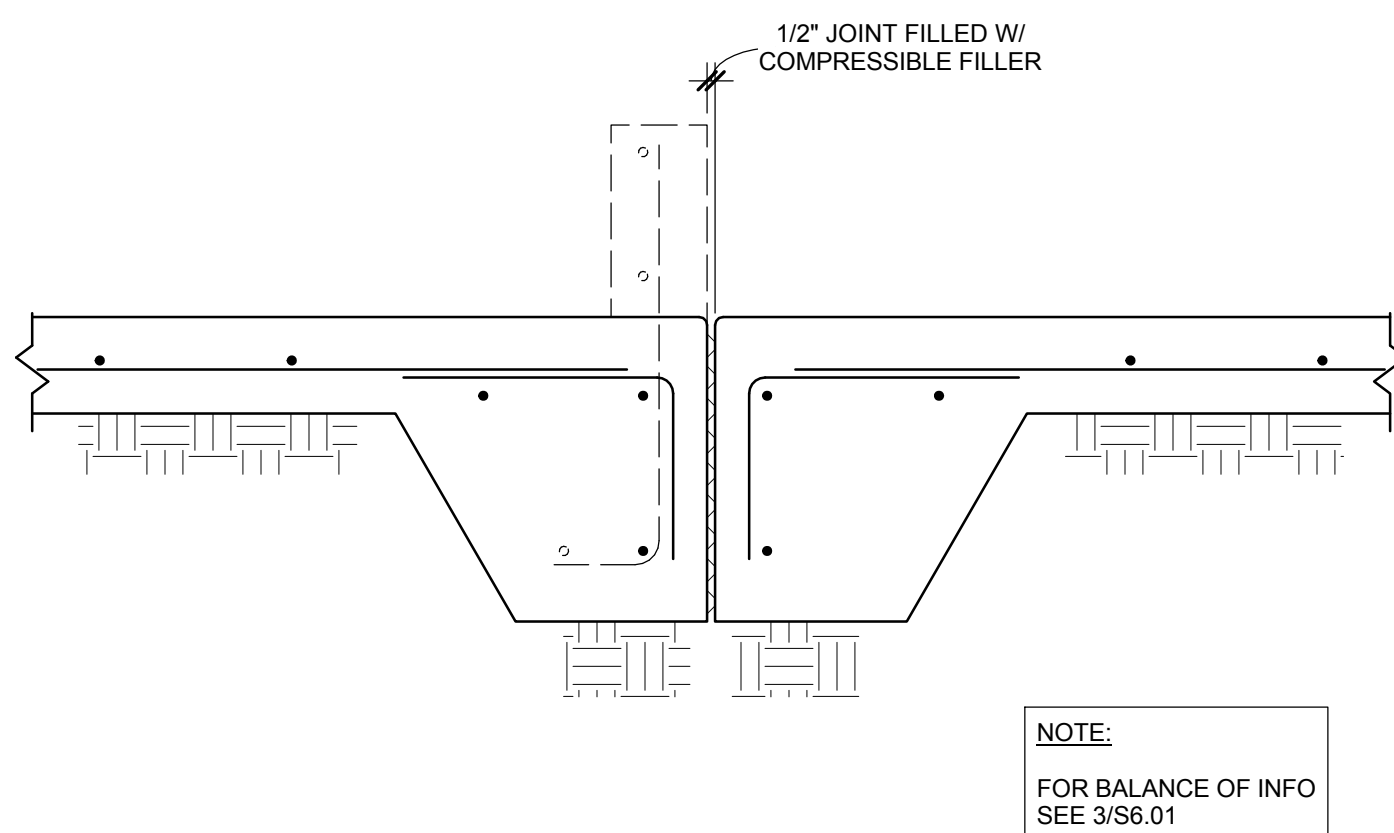
10 CONCRETE STAIRS ON GRADE
SCALE : 1" = 1'-0"



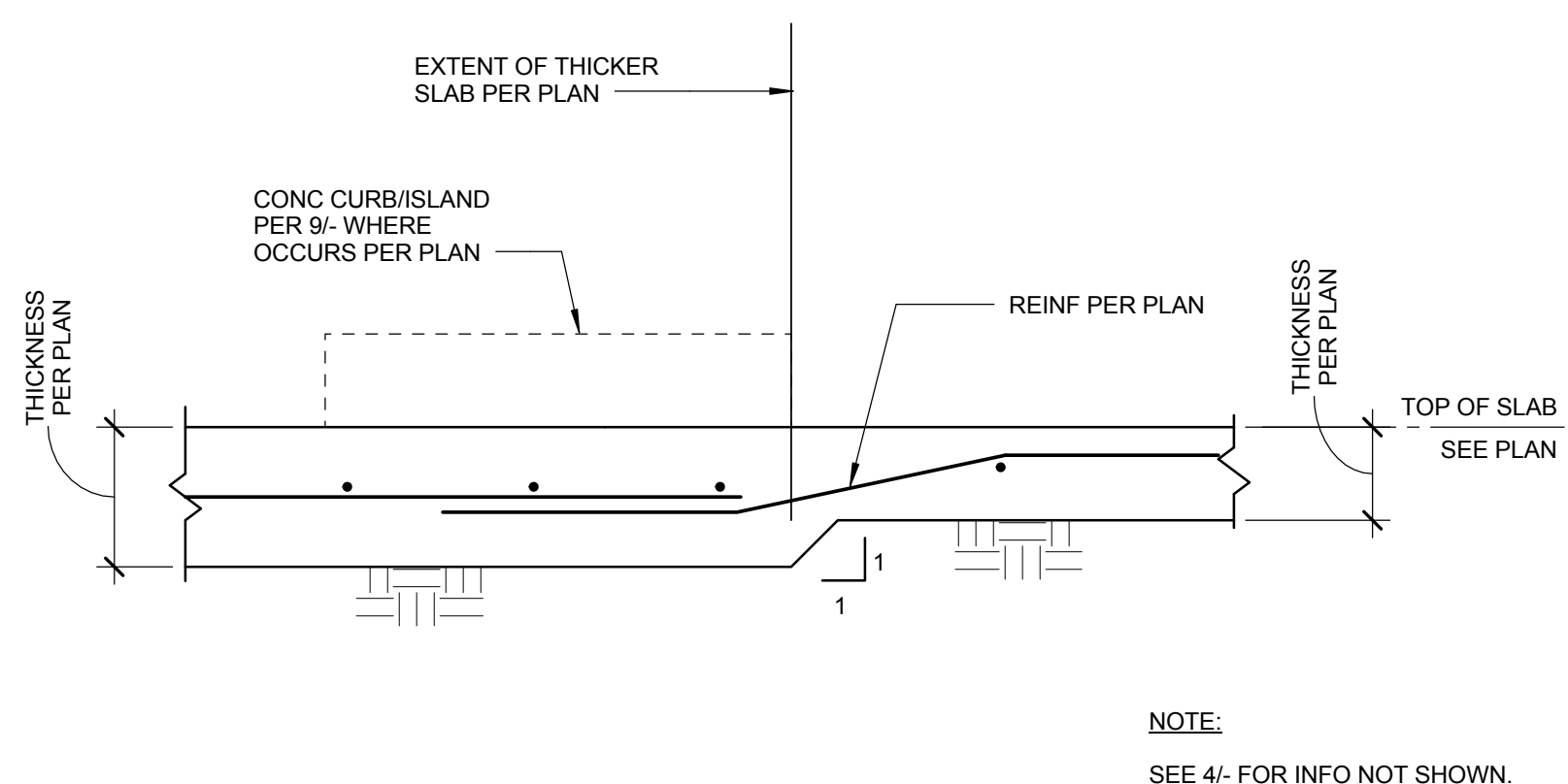
7 THICKENED SLAB ON GRADE
SCALE : 1" = 1'-0"



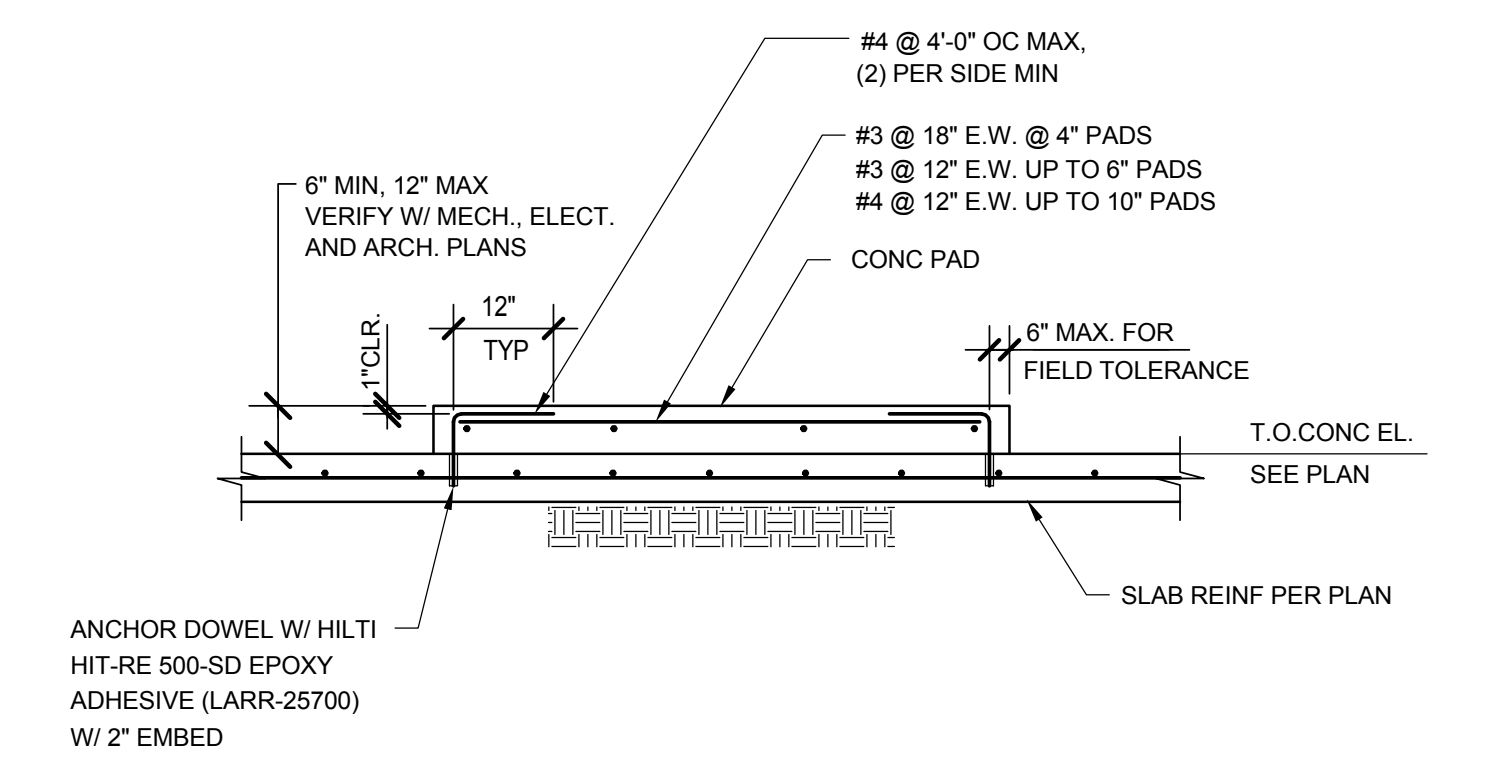
3 SLAB ON GRADE THICKENED EDGE
SCALE : 1" = 1'-0"



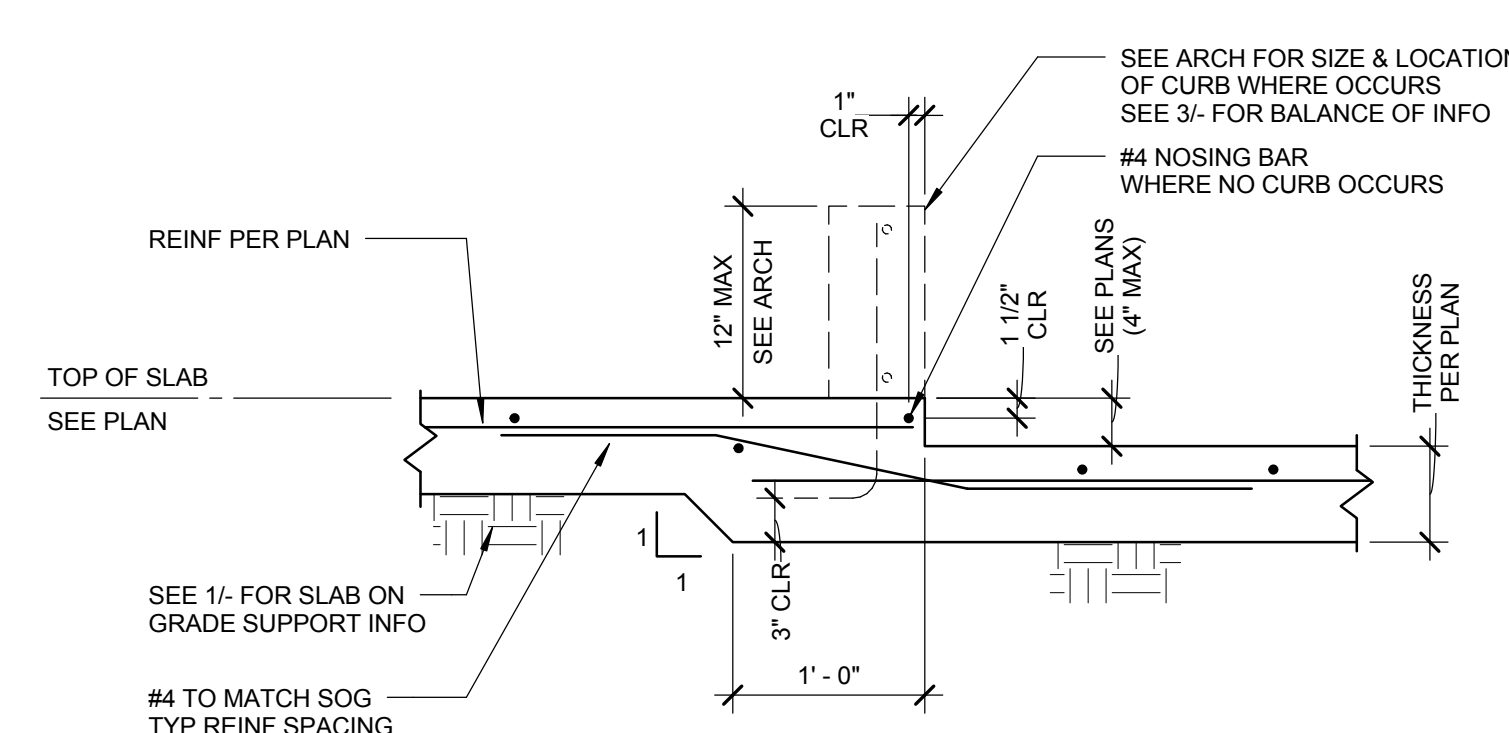
14 EXPANSION JOINT
SCALE : 1" = 1'-0"



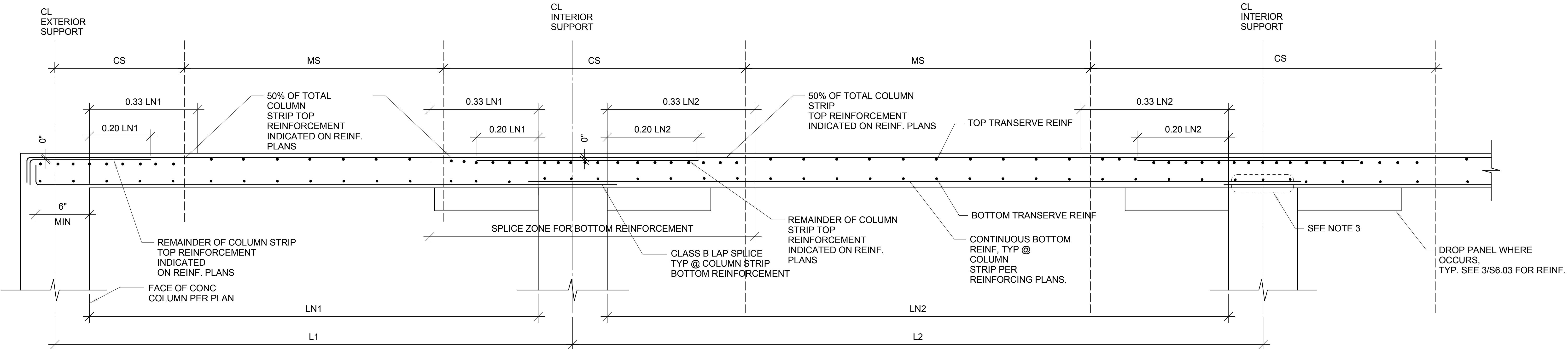
11 SLAB THICKNESS TRANSITION
SCALE : 1" = 1'-0"



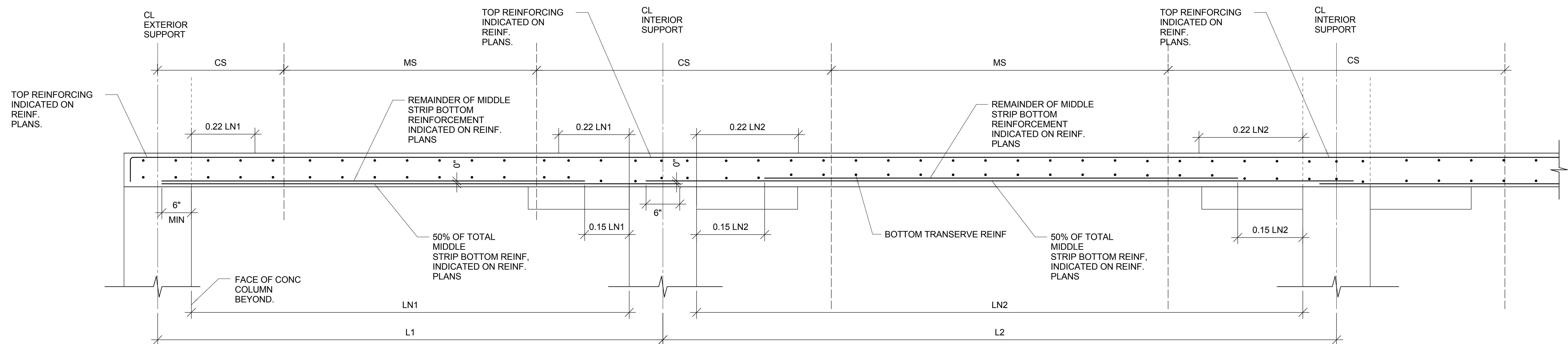
8 TYPICAL CONCRETE PAD AT SLAB ON GRADE
SCALE : 1" = 1'-0"



4 SLAB DEPRESSION
SCALE : 1" = 1'-0"



TYPICAL SECTION AT COLUMN STRIP (LOOKING NORTH - SOUTH)



TYPICAL SECTION MIDDLE STRIP (LOOKING NORTH-SOUTH)

NOTES:

- SLAB REINFORCING SEQUENCE:
 - NORTH/SOUTH BOTT BARS (FIRST) - SEE S2-0XXX
 - EAST/WEST BOTT BARS - SEE S2-0XXX
 - EAST/WEST TOP BARS - SEE S2-0XXX
 - NORTH/SOUTH TOP BARS (LAST) - SEE S2-0XXX
- MS - INDICATES MIDDLE STRIP
CS - INDICATES COLUMN STRIP
- AT LEAST TWO BOTTOM BARS MUST GO THRU THE COLUMN CONFINED CORE.
- SEE -/---- FOR CHORD REINFORCING AT EDGE OF CONCRETE SLAB.

1 TYPICAL TWO WAY SLAB REINFORCING DETAIL

SCALE : 1 1/2" = 1'-0"



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TYPICAL CONCRETE DETAILS

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ABBREVIATIONS			
AAV	AUTOMATIC AIR VENT	ID	INSIDE DIAMETER
ABV	ABOVE	IN	INCHES
AC	AIR CONDITIONING	IPLV	INTEGRATED PART LOAD VALUE
ACCEPT	ACCEPTANCE	JB	JUNCTION BOX
AQU	AIR CONDITIONING UNIT		
AD	ACCESS DOOR		
ADD	ADDITION	KW	KILOWATT
AF	AFTER FILTER	KWH	KILOWATT HOUR
AFF	ABOVE FINISHED FLOOR		
AFMS	AIR FLOW MEASURING STATION	L	LENGTH
AFUE	ANNUAL FUEL UTILIZATION EFFICIENCY	LAT	LEAVING AIR TEMPERATURE
AG	AIR GAP	LBS	POUNDS
AHU	AUTHORITY HAVING JURISDICTION	LBS	LEAVING DRY BULB
AHU	AIR HANDLING UNIT	LF	LINEAR FEET
AMB	AMBIENT	LP	LOW PRESSURE
	AMPERE	LPC	LOW PRESSURE CONDENSATE
AP	ACCESS PANEL	LPS	LOW PRESSURE STEAM
APPROX	APPROXIMATELY	LWB	LEAVING WET BULB
ARCH	ARCHITECT	LWT	LEAVING WATER TEMPERATURE
ARI	AMERICAN REFRIGERATION INSTITUTE	M	MOTOR
AS	SEPARATOR	MA	MIXED AIR
AUTO	AUTOMATIC	MAD	MIXED AIR DAMPER
AUX	AUXILIARY	MAX	MAXIMUM
		MBH	THOUSAND BTU PER HOUR
B	BOILER	MC	MECHANICAL CONTRACTOR
BAS	BUILDING MANAGEMENT SYSTEM	MCA	MINIMUM CIRCUIT AMPACITY
BDD	BACKDRAFT DAMPER	MCC	MOTOR CONTROL CENTER
BEL	BELOW	MD	MOTORIZED DAMPER
BHP	BRAKE HORSEPOWER	MECH	MECHANICAL
BOD	BOTTOM OF DUCT	MERV	MINIMUM EFFICIENCY RATING VALUE
BOP	BOTTOM OF PIPE	MFR	MANUFACTURER
BFP	BACKFLOW PREVENTER	MIN	MINIMUM
BSMT	BASEMENT	MOCOP	MAXIMUM OVER CURRENT
BTU	BRITISH THERMAL UNIT		
BTUH	BTU PER HOUR	MPC	MEDIUM PRESSURE CONDENSATE
BV	BALL VALVE OR BALANCING VALVE	MPS	MEDIUM PRESSURE STEAM
BVH	BUTTERFLY VALVE	MV	MANUAL AIR VENT
C	COMMON, CONDENSATE OR CONDUIT	(N)	NEW
CA	CONTROL AIR	N/A	NOT APPLICABLE
CAP	CAPACITY	NC	NORMALLY CLOSED
CAV	CONSTANT AIR VOLUME	NC	NOT IN CONTRACT
CB	CHILLED BEAM	NO	NORMALLY OPEN
CC	COOLING COIL OR CONTROLS CONTRACTOR	NOM	NOMINAL
CEG	CEILING EXHAUST GRILLE	NP	NET POSITIVE SUCTION HEAD
CER	CEILING EXHAUST REGISTER	NTS	NOT TO SCALE
CFF	CAP FOR FUTURE		
CFM	CUBIC FEET PER MINUTE	OA	OUTSIDE AIR
CFS	CUBIC FEET PER SECOND	OAD	OUTSIDE AIR DAMPER
CHWP	CHILLED WATER PUMP	OAT	OUTSIDE AIR TEMPERATURE
CHWR	CHILLED WATER RETURN	OBD	OPPOSED BLADE DAMPER
CHWS	CHILLED WATER SUPPLY	OCC	ON CENTER
CH	CHILLER	OC	OUTSIDE DIAMETER
CHV	CHECK VALVE	OCFI	OWNER FURNISHED CONTRACTOR
CI	CENTERLINE		
CLG	CEILING	OFOI	OWNER FURNISHED OWNER INSTALLED
CO	CLEANOUT	OPER	OPERATING
COL	COLUMN	P	PUMP OR PRESSURE OR POLE
COMP	COMPRESSOR	PC	PUMPED CONDENSATE
CONC	CONCRETE	PD	PRESSURE DROP
COND	CONDENSATE	PF	PREFILTER
CONN	CONNECTION	PG	PIPE GUIDE OR PRESSURE GAUGE
CONT	CONTINUATION	PHC	PHASE (ELECTRICAL)
CONTR	CONTRACTOR	PLBG	PREHEAT COIL
COP	COEFFICIENT OF PERFORMANCE	POC	POINT OF CONNECTION
CP	CONTROL PANEL OR CONDENSATE PUMP	POD	POINT OF DISCONNECTION
CR	CONDENSATE RETURN	PRESS	PRESSURE
CRR	CEILING RETURN REGISTER	PRV	PRESSURE REDUCING VALVE
CRG	CEILING RETURN GRILLE	PS	PRESSURE SENSOR
		PSI	POUNDS PER SQUARE INCH
CSD	CEILING SUPPLY DIFFUSER	PSIA	PSI ABSOLUTE
CTE	CONNECT TO EXISTING	PSIG	PSI GAUGE
CU FT	CUBIC FEET	QTY	QUANTITY
CU IN	CUBIC INCH	R	RISERS, RELOCATE OR RISE
CV	CONSTANT VOLUME OR CONTROL VALVE	RA	RETURN AIR
CW	COLD WATER	RAD	RETURN AIR DAMPER
CWP	CONDENSER WATER PUMP	RD	REFRIGERANT DISCHARGE
CWR	CONDENSER WATER RETURN	REF	REFRIGERANT
CWS	CONDENSER WATER SUPPLY	REF	REFRIGERATION
		REJ	REJECTION
D	DROP OR DRAIN	REOD	REQUIRED
DBT	DRY BULB TEMPERATURE	REV	REVISION OR REVOLUTIONS
DDC	DIRECT DIGITAL CONTROL	RF	RETURN FAN
DEFL	DEFLECTION	RH	RELATIVE HUMIDITY
DIA	DIAMETER	RHC	REHEAT COIL
DIFF	DIFFERENCE	RHT	RADIANT HEATER
DP	DIFFERENCE	RM	ROOM
DPT	DEW POINT TEMPERATURE	RM	REVOLUTIONS PER MINUTE
DS	DUCT SMOKE DETECTOR	RS	REFRIGERANT SUCTION
DV	DIAPHRAGM VALVE	RTU	ROOFTOP UNIT
DWG(S)	DRAWING(S)	S	SUPPLY OR SLOPE
DX	DIRECT EXPANSION	SA	SUPPLY AIR
		SCFM	CFM, STANDARD CONDITIONS
(E)	EXISTING	SD	SMOKE DAMPER
(ER)	EXISTING RELOCATED	SEER	SEASONAL ENERGY EFFICIENCY
EA	EXHAUST AIR OR EACH	SEN	SENSIBLE
EAD	EXHAUST AIR DAMPER	SF	SUPPLY FAN OR SQUARE FEET
EAT	ENTERING AIR TEMPERATURE	SHC	SQUARE HEAD COCK
EC	ELECTRICAL CONTRACTOR	SIU	SPLIT INDOOR UNIT
ECON	ECONOMIZER	SOU	SPLIT OUTDOOR UNIT
EDB	ENTERING DRY BULB TEMPERATURE	SP	STATIC PRESSURE
EER	ENERGY EFFICIENCY RATING	SPD	SPLITTER DAMPER
EF	EXHAUST FAN	SEC	SPECIFICATIONS
EFF	EFFICIENCY	SO IN	SQUARE INCH
EJ	EXPANSION JOINT	STD	STRAINER OR SOUND TRAP
EL	ELEVATION	STD	STANDARD
ELEC	ELECTRICAL	STRUCT	STRUCTURAL
EQUIP	EQUIPMENT	SV	STEAM VENT
ESP	EXTERNAL STATIC PRESSURE		
ET	EXPANSION TANK	T	THERMOSTAT OR THERMOMETER
EWB	ENTERING WET BULB TEMPERATURE	TC	TEMPERATURE CONTROL PANEL
EWT	ENTERING WATER TEMPERATURE	TDH	TOTAL DYNAMIC HEAD
EXH	EXHAUST	TEMP	TEMPERATURE
EXT	EXTERNAL	TI	TENANT IMPROVEMENT
		TRG	TRANSFER GRILLE
F	FAHRENHEIT OR FILTER	TS	TEMPERATURE SENSOR
FPB	FAN POWERED BOX	TSP	TOTAL STATIC PRESSURE
FC	FLEXIBLE CONNECTION OR FAIL CLOSED	TT	TEST TAP OR TEST TEE
FCU	FAN COIL UNIT	TXV	THERMAL EXPANSION VALVE
FD	FIRE DAMPER	(TYP)	TYPICAL
FF	FINAL FILTER OR FINISHED FLOOR		
FLOOR	FLOOR		
FO	FAIL OPEN		
FPI	FINS PER INCH		
FRM	FEET PER MINUTE		
FPS	FEET PER SECOND		
FSD	FIRE/SMOKE DAMPER		
FT	FOOT OR FEET		
G	GAS	U	HEAT TRANSFER COEFFICIENT
GA	GAUGE, GAGE	UC	UNDER CUT DOOR
GAL	GALLONS	UG	UNDERGROUND
GALV	GALVANIZED	UH	UNIT HEATER
GC	GAS COCK OR GENERAL CONTRACTOR	UON	UNLESS OTHERWISE NOTED
GLV	GLOBE VALVE		
GN	GENERAL NOTE	V	VENT OR VOLT OR VELOCITY
GPM	GALLONS PER MINUTE	VAV	VARIABLE AIR VOLUME
GND	GROUND	VB	VACUUM BREAKER
GV	GATE VALVE	VD	VOLUME DAMPER
		VEL	VELOCITY
H	HEIGHT	VERT	VERTICAL
HB	HOSE BIBB	VFD	VARIABLE FREQUENCY DRIVE
HC	HEATING COIL	VFM	VENTURI FLOW METER
HD	HEAD	VOL	VOLUME
HOR	HORIZONTAL	VTR	VENT THROUGH ROOF
HP	HIGH PRESSURE		
HP	HORSEPOWER	W	WASTE OR WIDTH OR WATTS
HPC	HEAT PUMP	WITH	WITH
HPC	HIGH PRESSURE CONDENSATE	W/O	WITHOUT
HPS	HIGH PRESSURE STEAM	WB	WET BULB TEMPERATURE
HR	HOURS	WC	WATER COLUMN
HRT	HEAT RECOVERY UNIT	WEG	WALL EXHAUST GRILLE
HRS	HUMIDITY SENSOR	WG	WATER GAUGE
HTR	HEATER	WOP	WORKING PRESSURE
HV	HOSE VALVE	WPD	WATER PRESSURE DROP
HVAC	HEATING, VENTILATING & AIR CONDITIONING	WRR	WALL RETURN REGISTER
		WSP	WATER-SOURCE HEAT PUMP
HW	HEATING WATER	WSR	WALL SUPPLY REGISTER
HWP	HEATING WATER PUMP	WT	WEIGHT
HWR	HEATING WATER RETURN		
HWS	HEATING WATER SUPPLY	XFMR	TRANSFORMER
HX	HEAT EXCHANGER	Z	ZONE
HZ	FREQUENCY (HERTZ)	ZD	ZONE DAMPER

NOTE: NOT ALL SYMBOLS OR ABBREVIATIONS ARE APPLICABLE TO THIS PROJECT. REFER TO DETAILS AND NOTES FOR MOUNTING HEIGHTS.

HVAC LEGEND			
GENERAL		VALVES & GAUGES	
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	NEW WORK		BALL VALVE
	EXISTING WORK TO REMAIN		BUTTERFLY VALVE
	EXISTING WORK TO BE REMOVED		GATE VALVE
	RELOCATE EXISTING		BALANCING VALVE
	CENTER LINE		GLOBE VALVE
	POINT OF CONNECTION OR POINT OF DISCONNECTION		CHECK VALVE
	DETAIL 1, DRAWING M-1		ANGLE GLOBE VALVE
	SECTION A, DRAWING M-1		PRESSURE AND TEMPERATURE RELIEF VALVE
	ELEVATION 1, DRAWING M-1		2-WAY CONTROL VALVE (BALL)
	RISER IDENTIFICATION		2-WAY CONTROL VALVE (BUTTERFLY)
	EXHAUST #1		3-WAY CONTROL VALVE
	EQUIPMENT IDENTIFICATION		SOLENOID CONTROL VALVE
	HEAT PUMP UNIT #1		PRESSURE REDUCING VALVE (PRV)
	KITCHEN EQUIPMENT TAG		DIAPHRAGM VALVE
	NUMBER OR QUANTITY		LOCK SHIELD VALVE
	KEYED NOTE		NEEDLE VALVE
	DIRECTION OF TRANSFER		QUICK OPENING VALVE
	ARIFLOW (150 CFM)		VENTURI FLOW METER
	78 DEGREES FAHRENHEIT		PRESSURE GAUGE
			PRESS. GAUGE WITH COCK AND SNUBBER
			THERMOMETER
			TEMPERATURE SENSOR IN THERMOWELL
PIPING		HYDRONIC	
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	DIRECTION OF SLOPE		CHILLED WATER SUPPLY
	DIRECTION OF FLOW		CHILLED WATER RETURN
	PIPE UP		HEATING WATER SUPPLY
	PIPE DOWN		HEATING WATER RETURN
	PIPE DROP/PIPE RISE		CONDENSER WATER SUPPLY
	TOP CONNECTION - BRANCH LINE		CONDENSER WATER RETURN
	BOTTOM CONNECTION - BRANCH LINE		
	PIPE ANCHOR		
	TEE UP		
	TEE DOWN		
	STRAINER		
	STRAINER WITH BLOW OFF HOSE BIBB		
	INLINE PUMP		
	REQUIRED		
	BASE MOUNTED PUMP		
	TEST TAP (PETE'S PLUG)		
	MANUAL AIR VENT		
	AUTOMATIC AIR VENT		
	VACUUM BREAKER		
	VENT THRU ROOF		
	PIPE GUIDE		
	EXPANSION JOINT		
	FLEXIBLE CONNECTOR		
	UNION		
	CAPPED OR PLUGGED TEE		
	BLIND FLANGE, CAP		
	CONCENTRIC REDUCER		
	EXPANSION LOOP		
	VALVE ON RISE		
	PIPE SIZE (DIAMETER IN INCHES)		
STEAM		MISCELLANEOUS	
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	HIGH PRESSURE STEAM		REFRIGERANT LIQUID/DISCHARGE
	MEDIUM PRESSURE STEAM		REFRIGERANT SUCTION
	LOW PRESSURE STEAM		PNEUMATIC CONTROL VALVE
	HIGH PRESSURE CONDENSATE		AIR LINE
	MEDIUM PRESSURE CONDENSATE		SOUND TRAP
	LOW PRESSURE CONDENSATE		COOLING COIL
	GRAVITY OR PUMPED CONDENSATE		HEATING COIL
	CONDENSATE RETURN		RADIANT HEATER
	STEAM VENT PIPE		BASEBOARD HEATER
	STEAM TRAP		CHILLED BEAM
	BOILER BLOW DOWN		THRUST BLOCK
	ECCENTRIC REDUCER (STEAM PIPE)		
SYMBOL (DOUBLE LINE)		SYMBOL (SINGLE LINE)	
DESCRIPTION		DESCRIPTION	

HVAC DRAWING LIST	
M-000	MECHANICAL LEAD SHEET
M-001	MECHANICAL SCHEDULES
M-002	MECHANICAL SCHEDULES
M-003	MECHANICAL SCHEDULES
M-004	MECHANICAL SCHEDULES
M-005	BASEMENT MECHANICAL ZONING PLAN
M-006	1ST FLOOR MECHANICAL ZONING PLAN
M-007	2ND FLOOR MECHANICAL ZONING PLAN
M-008	3RD FLOOR MECHANICAL ZONING PLAN
M-100	MECHANICAL SITE PLAN
M-200N	BASEMENT FLOOR PLAN NORTH - MECHANICAL
M-200S	BASEMENT FLOOR PLAN SOUTH - MECHANICAL
M-201N	1ST FLOOR PLAN NORTH - MECHANICAL
M-201S	1ST FLOOR PLAN SOUTH - MECHANICAL
M-202N	2ND FLOOR PLAN NORTH - MECHANICAL
M-202S	2ND FLOOR PLAN SOUTH - MECHANICAL
M-203N	ROOF PLAN NORTH - MECHANICAL
M-203S	3RD FLOOR PLAN SOUTH - MECHANICAL
M-204S	ROOF PLAN SOUTH - MECHANICAL
M-500	MECHANICAL DETAILS
M-501	MECHANICAL DETAILS
M-502	MECHANICAL DETAILS
M-503	MECHANICAL DETAILS
M-700	MECHANICAL AIR RISERS
M-701	MECHANICAL WATER RISERS

SOUND TRAP SCHEDULE															
TAG	MANUFACTURER	MODEL NUMBER	EQUIPMENT SERVED	AIRFLOW (CFM)	PRESSURE DROP (IN. WG.)	DYNAMIC INSERTION LOSS (dB re: 10-12 W) PER OCTAVE BAND								SIZE (L'xW'xH')	NOTES
						63 HZ	125 HZ	250 HZ	500 HZ	1000 HZ	2000 HZ	4000 HZ	8000 HZ		
ST-1	-		AHU-1 SUPPLY	27,600	-	-	-	-	-	-	-	-	-	1	
ST-2			AHU-1 RETURN	6,300											
ST-3			AHU-2 SUPPLY	38,600											
ST-4			AHU-2 RETURN	7,500											
ST-5			AHU-3 SUPPLY	30,150											
ST-6			AHU-3 RETURN	7,750											
NOTES															
1. DYNAMIC INSERTION LOSS VALUES PER NVLAP-ACCREDITED LABORATORY TESTS IN ACCORDANCE WITH ASTM E477-99 AT A FACE VELOCITY OF 1000 FPM.															
2. PROVIDE SOUND TRAP WITH ENCAPSULATED LINING FOR (HOSPITAL / CLEANROOM) APPLICATION (IAC CLEAN-FLOW SERIES OR APPROVED EQUAL).															
3. PROVIDE PACKLESS TYPE SOUND TRAP (IAC ULTRA PALS SERIES OR APPROVED EQUAL).															
4. PROVIDE SOUND TRAP WITH 100% RECYCLED, NATURAL FIBER FILL MATERIAL (IAC ULTRA/GREEN SERIES OR APPROVED EQUAL).															

LABORATORY EXHAUST FAN SCHEDULE																				
TAG	MANUFACTURER	MODEL NUMBER	LOCATION	AREA SERVED	DISCH	TYPE	DRIVE	AIR FLOW (CFM)	SP (IN.WG.)	MIN. EFF. (%)	FAN RPM	MTR RPM	ELECTRICAL					UNIT SIZE (L"xW"xH")	OPER. WT. (LBS)	NOTES
													BHP	No./HP	V/PH	VFD (Y/N)	EMERG POWER (Y/N)			
EF-1 - A,B,C	STROBIC	TS2L075A12	ROOF	HOOD EXHAUST/LAB EXHAUST		STROBIC 3 FAN	BELT	21,000	-	-	-	-	-	3@7.5 HP	460/3	Y	Y	-	-	1,2,3,4,5,6,7
EF-2 - A,B,C,D	STROBIC	TS3S150D9	ROOF	HOOD EXHAUST/LAB EXHAUST		STROBIC 4 FAN	BELT	48,000						4@15 HP	460/3	Y	Y			1,2,3,4,5,6,7
NOTES																				
1. COORDINATE WITH ELECTRICAL FOR POWER AND DISCONNECT AS REQUIRED.																				
2. VARIABLE FREQUENCY DRIVE (VFD) PROVIDED BY MECHANICAL WIRED BY ELECTRICAL.																				
3. PROVIDE NEMA PREMIUM EFFICIENCY MOTOR.																				
4. PROVIDE NON-OVERLOADING MOTOR.																				
5. PROVIDE INVERTER READY MOTOR WITH SHAFT GROUNDING SYSTEM.																				
6. PROVIDE VIBRATION ISOLATION AND SEISMIC RESTRAINT PER SPECIFICATIONS.																				
7. PROVIDE BYPASS AIR PLENUM BASE.																				

LAB CAV SCHEDULE																						
TAG	MANUFACTURER	MODEL NUMBER	SPACE SERVED	INLET SIZE (IN)	AIRFLOW			AIR PD (IN. WG.)	HW REHEAT COIL								CONTROL VALVE		OPER. WT. (LBS)	NOTES		
					COOL MAX (CFM)	COOL MIN (CFM)	HEAT MAX (CFM)		CAP (MBH)	LAT (°F)	EAT (°F)	EWI (°F)	LWT (°F)	FLOW (GPM)	WPD (FT. WG.)	ROWS	CONN (IN.)	PORTS			CV	
CAV-NB-1	-	-	ENERGY LAB	-	1,420	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-NB-2			ENVIRONMENTAL LAB		2,620	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-NB-3			HYDRAULICS LAB		1,415	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-NB-4			FLUID MECH LAB		1,515	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-N1-1			FABRICATION SHOP		1,035	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-N1-2			INCUBATOR SPACE		5,600	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-N2-1			VIRO BIOENGINEERING		4,520	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-N2-2			VIROMICS BSL2		1,800	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-SB-1			MECH MAT		2,860	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-SB-2			SOILS LAB		1,580	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-SB-3			SB RESEARCH LAB		3,310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-SB-4			INSTRUMENT ROOM		1,250																	
N/A			NOT USED																			
CAV-S1-1			CREATIVE DESIGN		4,510	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S1-2			CREATIVE D FABRICAION		2,210	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S1-3			ORG STORAGE		805	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S1-4			MINI BAJA		745	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S1-5			NANO FAB CLEAN ROOM		1205	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S1-6			NANO FAB RESEARCH		2055	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S1-7			RECEIVING		615	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S2-1			S2 RESEARCH LAB WEST		5,500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S2-2			S2 RESEARCH LAB WEST		5,500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S2-3			S2 RESEARCH LAB EAST		5,950	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S3-1			S3 RESEARCH LAB WEST		3,425	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S3-2			S3 RESEARCH LAB WEST		3,425	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
CAV-S3-3			S3 RESEARCH LAB EAST		5,625	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-NB-1			ENERGY LAB		1,420	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-NB-2			ENVIRONMENTAL LAB		2,620	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-NB-3			HYDRAULICS LAB		1,415	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-NB-4			FLUID MECH LAB		1,515	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-N1-1			FABRICATION SHOP		1,035	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-N1-2			INCUBATOR SPACE		5,600	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-N2-1			VIRO BIOENGINEERING		4,520	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-N2-2			VIROMICS BSL2		1,800	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-SB-1			MECH MAT		2,860	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-SB-2			SOILS LAB		1,580	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-SB-3			SB RESEARCH LAB		3,310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-SB-4			INSTRUMENT ROOM		1,250																	
ECAV-S1-1			CREATIVE DESIGN		4,510	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S1-2			CREATIVE D FABRICAION		2,210	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S1-3			STUDENT ORG STORAGE 0		805	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S1-4			STUDENT ORG MINI BAJA		745	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S1-5			NANO FAB CLEAN ROOM		1205	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S1-6			NANO FAB RESEARCH		2055	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S1-7			RECEIVING		615	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S2-1			S2 RESEARCH LAB WEST		5,500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S2-2			S2 RESEARCH LAB WEST		5,500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S2-3			S2 RESEARCH LAB EAST		5,950	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S3-1			S3 RESEARCH LAB WEST		3,425	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S3-2			S3 RESEARCH LAB WEST		3,425	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
ECAV-S3-3			S3 RESEARCH LAB EAST		5,625	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
NOTES																						

FUME HOOD VARIABLE AIR VOLUME DAMPER SCHEDULE																											
TAG	FUME HOOD TAG	MANUFACTURER	MODEL NUMBER	SPACE SERVED	INLET SIZE (IN)	AIRFLOW CFM		DAMPER PD (IN.WG.)	VALVE MOUNTING	FAILURE POSITION	ELECTRIC V/PH	EMERG POWER (Y/N)	OPER. WT. (LBS)	NOTES													
						MAX	MIN																				
ECV-NB-1	-	ACCUTROL	-	TEACHING LABS	-	800	300	-	-	-	-	Y	-	1,2,3,4													
ECV-NB-2		ACCUTROL		TEACHING LABS		800	300					Y		1,2,3,4													
ECV-N1-1		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-N1-2		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-N2-1		ACCUTROL		VIRO BIOENGINEERING		800	300					Y		1,2,3,4													
ECV-N2-2		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-N2-3		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-N2-4		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-SB-1		ACCUTROL		MECH MAT		800	300					Y		1,2,3,4													
ECV-SB-2		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-SB-3		ACCUTROL		SB RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-SB-4		ACCUTROL		SB RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-SB-5		ACCUTROL		INSTRUMENT ROOM		800	300					Y		1,2,3,4													
ECV-SB-6		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S1-1		ACCUTROL		CREATIVE DESIGN LAB		800	300					Y		1,2,3,4													
ECV-S1-2		ACCUTROL		CREATIVE DESIGN LAB		800	300					Y		1,2,3,4													
ECV-S1-3		ACCUTROL		NANO FAB		800	300					Y		1,2,3,4													
ECV-S1-4		ACCUTROL		NANO FAB		800	300					Y		1,2,3,4													
ECV-S1-5		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S1-6		ACCUTROL		NANO RESEARCH		800	300					Y		1,2,3,4													
ECV-S2-1		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-2		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-3		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-4		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-5		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-6		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-7		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-8		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-9		ACCUTROL		S2 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-10		ACCUTROL		S2 RESEAROH LAB WEST		800	300					Y		1,2,3,4													
ECV-S2-11		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S2-12		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S2-13		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S2-14		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S2-15		ACCUTROL		S2 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S2-16		ACCUTROL		S2 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S2-17		ACCUTROL		S2 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S2-18		ACCUTROL		S2 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S2-19		ACCUTROL		S2 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S2-20		ACCUTROL		S2 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S2-21		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S2-22		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S3-1		ACCUTROL		S3 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S3-2		ACCUTROL		S3 RESEAROH LAB WEST		800	300					Y		1,2,3,4													
ECV-S3-3		ACCUTROL		S3 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S3-4		ACCUTROL		S3 RESEARCH LAB WEST		800	300					Y		1,2,3,4													
ECV-S3-5		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S3-6		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S3-7		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S3-8		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S3-9		ACCUTROL		S3 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S3-10		ACCUTROL		S3 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S3-11		ACCUTROL		S3 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S3-12		ACCUTROL		S3 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S3-13		ACCUTROL		S3 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S3-14		ACCUTROL		S3 RESEARCH LAB EAST		800	300					Y		1,2,3,4													
ECV-S3-15		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
ECV-S3-16		ACCUTROL		FUTURE		800	300					Y		1,2,3,4													
NOTES																											
1. ELECTRONIC PRESSURE INDEPEENT LINEAR AIRFLOW CONTROL VALVE.																											
2. PROVIDE SASH POSTION SENSOR AND REDUCE AIRFLOW TO MINIMUM WHEN CLOSED.																											
3. VALVE CONSTRUCTED OF 316L STAINLESS STEEL WITH PHENOLIC COATING.																											
4. COORDINATE DAMPER WITH BUILDING AUTOMATION SYSTEM.																											



TWO PIPE COOLING FAN COIL UNIT SCHEDULE

TAG	MANUFACTURER	MODEL NUMBER	LOCATION	SUPPLY AIR (CFM)	OSA (CFM)	ESP (IN.WG.)	COOLING COIL										AIR FILTER		ELECTRICAL					UNIT SIZE (L'xW'xH')	OPER. WT. (LBS)	NOTES
							CAP TOTAL (MBH)	CAP SENS (MBH)	EAT DB/WB (°F)	LAT DB/WB (°F)	EWT (°F)	LWT (°F)	FLOW (GPM)	WPD (FT.WG.)	ROWS/ FPI	TYPE	MERV	BHP	HP	V/PH	MOTOR	EMERG POWER (Y/N)				
FCU-NB-1	IEC	CPY	NORTH BASE MACH RM	625		-	-	-	-	-	-	-	-	-	-	-	-	-	-	120/1	ECM	-	-	-	1,3,5,7	
FCU-NB-2	IEC	CPY	NORTH BASE ELEC. RM	575																120/1	ECM	Y			1,3,5,7	
FCU-NB-3	IEC	CPY	NORTH BASE DATA RM	500																120/1	ECM	Y			1,3,5,7	
FCU-NB-4	IEC	CPY	MECHANICAL ROOM	1000																120/1	ECM				1,3,5,7	
FCU-SB-1	IEC	CPY	SOUTH BASE DATA RM.	455																120/1	ECM				1,3,5,7	
FCU-SB-2	IEC	CPY	SOUTH BASE MACH RM	560																120/1	ECM				1,3,5,7	
FCU-SB-3	IEC	CPY	SOUTH BASE ELEC. RM	595																120/1	ECM				1,3,5,7	
FCU-SB-4	IEC	CPY	SOUTH BASE ELEC. RM	675																120/1	ECM				1,3,5,7	
FCU-N1-1	IEC	CPY	NORTH 1ST FL DATA RM	650																120/1	ECM	Y			1,3,5,7	
FCU-N1-2	IEC	CPY	NORTH 1ST FL ELEC RM	575																120/1	ECM				1,3,5,7	
FCU-S1-1	IEC	CPY	SOUTH 1ST FL ELEC RM	675																120/1	ECM				1,3,5,7	
FCU-S1-2	IEC	CPY	SOUTH 1ST FL DATA RM	500																120/1	ECM	Y			1,3,5,7	
FCU-N2-1	IEC	CPY	NORTH 2ND FL DATA RM	625																120/1	ECM	Y			1,3,5,7	
FCU-N2-2	IEC	CPY	NORTH 2ND FL ELEC RM	575																120/1	ECM				1,3,5,7	
FCU-S2-1	IEC	CPY	SOUTH 2ND FL ELEC RM	575																120/1	ECM				1,3,5,7	
FCU-S2-2	IEC	CPY	SOUTH 2ND FL DATA RM	535																120/1	ECM	Y			1,3,5,7	
FCU-S3-1	IEC	CPY	SOUTH 3RD FL ELEC RM	575																120/1	ECM				1,3,5,7	
FCU-S3-2	IEC	CPY	SOUTH 3RD FL DATA RM	550																120/1	ECM	Y			1,3,5,7	

- NOTES
- COORDINATE WITH ELECTRICAL FOR POWER AND DISCONNECT AS REQUIRED.
 - PROVIDE 100% MODULATING (ENTHALPY / TEMPERATURE) ECONOMIZER.
 - PROVIDE VIBRATION ISOLATION PER SPECIFICATIONS.
 - PROVIDE SMOKE DETECTOR IN SUPPLY AIR FOR AUTOMATIC UNIT SHUTDOWN ABOVE 2000 CFM.
 - PROVIDE NEMA PREMIUM EFFICIENCY MOTOR.
 - PROVIDE ECM (ELECTRONICALLY COMMUTATED MOTOR) WITH FACTORY MOUNTED POTENTIOMETER FOR SPEED ADJUSTMENT.
 - PROVIDE CONDENSATE PUMP WITH 120 V / 1PH CONNECTION.
 - PROVIDE SECONDARY DRIP PAN. REFER TO PLUMBING DRAWING FOR DRAIN DISCHARGE.
 - PROVIDE SECONDARY DRIP PAN WITH MOISTURE DETECTOR FOR FCU SHUTDOWN.
 - CHILLED WATER CONTAINS xx% PROPYLENE GLYCOL.

EXHAUST FAN SCHEDULE

TAG	MANUFACTURER	MODEL NUMBER	LOCATION	AREA SERVED	DISCH	TYPE	DRIVE	AIR FLOW (CFM)	SP (IN.WG.)	MIN. EFF. (%)	FAN RPM	MTR RPM	ELECTRICAL					UNIT SIZE (L'xW'xH')	OPER. WT. (LBS)	NOTES
													BHP	HP	V/PH	VFD (Y/N)	EMERG POWER (Y/N)			
TEF-1	GREENHECK	-	-	-		UTILITY	BELT	-	-	-	-	-	-	1.5	-	NO	N	-	-	1,3,4
TEF-2	GREENHECK					UTILITY	BELT							1.5		NO	N			1,3,4
GX-1	GREENHECK					UTILITY	BELT							1.5		NO	Y			1,3,4

- NOTES
- COORDINATE WITH ELECTRICAL FOR POWER AND DISCONNECT AS REQUIRED.
 - VARIABLE FREQUENCY DRIVE (VFD) PROVIDED BY MECHANICAL. WIRED BY ELECTRICAL.
 - PROVIDE NEMA PREMIUM EFFICIENCY MOTOR.
 - PROVIDE NON-OVERLOADING MOTOR.
 - PROVIDE INVERTER READY MOTOR WITH SHAFT GROUNDING SYSTEM.
 - PROVIDE VIBRATION ISOLATION AND SEISMIC RESTRAINT PER SPECIFICATIONS.
 - PROVIDE BYPASS AIR PLENUM BASE.

FLASH TANK SCHEDULE

TAG	MANUFACTURER	MODEL NUMBER	LOCATION	SERVICE	CAPACITY (LBS/HR)	SIZE DIA. X LENGTH	ASME STAMPED PSI	OPERATING WEIGHT (LBS)	NOTES
FT-1	SPIRAX SARCO	PMO---	BASEMENT	STEAM CP-1	TBD	TBD			

NOTES

STEAM PRV

TAG	MANUFACTURER	MODEL NUMBER	LOCATION	SERVICE	CAPACITY (LBS/HR)	PRESSURE			PIPING		NOTES
						INLET PRESSURE	OUTLET PRESSURE	LBS/HR	INLET DIAMETER	OUTLET DIAMETER	
PRV-1A	SPIRAX SARCO		BASEMENT	STEAM	3000	150	15		2-1/2"	6"	
PRV-1B	SPIRAX SARCO		BASEMENT	STEAM	3000	150	15		2-1/2"	6"	

NOTES

CONDENSATE PUMP SCHEDULE

TAG	MANUFACTURER	MODEL NUMBER	SYSTEM SERVED	TYPE	RECEIVER CAPACITY (GALLONS)	HEAD MAX TEMP. (DEG F)	No.	GPM EACH	PSIG	ELECTRICAL				EMERG POWER (Y/N)	OPER. WT. (LBS)	NOTES
										HP	RPM	V/PH	VFD			
CP-1	STERLING	-	STEAM SYSTEM	DUPLEX		212	2	-	40	1	1750	120/1/60		N	-	1,4,8,9

- NOTES
- COORDINATE WITH ELECTRICAL FOR POWER AND DISCONNECT AS REQUIRED.
 - PROVIDE NON-OVERLOADING NEMA PREMIUM EFFICIENCY, INVERTER READY MOTOR.
 - FOR PUMPS LOCATED OUTDOORS OR IN DUSTY ENVIRONMENTS PROVIDE TOTALLY ENCLOSED FAN COOLED (TEFC) MOTORS.
 - FOR PUMPS LOCATED INDOORS PROVIDE OPEN DRIPPROOF (ODP) MOTORS.
 - PROVIDE VARIABLE FREQUENCY DRIVE (VFD) BY MECHANICAL AND WIRED BY ELECTRICAL.
 - PROVIDE MOTOR SHAFT GROUNDING SYSTEM FOR MOTOR CONTROLLED BY VFD.
 - PROVIDE INLET SUCTION DIFFUSER OR STRAINER. SEE PIPING DESIGN.
 - PROVIDE VIBRATION ISOLATION AND SEISMIC RESTRAINT PER SPECIFICATIONS.
 - PUMPS TO RUN LEAD/LAG.
 - PUMPS TO RUN IN PARALLEL.
 - PUMP TO BE PROVIDED, FACTORY INSTALLED AND WIRED BY BOILER MANUFACTURER.
 - PUMP TO BE USED IN SYSTEM CONTAINING xx% PROPYLENE GLYCOL.

AIR SEPARATOR SCHEDULE

TAG	MANUFACTURER	MODEL NUMBER	LOCATION	SYSTEM SERVED	FLOW (GPM)	PRESSURE DROP (FT.WG.)	CONN. SIZE (IN)	INTEGRAL STRAINER (Y/N)	SIZE DIA x H (IN)	OPER. WT. (LBS)	NOTES
AS-1	-	-	-	HW	300	-	-	-	-	-	1,2,3,4
AS-2	-	-	-	SCHW	100	-	-	-	-	-	1,2,3,4

- NOTES
- PROVIDE SEPARATOR RATED FOR ASME WORKING PRESSURE OF 125 PSIG.
 - PROVIDE DRAIN LEG WITH BALL VALVE SHUTOFF AND ROUTE DRAIN TO FLOOR SINK.
 - PROVIDE WITH AUTOMATIC AIR VENT AND ROUTE TO FLOOR SINK.
 - PROVIDE FLOOR STAND WITH SEISMIC RESTRAINT. ANCHOR TO HOUSEKEEPING PAD.
 - SUSPEND FROM STRUCTURE ABOVE.

WATER-TO-WATER HEAT EXCHANGER SCHEDULE																					
TAG	MANUFACTURER	MODEL NUMBER	LOCATION	SYSTEM SERVED	TYPE (PLATE&FRAME)	QUANTITY (PLATES)	AREA (SQ.FT.)	CAPACITY (MBH)	HOT SIDE					COLD SIDE					UNIT SIZE (L"xW"xH")	OPER. WT. (LBS)	NOTES
									FLOW (GPM)	P.D. (FT.WG.)	EWT (°F)	LWT (°F)	PASSES	FLOW (GPM)	P.D. (FT.WG.)	EWT (°F)	LWT (°F)	PASSES			
PHX-1	BELL & GOSSETT		BASEMENT	PROCESS COOLING LOOP	PLATE&FRAME			TBD	100		75	65				42	TBD			1,5,6,7,9,10	
PHX-2	BELL & GOSSETT		BASEMENT	PROCESS COOLING LOOP	PLATE&FRAME			TBD	100		75	65				42	TBD			1,5,6,7,9,10	
NOTES																					
1. PROVIDE UNIT RATED FOR ASME WORKING PRESSURE OF 125 PSIG.																					
2. FOULING FACTOR NOT TO EXCEED 0.002 HR. FT ² . °F/BTU.																					
3. PLATES ALLOY 304/0.40 MM																					
4. PROVIDE MOUNTING LEGS.																					
5. PROVIDE SEISMIC STAND DESIGNED BY A LICENSED STRUCTURAL ENGINEER.																					
6. PERFORMANCE BASED ON COLD-SIDE WATER																					
7. GASKET MATERIAL SHALL BE (NBRB).																					
8. INSULATE EXTERIOR SURFACE AS REQUIRED PER SPECIFICATION.																					
9. PROCESS CHW HX RUN/STANDBY																					
10. ASSUMED 100 GPM ALLOWANCE AT SCHEMATIC DESIGN. FLUID FLOW TBD BY LAB CONSULTANT.																					

DUCT MOUNTED COIL SCHEDULE																					
TAG	AIR HANDLER TAG	MANUFACTURER	MODEL NUMBER	LOCATION	CAPACITY		AIR						WATER					CONTROL VALVE (2-W/3-W)	SIZE (L"xH")	OPER. WT. (LBS)	NOTES
					TOTAL (MBH)	SENSIBLE (MBH)	FLOW (CFM)	VEL. (FPM)	APD (IN.WG.)	EAT (DB °F)	EAT (WB °F)	LAT (DB °F)	LAT (WB °F)	EWT (°F)	LWT (°F)	FLOW (GPM)	WPD (FT.WG.)				
CC-1	-	-	-	CLEAN ROOM	-	-	2000	-	-	-	-	-	-	-	-	-	-	2-W	-	-	1
HC-1				CLEAN ROOM			2000											2-W			
NOTES																					
1. COPPER TUBES AND ALUMINUM FINs.																					
2. CHILLED WATER CONTAINS xx% PROPYLENE GLYCOL.																					

TAG	MANUFACTURER	MODEL NUMBER	SYSTEM SERVED	TYPE	TANK VOL. (GAL)	ACCEPT. VOL. (GAL)	SYSTEM VOL. (GAL)	FILL PRESS. (PSIG)	OPER. PRESS. (PSIG)	RELIEF PRESS. (PSIG)	MIN. TEMP. (°F)	MAX. TEMP. (°F)	SYSTEM CONN. (IN.)	SIZE DIA x H (IN)	OPER. WT. (LBS)	NOTES
ET-1	-	-	HW	BLADDER	-	-	-									1,2,3
ET-2			SCHW	BLADDER												1,2,3
NOTES																
1. PROVIDE TANK RATED FOR ASME WORKING PRESSURE OF 125 PSIG.																
2. PROVIDE TANK WITH VERTICAL CONFIGURATION.																
3. ANCHOR TO HOUSEKEEPING PAD. COORDINATE WITH STRUCTURAL DESIGN.																

DIFFUSER AND GRILLE SCHEDULE									
TAG	MANUFACTURER	MODEL NUMBER	DESCRIPTION	FACE TYPE	FACE SIZE (INCHES)	COLOR (NOTE #1)	MATERIAL	OBD	NOTES
SA									
RA									
EA									
NOTES									
1. MAXIMUM TOTAL PRESSURE DROP SHALL NOT EXCEED 0.15" WG WITH DUCT TRANSITION.									
2. MAXIMUM NC LEVEL SHALL BE (20 / 25 / 30).									
3. ALL VISIBLE SURFACES AND DUCTWORK BEHIND FACE SHALL BE PAINTED FLAT BLACK.									
4. COORDINATE WITH ARCHITECTURAL REFLECTED CEILING PLANS FOR BORDER TYPES.									
5. NECK SIZE AND CFM SHOWN ARE ON PLANS (EXAMPLE: SA12x12-400 REFERS TO TAG "SA" WITH 12x12 NECK AND 400 CFM).									
6. PROVIDE RECTANGULAR/SQUARE TO ROUND TRANSITION AS REQUIRED AND SIZED FOR MAXIMUM 0.01" WG TOTAL PRESSURE DROP.									
7. ADJUSTABLE HORIZONTAL / VERTICAL DISCHARGE									
8. ANY GRILLE/DIFFUSER MOUNTED DIRECTLY ON EXPOSED DUCT SHALL MATCH COLOR OF DUCT (IF PAINTED) OR SHALL HAVE CLEAR ANODIZED FINISH (IF DUCT IS UNPAINTED).									
9. PROVIDE BORDER TYPE (11 / 22 / xx) (WITH BORDER TYPE 2CRA FOR TWO-SLOT DIFFUSERS).									
10. PLENUM CONNECTION SIZE, SLOT LENGTH, AND CFM SHOWN ON PLANS (EXAMPLE: SN8-120-2000 REFERS TO TAG "SN" WITH 8" ROUND CONNECTION, 120" CONTINUOUS SLOT LENGTH AND 2000 CFM).									
11. PROVIDE MANUFACTURER'S INSULATED PLENUM.									
12. PROVIDE LIGHT SHIELD FOR UNDUCTED SLOT RETURN GRILLES.									
13. GRILLE SHALL BE OF ENTIRELY NON-FERROUS CONSTRUCTION FOR MRI IMAGING EQUIPMENT ROOM APPLICATION.									

PUMP SCHEDULE																
TAG	MANUFACTURER	MODEL NUMBER	SYSTEM SERVED (TAG)	TYPE	FLOW (GPM)	HEAD (FT. WG.)	MIN. EFF. (%)	NPSH (FT.)	RPM	ELECTRICAL				EMERG POWER (Y/N)	OPER. WT. (LBS)	NOTES
										BHP	HP	V/PH	VFD			
CHWP-1	BELL & GOSSETT	-	CHW LOOP	BASE-MOUNTED	330	-	-	-	-	-	7.5	460/3	Y	Y	-	1,2,3,4,5,6,7,8
CHWP-2	BELL & GOSSETT	-	CHW LOOP	BASE-MOUNTED	330	-	-	-	-	-	7.5	460/3	Y	N	-	1,2,3,4,5,6,7,8
SCHWP-1	BELL & GOSSETT	-	SCHW	BASE-MOUNTED	100	-	-	-	-	-	3	460/3	Y	Y	-	1,2,3,4,5,6,7,9,10
SCHWP-2	BELL & GOSSETT	-	SCHW	BASE-MOUNTED	100	-	-	-	-	-	3	460/3	Y	Y	-	1,2,3,4,5,6,7,9,10
HWP-1	BELL & GOSSETT	-	HW LOOP	BASE-MOUNTED	300	-	-	-	-	-	7.5	460/3	N	N	-	1,2,3,4,5,6,7,8
HWP-2	BELL & GOSSETT	-	HW LOOP	BASE-MOUNTED	300	-	-	-	-	-	7.5	460/3	N	N	-	1,2,3,4,5,6,7,8
NOTES																
1. COORDINATE WITH ELECTRICAL FOR POWER AND DISCONNECT AS REQUIRED.																
2. PROVIDE NON-OVERLOADING NEMA PREMIUM EFFICIENCY, INVERTER READY MOTOR.																
3. FOR PUMPS LOCATED INDOORS PROVIDE OPEN DRIPPROOF (ODP) MOTORS.																
4. PROVIDE VARIABLE FREQUENCY DRIVE (VFD) BY MECHANICAL AND WIRED BY ELECTRICAL.																
5. PROVIDE MOTOR SHAFT GROUNDING SYSTEM FOR MOTOR CONTROLLED BY VFD.																
6. PROVIDE INLET SUCTION DIFFUSER OR STRAINER. SEE PIPING DESIGN.																
7. PROVIDE VIBRATION ISOLATION AND SEISMIC RESTRAINT PER SPECIFICATIONS.																
8. PUMPS TO RUN LEAD/LAG.																
9. PROCESS CHW PUMP RUN/STANDBY																
10. ASSUMED 100 GPM ALLOWANCE AT SCHEMATIC DESIGN. FLUID FLOW TBD BY LAB CONSULTANT.																

FILTER SCHEDULE											
TAG	MANUFACTURER	MODEL NUMBER	SYSTEM SERVED	TYPE	MIN. EFF. (MERV)	AIR FLOW (CFM)	CLEAN PD (IN.WG.)	DIRTY PD (IN.WG.)	FACE VELOCITY (FPM)	OVERALL BANK SIZE (W"xH"xD")	NOTES
F-1	-		CLEAN ROOM SUPPLY	1"	TBD	-	0.2	0.5	500	-	1,2, 4
FF			CLEAN RM FAN FILTER UNITS		HEPA						5
NOTES											
1. FILTER EFFICIENCY SHALL BE PER ASHRAE STANDARD 52.2.											
2. REFER TO SPECIFICATIONS FOR FILTER FRAMES AND HOUSING TYPES TO BE PROVIDED.											
3. PROVIDE BAG-IN, BAG-OUT FILTER HOUSING.											
4. PROVIDE MAGNEHELIC DIFFERENTIAL PRESSURE GAUGE ACROSS FILTER BANK.											
5. PROVIDE 25% COVERAGE BY FAN FILTER UNITS IN NANOFAB CLEANROOM. TAG NOT SHOW ON PLANS, MAKE ALLOWANCE AS REQUIRED FOR QUANTITY BASED ON FLOOR PLAN AREA OF CLEANROOM											

ELECTRIC HUMIDIFIER SCHEDULE												
TAG	MANUFACTURER	HUMIDIFIER MODEL NUMBER	DISPERSION TUBE MODEL NUMBER	LOCATION	STEAM CAPACITY (LBS/HR)	DUCT AIRFLOW (CFM)	ELECTRICAL			UNIT SIZE ("L"xW"xH")	OPER. WT. (LBS)	NOTES
							KW	MCA (A)	V/PH			
EH-1	Dri-Steam	VLC-21-1	-	CLEAN ROOM SUPPLY DUCT	50	2000	21	-	480/3	-	-	1,2,3,4
NOTES												
1. COORDINATE WITH ELECTRICAL FOR POWER AND DISCONNECT AS REQUIRED.												
2. PROVIDE WITH HUMIDISTAT CONTROL. HUMIDIFIER SHALL BE ALL STAINLESS STEEL SUITABLE FOR DI/RO WATER.												
3. PROVIDE DRAIN COOLER WITH DOMESTIC WATER CONNECTION AND ROUTE DRAIN TO FLOOR SINK.												
4. LOCATE DISPERSION TUBE IN STRAIGHT DUCT SECTION WITH A MINIMUM OF 3 FEET ABSORPTION DISTANCE.												

SHELL AND TUBE HEAT EXCHANGER SCHEDULE																		
TAG	MANUFACTURER	MODEL NUMBER	LOCATION	TYPE (PLATE&FRAME)	HEATING CAPACITY (MBH)	WATER SIDE					STEAM SIDE					UNIT SIZE (L"xW"xH")	OPER. WT. (LBS)	NOTES
						FLOW (GPM)	P.D. (FT.WG.)	EWIT ("F)	LWT ("F)	PASSES	FLUID	LBS/HR	PSIG	PD	FLUID			
HX-1	BELL & GOSSETT	SU.....	BASEMENT	U-TUBE - 4 PASS		300		160	180	4	WATER	3000	15	TBD	STEAM			
HX-2	BELL & GOSSETT	SU.....	BASEMENT	U-TUBE - 4 PASS		300		160	180	4	WATER	3000	15	TBD	STEAM			
NOTES																		



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MECHANICAL SCHEDULES

SCHEMATIC DESIGN

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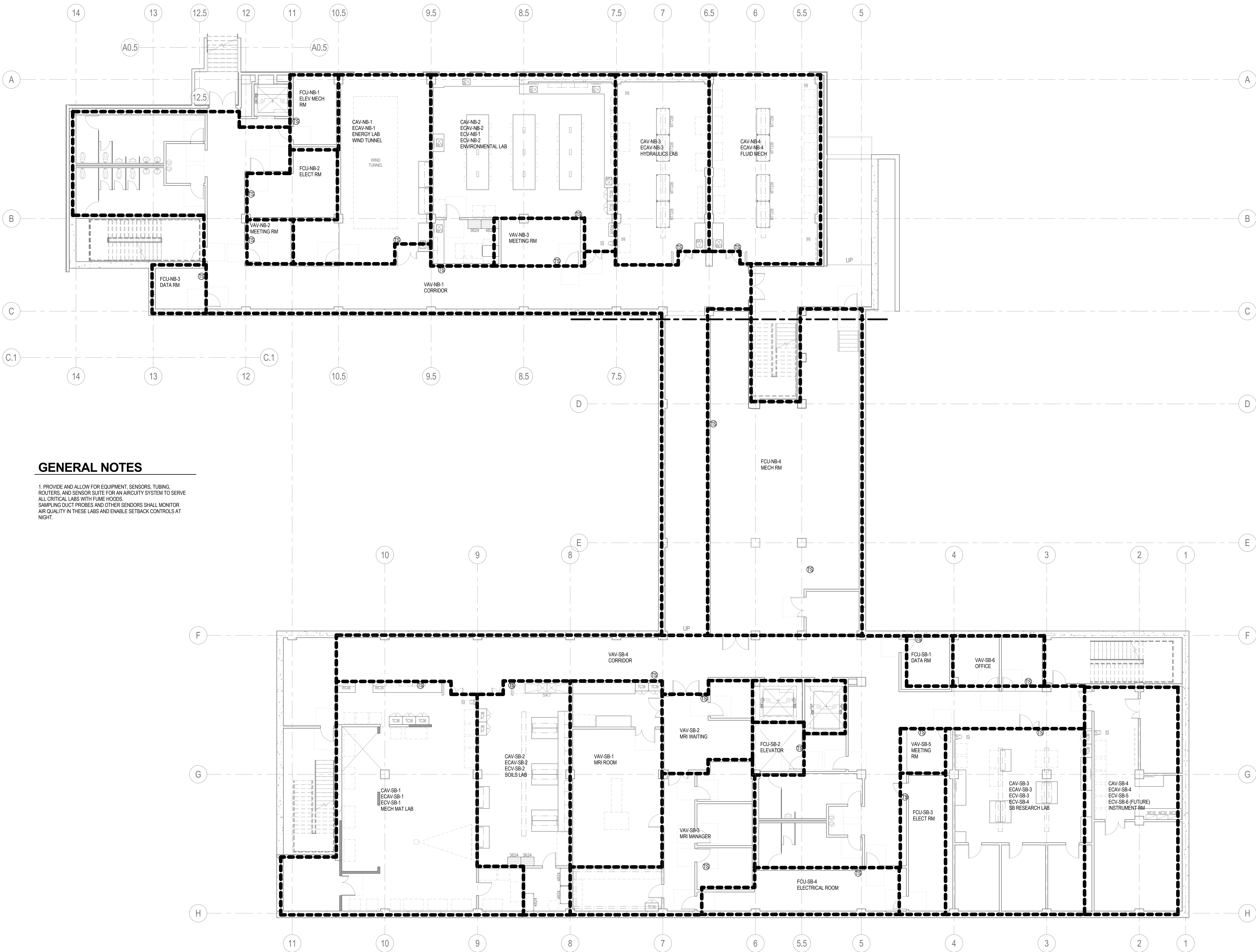
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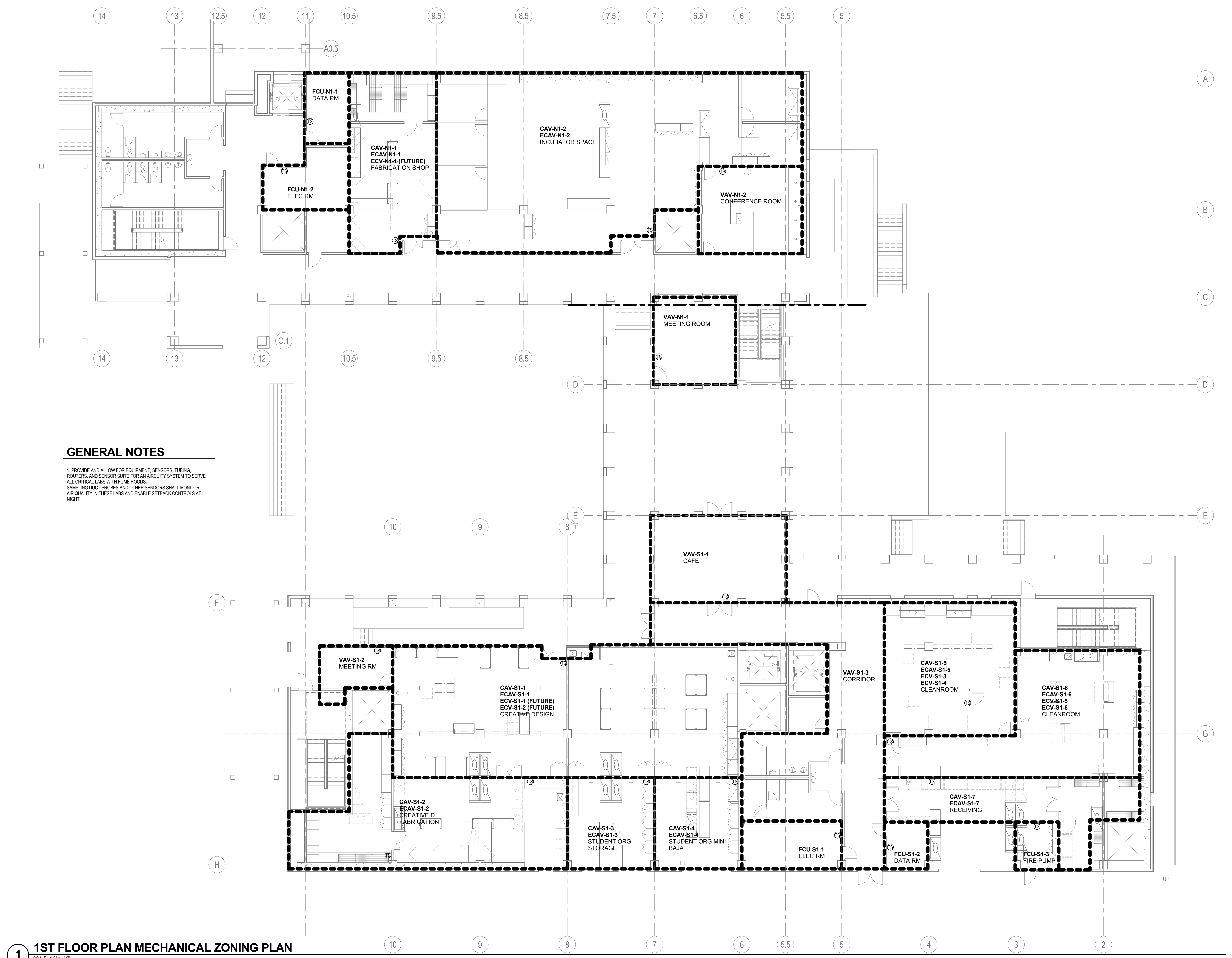
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GENERAL NOTES

1. PROVIDE AND ALLOW FOR EQUIPMENT, SENSORS, TUBING, ROUTERS, AND SENSOR SUITE FOR AN AIRCUTY SYSTEM TO SERVE ALL CRITICAL LABS WITH FUME HOODS. SAMPLING DUCT PROBES AND OTHER SENSORS SHALL MONITOR AIR QUALITY IN THESE LABS AND ENABLE SETBACK CONTROLS AT NIGHT.



GENERAL NOTES

1. PROVIDE AND ALLOW FOR EQUIPMENT, SENSORS, TUBING, ROUTERS, AND SENSOR SUITE FOR AN AIRCUTY SYSTEM TO SERVE ALL CRITICAL LABS WITH FUME HOODS. SAMPLING DUCT PROBES AND OTHER SENSORS SHALL MONITOR AIR QUALITY IN THESE LABS AND ENABLE SETBACK CONTROLS AT NIGHT.



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1ST FLOOR MECHANICAL ZONING PLAN

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RESEARCH

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GENERAL NOTES

1. PROVIDE AND ALLOW FOR EQUIPMENT, SENSORS, TUBING, ROUTERS, AND SENSOR SUITE FOR AN AIRCUTY SYSTEM TO SERVE ALL CRITICAL LABS WITH FUME HOODS. SAMPLING DUCT PROBES AND OTHER SENDORS SHALL MONITOR AIR QUALITY IN THESE LABS AND ENABLE SETBACK CONTROLS AT NIGHT.

1 2ND FLOOR PLAN - MECHANICAL ZONING PLAN
SCALE: 1/8" = 1'-0"



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2ND FLOOR MECHANICAL ZONING PLAN

SCHEMATIC DESIGN

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5500 Campanile Drive San Diego, CA 92182

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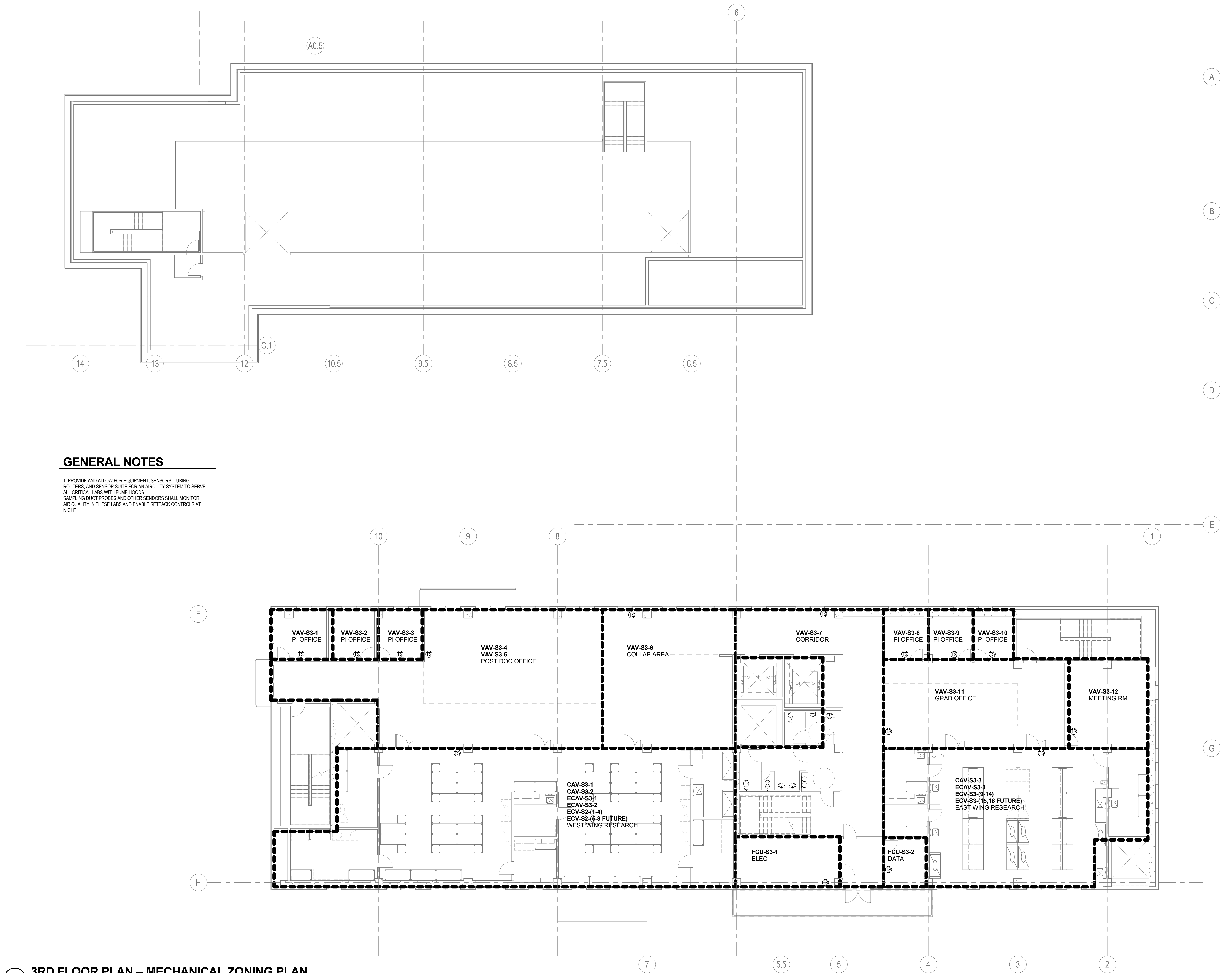


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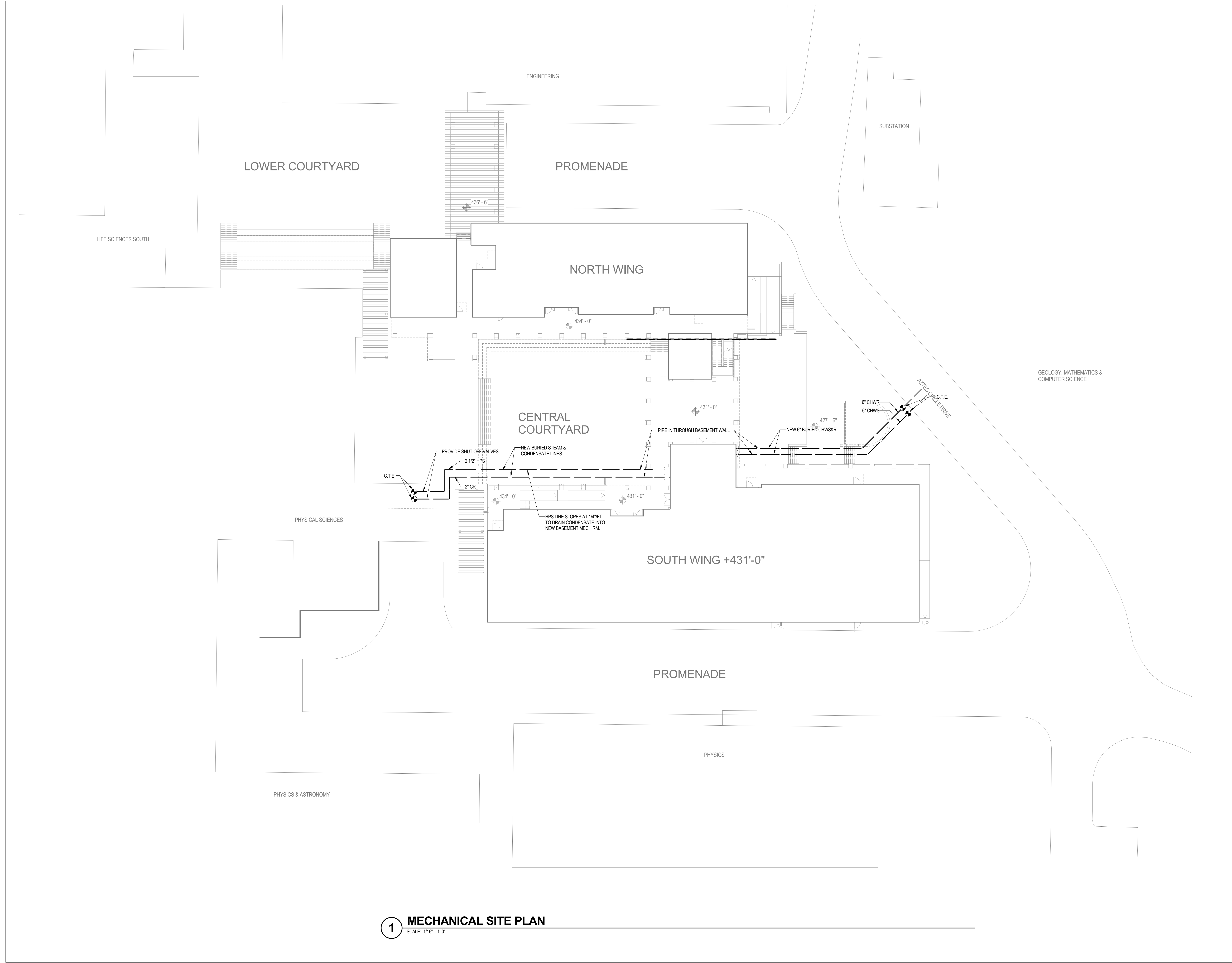
San Diego State University

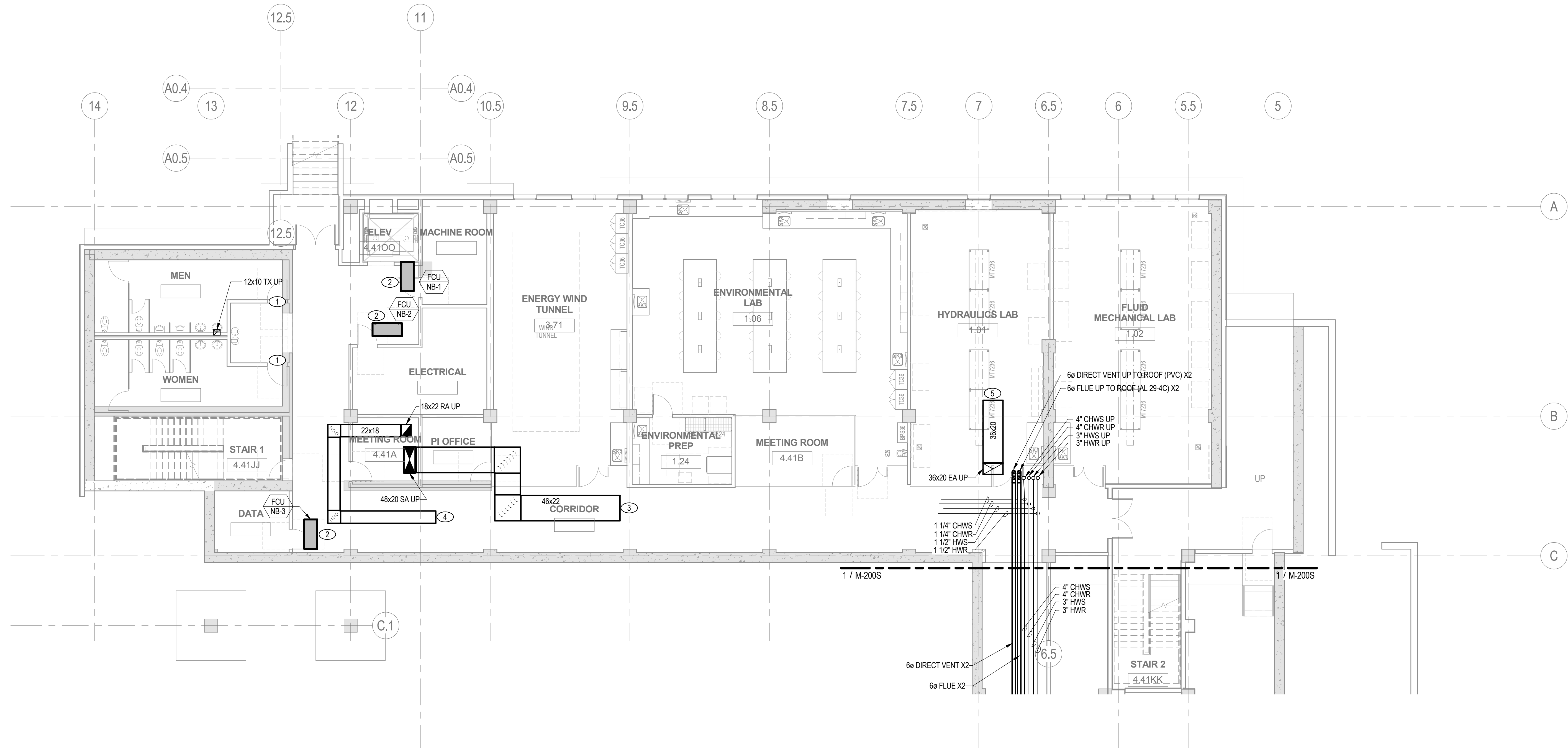
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1 3RD FLOOR PLAN – MECHANICAL ZONING PLAN
SCALE: 1/8" = 1'-0"





1 BASEMENT FLOOR PLAN NORTH - MECHANICAL
SCALE: 1/8" = 1'-0"

SHEET NOTES

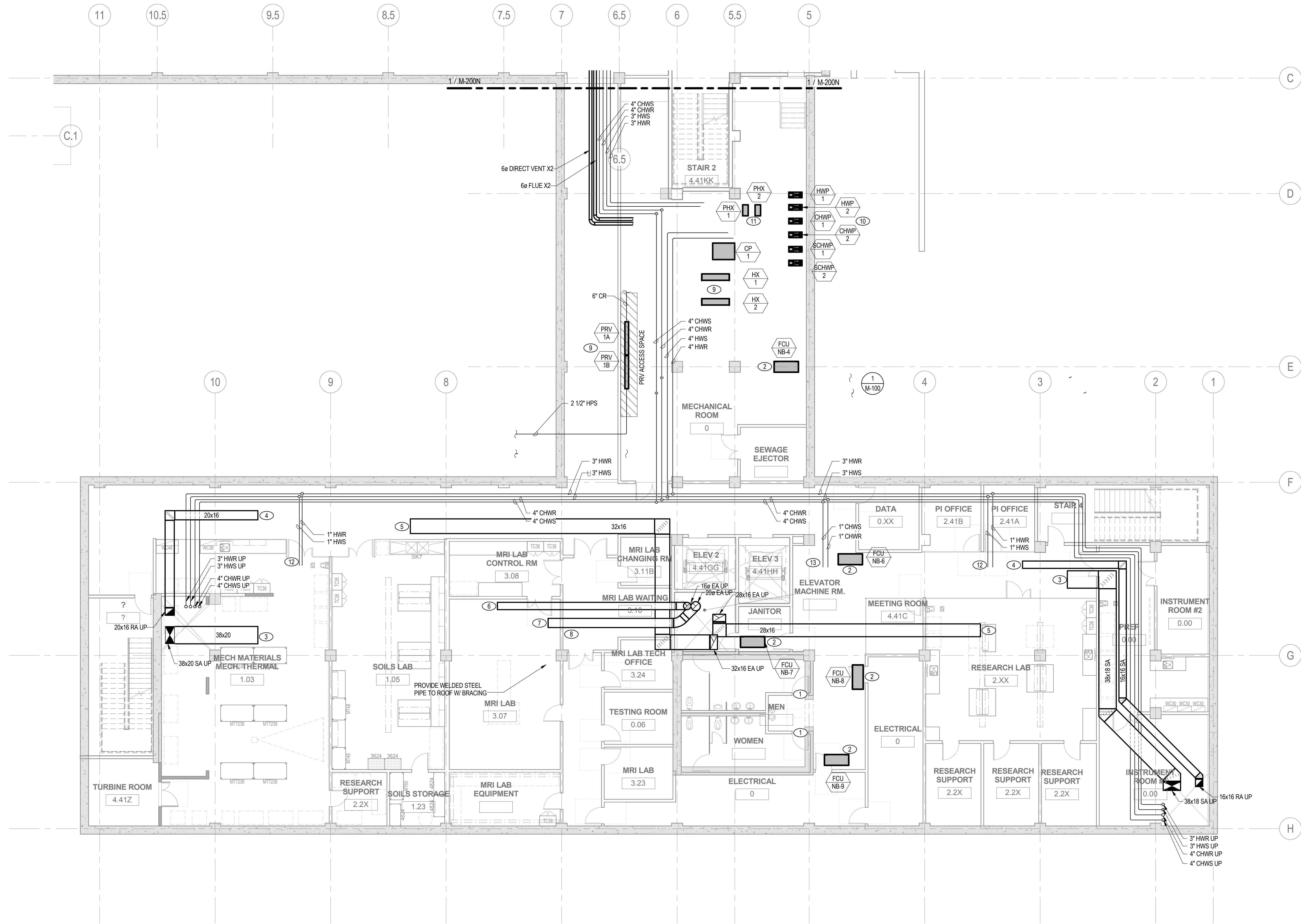
- A. REFER TO ZONING PLAN ON DWG. M-005 FOR PROPOSED ZONING STRATEGY AND ALLOWANCES

GENERAL NOTES

1. PROCESS PIPING IS NOT SHOWN ON THESE PLANS. ROUTING FOR ANY PROCESS PIPING SHALL BE PROVIDED AND COORDINATED IN DETAILED DESIGN PHASE. CONTRACTOR SHALL ALLOW FOR 100 GPM OF PROCESS FLOW (EXACT FLOW TBD IN DD PHASE). ALLOW FOR A 3" PROCESS MAIN AND A 2-1/2" RISER ON THE NORTH AND SOUTH WINGS RESPECTIVELY TO SERVE PROCESS WATER NEEDS.

KEYED NOTES

1. PROVIDE 28" X 12" TRANSFER DUCT ACROSS WALL TO BATHROOM WITH FSD. PROVIDE WITH 90 DEGREE ELBOW & 1" A.L. FOR SOUND ATTENUATION.
2. MAKE ALLOWANCE FOR SUPPLY AND RETURN DUCTWORK. CONNECT FANS TO/FROM FCU AND ALLOW FOR FSD AT RATED WALL PENETRATIONS TO SPACES SERVED.
3. SUPPLY MAIN SHALL BE ROUTED IN CEILING TO SERVE ALL ZONES ON THIS FLOOR. SEE RISER DIAGRAM ON DWG. M-700 FOR ZONES SERVED AND ASSOCIATED CAV AND VAV REQUIRED.
4. RETURN DUCTWORK SHALL BE ROUTED IN CEILING TO SERVE ALL NON-LAB SPACES RETURN AIR NEEDS. SEE RISER DIAGRAM ON DWG. M-700 FOR ALL ZONES SERVED.
5. EXHAUST AIR DUCT SHALL BE ROUTED IN CEILING TO SERVE ALL LAB EXHAUST AND FUME HOOD EXHAUST AT THIS LEVEL AND IN THIS WING. EXHAUST RISER SHALL BE ROUTED INDIVIDUALLY TO ROOF AND MANIFOLDS TO MAIN EXHAUST DUCT AT ROOF LEVEL.



1 BASEMENT FLOOR PLAN SOUTH - MECHANICAL

SCALE: 1/8" = 1'-0"

SHEET NOTES

- A. REFER TO ZONING PLAN ON DWG. M-005 FOR PROPOSED ZONING STRATEGY AND ALLOWANCES.

GENERAL NOTES

1. PROCESS PIPING IS NOT SHOWN ON THESE PLANS. ROUTING FOR ANY PROCESS PIPING SHALL BE PROVIDED AND COORDINATED IN DETAILED DESIGN PHASE. CONTRACTOR SHALL ALLOW FOR 100 GPM OF PROCESS FLOW (EXACT FLOW TBD IN DD PHASE). ALLOW FOR A 3" PROCESS MAIN AND A 2-1/2" RISER ON THE NORTH AND SOUTH WINGS RESPECTIVELY TO SERVE PROCESS WATER NEEDS.

KEYED NOTES

1. PROVIDE 26" X 12" TRANSFER DUCT ACROSS WALL TO BATHROOM WITH FSD. PROVIDE WITH 90 DEGREE ELBOW & 1" A.L. FOR SOUND ATTENUATION.
2. MAKE ALLOWANCE FOR SUPPLY AND RETURN DUCTWORK. CONNECT FANS TO/FROM FCU AND ALLOW FOR FSD AT RATED WALL PENETRATIONS TO SPACES SERVED.
3. SUPPLY MAIN SHALL BE ROUTED IN CEILING TO SERVE ALL ZONES ON THIS FLOOR. SEE RISER DIAGRAM ON DWG. M-700 FOR ZONES SERVED AND ASSOCIATED CAV AND VAV REQUIRED.
4. RETURN DUCTWORK SHALL BE ROUTED IN CEILING TO SERVE ALL NON-LAB SPACES RETURN AIR NEEDS. SEE RISER DIAGRAM ON DWG. M-700 FOR ALL ZONES SERVED.
5. EXHAUST AIR DUCT SHALL BE ROUTED IN CEILING TO SERVE ALL LAB EXHAUST AND FUME HOOD EXHAUST AT THIS LEVEL AND IN THIS WING. EXHAUST RISER SHALL BE ROUTED INDIVIDUALLY TO ROOF AND MANIFOLDS TO MAIN EXHAUST DUCT AT ROOF LEVEL.
6. MRI ROOM EXHAUST, TERMINATE WITH EXHAUST GRILLE ON CEILING. ALL DUCTWORK ROUTED IN MRI RM. SHALL BE ALUMINUM.
7. MRI QUENCH DUCT. TERMINATE AT MAGNET QUENCH FLANGE. PROVIDE NON-MAGNETIC STAINLESS STEEL CONSTRUCTION.
8. PROVIDE RF WAVE GUIDE AT ANY PENETRATION THROUGH MRI RM WALL.
9. SEE DETAIL 4 ON DRAWING M-501 FOR HEAT TRANSFER PACKAGE.
10. SEE DETAIL 12 ON DRAWING M-500 FOR BASE MOUNTED END SUCTION PUMPS ON VIBRATION ISOLATOR.
11. SEE DETAIL 3 ON DRAWING M-501 FOR PLATE AND FRAME HEAT EXCHANGERS.
12. ROUTE HW PIPES OUT TO SERVE ALL TERMINAL VAV BOXES AND LAB CONSTANT FLOW SUPPLY VALVES HEATING COILS.
13. ROUTE ALL CHW PIPES OUT TO SERVE ALL FCU'S



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BASEMENT FLOOR PLAN SOUTH - MECHANICAL

SCHEMATIC DESIGN

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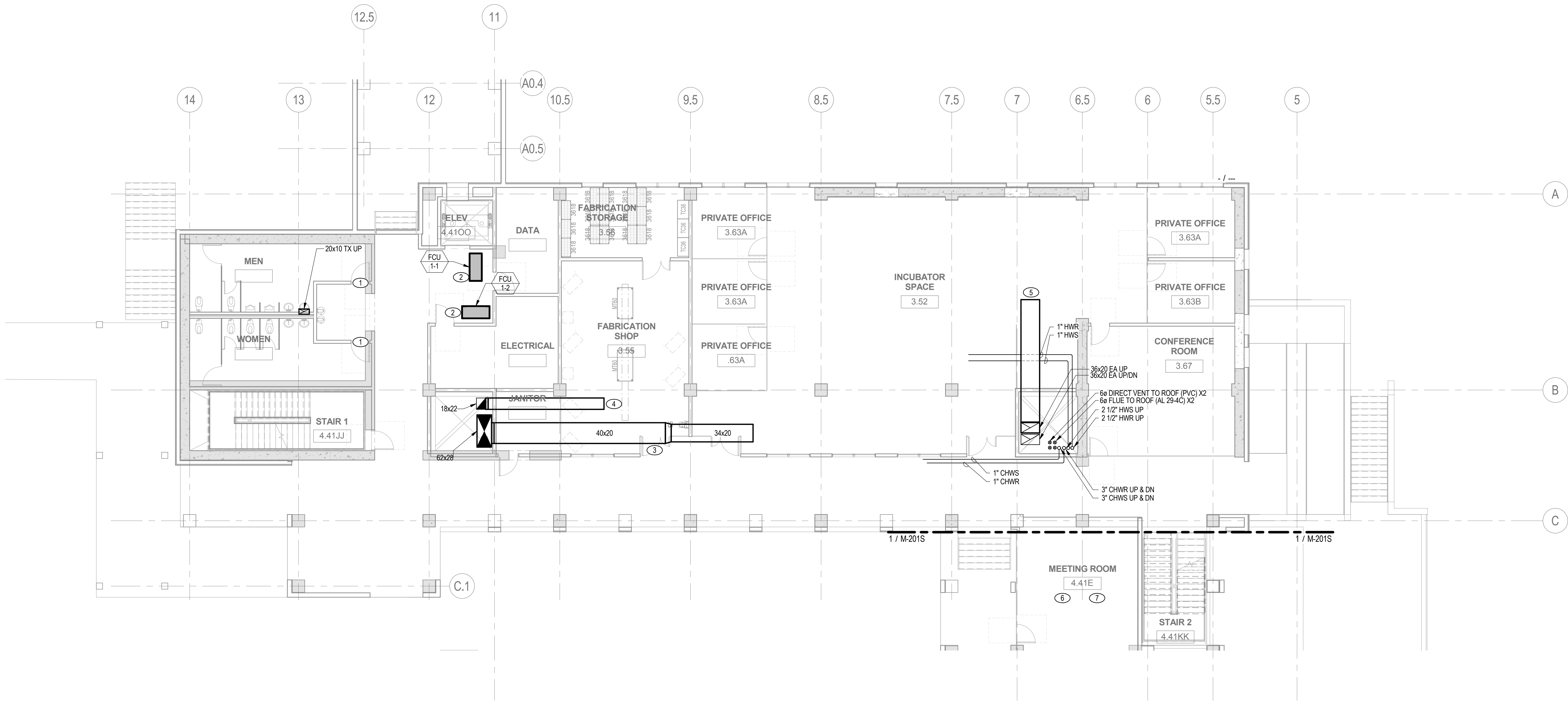
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1 1ST FLOOR PLAN NORTH - MECHANICAL
SCALE: 1/8" = 1'-0"

SHEET NOTES

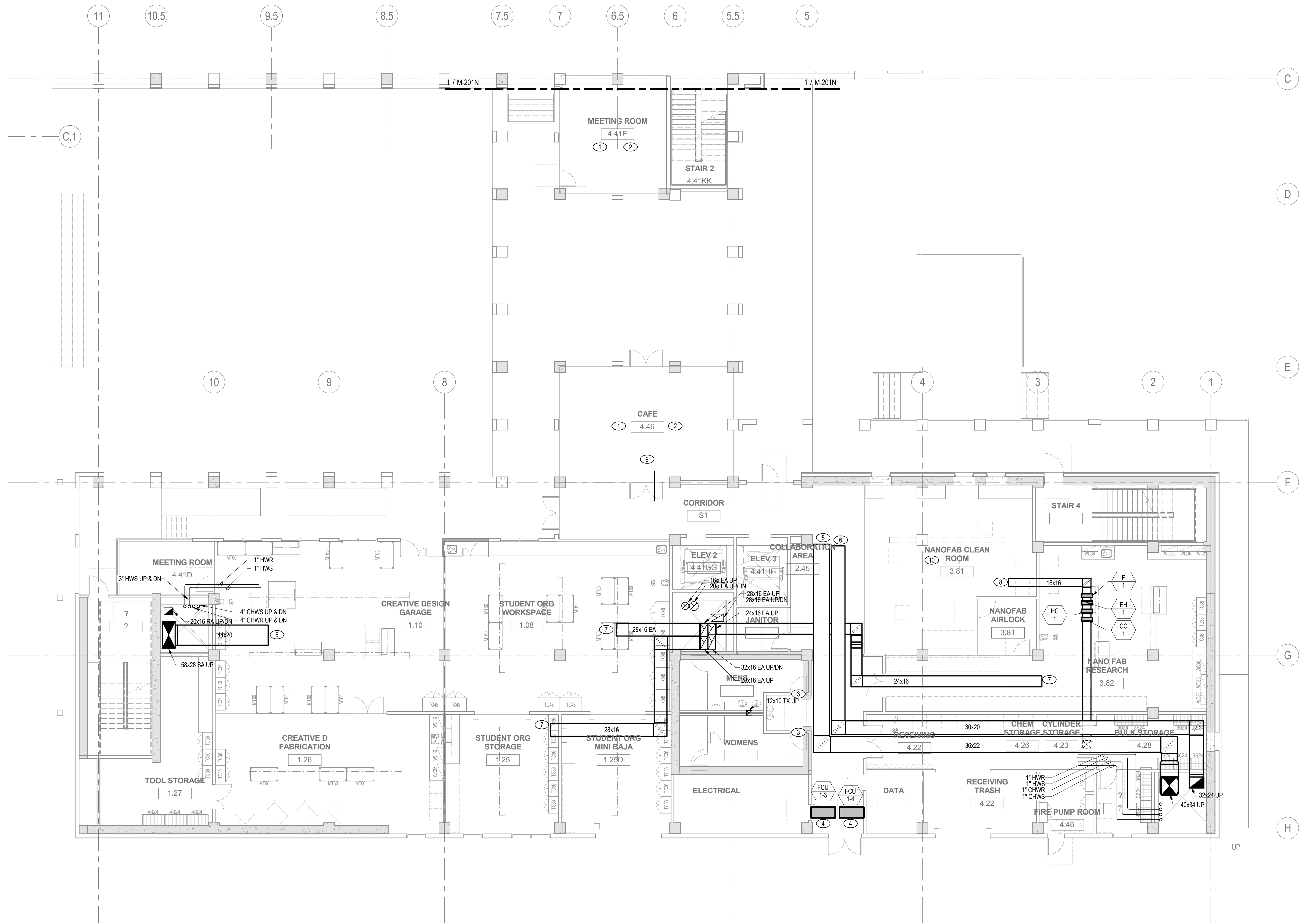
A. REFER TO ZONING PLAN ON DWG. M-006 FOR PROPOSED ZONING STRATEGY AND ALLOWANCES

GENERAL NOTES

1. PROCESS PIPING IS NOT SHOWN ON THESE PLANS. ROUTING FOR ANY PROCESS PIPING SHALL BE PROVIDED AND COORDINATED IN DETAILED DESIGN PHASE. CONTRACTOR SHALL ALLOW FOR 100 GPM OF PROCESS FLOW (EXACT FLOW TBD IN DD PHASE). ALLOW FOR A 3" PROCESS MAIN AND A 2-1/2" RISER ON THE NORTH AND SOUTH WINGS RESPECTIVELY TO SERVE PROCESS WATER NEEDS.

KEYED NOTES

- 1 PROVIDE 28" X 12" TRANSFER DUCT ACROSS WALL TO BATHROOM WITH FSD. PROVIDE WITH 90 DEGREE ELBOW & 1" A.L. FOR SOUND ATTENUATION.
- 2 MAKE ALLOWANCE FOR SUPPLY AND RETURN DUCTWORK. CONNECT FANS TO/FROM FCU AND ALLOW FOR FSD AT RATED WALL PENETRATIONS TO SPACES SERVED.
- 3 SUPPLY MAIN SHALL BE ROUTED IN CEILING TO SERVE ALL ZONES ON THIS FLOOR. SEE RISER DIAGRAM ON DWG. M-700 FOR ZONES SERVED AND ASSOCIATED CAV AND VAV REQUIRED.
- 4 RETURN DUCTWORK SHALL BE ROUTED IN CEILING TO SERVE ALL NON-LAB SPACES RETURN AIR NEEDS. SEE RISER DIAGRAM ON DWG. M-700 FOR ALL ZONES SERVED.
- 5 EXHAUST AIR DUCT SHALL BE ROUTED IN CEILING TO SERVE ALL LAB EXHAUST AND FUME HOOD EXHAUST AT THIS LEVEL AND IN THIS WING. EXHAUST RISER SHALL BE ROUTED INDIVIDUALLY TO ROOF AND MANIFOLDS TO MAIN EXHAUST DUCT AT ROOF LEVEL.
- 6 THIS ROOM SHALL BE PROVIDED WITH WINDOW SWITCHES TO SHUT OFF VAV BOX WHEN WINDOWS ARE OPENED.
- 7 THIS ROOM SHALL BE PROVIDED WITH CEILING FANS TO RUN WHEN THE WINDOWS ARE OPENED.



1 1ST FLOOR PLAN SOUTH - MECHANICAL
SCALE: 1/8" = 1'-0"

SHEET NOTES

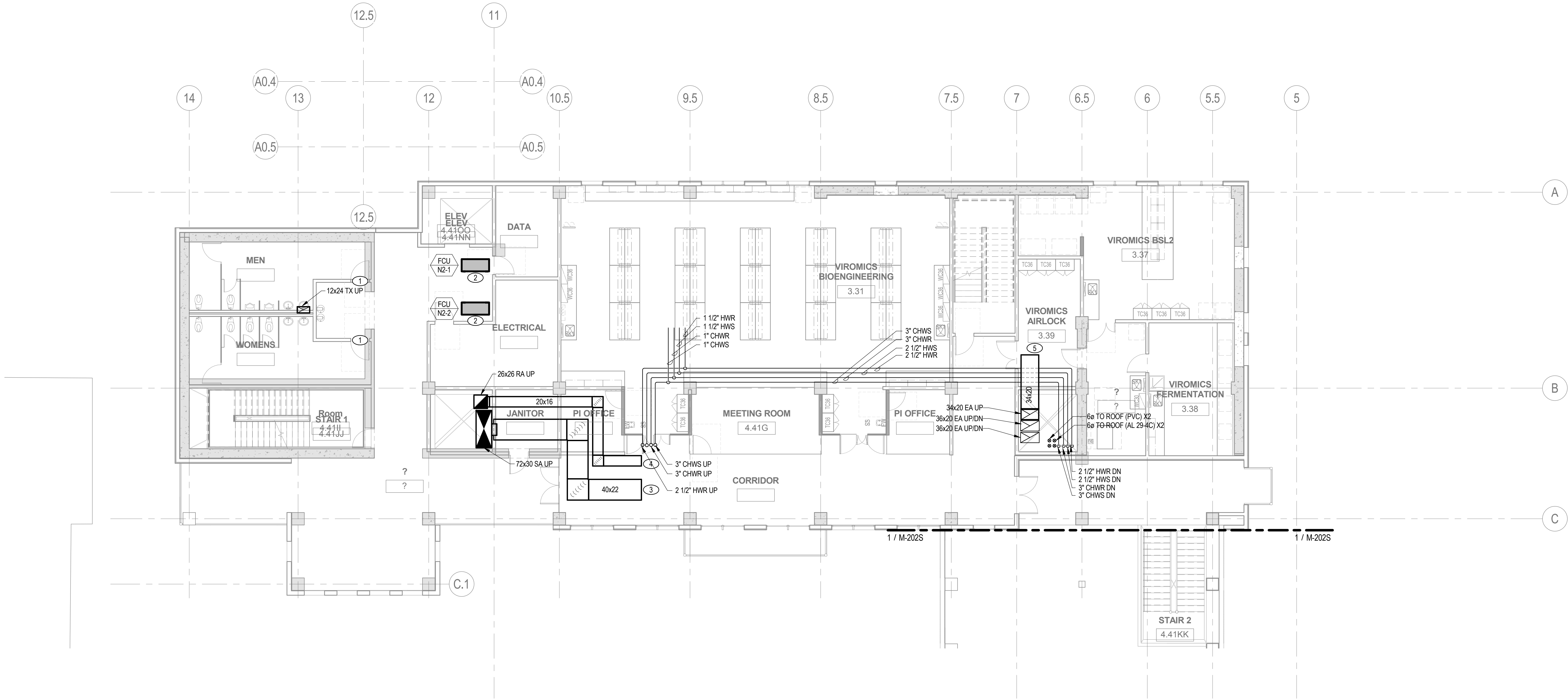
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GENERAL NOTES

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KEYED NOTES

- THIS ROOM SHALL BE PROVIDED WITH WINDOW SWITCHES TO SHUT OFF VAV BOX WHEN WINDOWS ARE OPENED.
- THIS ROOM SHALL BE PROVIDED WITH CEILING FANS TO RUN WHEN THE WINDOWS ARE OPENED.
- PROVIDE 26" X 12" TRANSFER DUCT ACROSS WALL TO BATHROOM WITH FSD. PROVIDE WITH 90 DEGREE ELBOW & 1" A.L. FOR SOUND ATTENUATION.
- MAKE ALLOWANCE FOR SUPPLY AND RETURN DUCTWORK, CONNECT FANS TO/FROM FCU AND ALLOW FOR FSD AT RATED WALL PENETRATIONS TO SPACES SERVED.
- SUPPLY MAIN SHALL BE ROUTED IN CEILING TO SERVE ALL ZONES ON THIS FLOOR. SEE RISER DIAGRAM ON DWG. M-700 FOR ZONES SERVED AND ASSOCIATED CAV AND VAV REQUIRED.
- RETURN DUCTWORK SHALL BE ROUTED IN CEILING TO SERVE ALL NON-LAB SPACES RETURN AIR NEEDS. SEE RISER DIAGRAM ON DWG. M-700 FOR ALL ZONES SERVED.
- EXHAUST AIR DUCT SHALL BE ROUTED IN CEILING TO SERVE ALL LAB EXHAUST AND FUME HOOD EXHAUST AT THIS LEVEL, AND IN THIS WING. EXHAUST RISER SHALL BE ROUTED INDIVIDUALLY TO ROOF AND MANIFOLDS TO MAIN EXHAUST DUCT AT ROOF LEVEL.
- CLEANROOM FRESH AIR SUPPLY DUCT WITH DUCT MOUNTED COOLING COIL, HEATING COIL, AND ELECTRIC STEAM HUMIDIFIER.
- PROVIDE 25% COVERAGE IN THE CLEAN BAYS. PROVIDE LOW WALL RETURNS. RETURN THE AIR TO THE CEILING SPACE THROUGH THE GREY ROOMS. PROVIDE 32 FAN FILTER UNITS IN THE CLEANROOM CEILING.



1 2ND FLOOR PLAN NORTH - MECHANICAL
SCALE: 1/8" = 1'-0"

SHEET NOTES

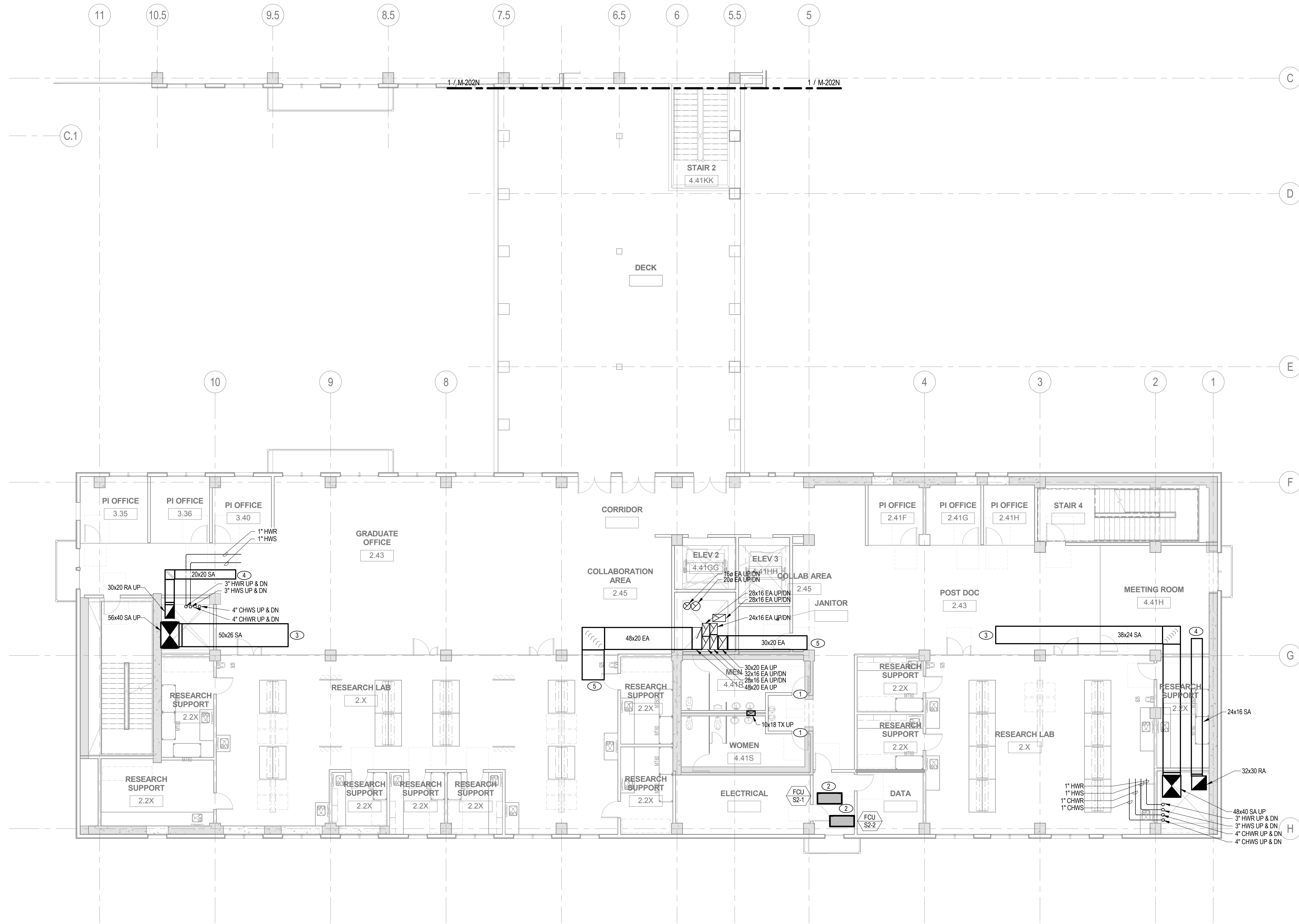
A. REFER TO ZONING PLAN ON DWG. M-007 FOR PROPOSED ZONING STRATEGY AND ALLOWANCES

GENERAL NOTES

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KEYED NOTES

1. PROVIDE 28" X 12" TRANSFER DUCT ACROSS WALL TO BATHROOM WITH FSD. PROVIDE WITH 90 DEGREE ELBOW & 1" A.L. FOR SOUND ATTENUATION.
2. MAKE ALLOWANCE FOR SUPPLY AND RETURN DUCTWORK. CONNECT FANS TO/FROM FCU AND ALLOW FOR FSD AT RATED WALL PENETRATIONS TO SPACES SERVED.
3. SUPPLY MAIN SHALL BE ROUTED IN CEILING TO SERVE ALL ZONES ON THIS FLOOR. SEE RISER DIAGRAM ON DWG. M-700 FOR ZONES SERVED AND ASSOCIATED CAV AND VAV REQUIRED.
4. RETURN DUCTWORK SHALL BE ROUTED IN CEILING TO SERVE ALL NON-LAB SPACES RETURN AIR NEEDS. SEE RISER DIAGRAM ON DWG. M-700 FOR ALL ZONES SERVED.
5. EXHAUST AIR DUCT SHALL BE ROUTED IN CEILING TO SERVE ALL LAB EXHAUST AND FUME HOOD EXHAUST AT THIS LEVEL AND IN THIS WING. EXHAUST RISER SHALL BE ROUTED INDIVIDUALLY TO ROOF AND MANIFOLDS TO MAIN EXHAUST DUCT AT ROOF LEVEL.



1 2ND FLOOR PLAN SOUTH - MECHANICAL
SCALE: 1/8" = 1'-0"

SHEET NOTES

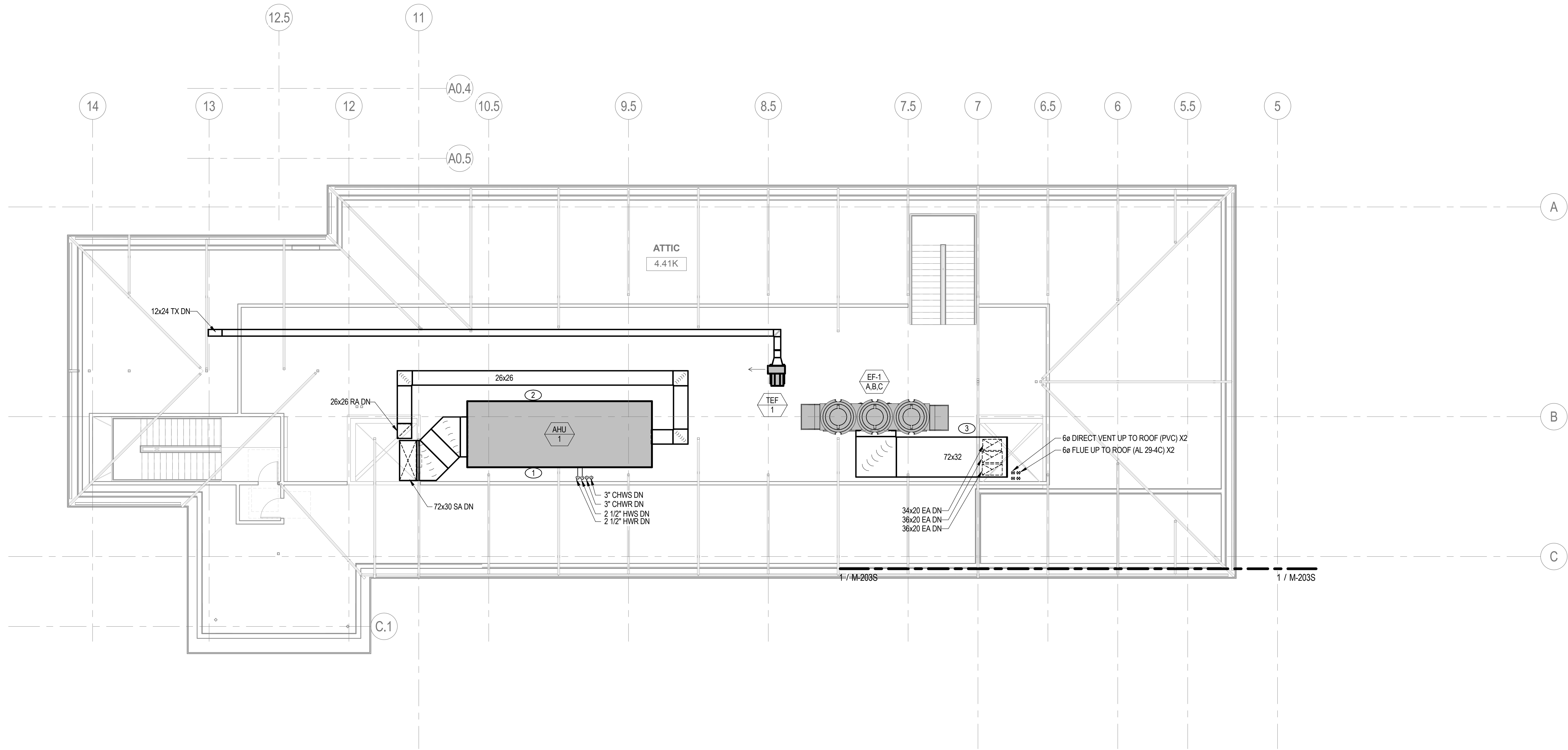
A. REFER TO ZONING PLAN ON DWG. M-007 FOR PROPOSED ZONING STRATEGY AND ALLOWANCES

GENERAL NOTES

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KEYED NOTES

1. PROVIDE 26" X 12" TRANSFER DUCT ACROSS WALL TO BATHROOM WITH FSD. PROVIDE WITH 90 DEGREE ELBOW & 1" A.L. FOR SOUND ATTENUATION.
2. MAKE ALLOWANCE FOR SUPPLY AND RETURN DUCTWORK. CONNECT FANS TO FROM FOU AND ALLOW FOR FSD AT RATED WALL PENETRATIONS TO SPACES SERVED.
3. SUPPLY MAIN SHALL BE ROUTED IN CEILING TO SERVE ALL ZONES ON THIS FLOOR. SEE RISER DIAGRAM ON DWG. M-700 FOR ZONES SERVED AND ASSOCIATED CAV AND VAV REQUIRED.
4. RETURN DUCTWORK SHALL BE ROUTED IN CEILING TO SERVE ALL NON-LAB SPACES RETURN AIR NEEDS. SEE RISER DIAGRAM ON DWG. M-700 FOR ALL ZONES SERVED.
5. EXHAUST AIR DUCT SHALL BE ROUTED IN CEILING TO SERVE ALL LAB EXHAUST AND FUME HOOD EXHAUST AT THIS LEVEL AND IN THIS WING. EXHAUST RISER SHALL BE ROUTED INDIVIDUALLY TO ROOF AND MANIFOLDS TO MAIN EXHAUST DUCT AT ROOF LEVEL.



1 3RD FLOOR PLAN NORTH – MECHANICAL
SCALE: 1/8" = 1'-0"

- KEYED NOTES**
- 1 OUTSIDE AIR TO BE DRAWN IN UNDER ROOF OVERHANG. ALLOW SUFFICIENT SPACE FOR ADEQUATE AIR CIRCULATION PER AHU MANUFACTURERS RECOMMENDATIONS.
 - 2 RELIEF AIR LOUVER
 - 3 MANIFOLD ALL EXH DUCTS FORM LABS TO MAIN MUCT AT ROOF (TOP OF SHAFT).